

COMPLETE MODULI SPACES OF COBLE SURFACES

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ABSTRACT. We describe a modular KSBA stable pair compactification of the moduli space of Coble surfaces and compare it to a semitoroidal compactification of the corresponding period domain. We further describe KSBA stable limits in terms of integral affine structures.

CONTENTS

1. Introduction	3
1A. Overview	3
2. Surfaces and Moduli	5
2A. Surfaces: K3, Enriques, and rational	5
2B. K3 double covers and their involutions	7
2C. Numerical polarizations and Noether–Lefschetz loci	9
3. Coble Surfaces	10
3A. Basics of Coble surfaces	10
3B. K3 covers	10
3C. Calculations	12
4. Lattices and Moduli	13
4A. Lattice Theory	13
4B. Root systems and root lattices	15
4C. Discriminant forms and the genus	18
4D. Embeddings, overlattices, and gluing	21
4E. Special lattices and transformations	26
4F. Invariant and coinvariant lattices	30
4G. Lattice Summary	31
4H. Coble lattices	32
4I. Nodal Enriques Surfaces	33
4J. Moduli	33
4K. Moduli Spaces Summary	34
4L. GIT Discussion	35
4M. Horikawa model	35
4N. Period Domains	35
4O. KSBA Spaces	36
5. Period Domains and Compactifications	36
5A. Hyperbolic reflection groups and Vinberg’s algorithm	36
5B. Foldings of Dynkin diagrams	40
5C. Toroidal and semitoroidal compactifications	41
5D. Scattone’s method for cusp enumeration	43
6. Degenerations	46
6A. Kulikov models and limiting mixed Hodge structures	46
6B. Anticanonical pairs, charge, and MMP singularities	47
6C. Divisor models and dlt degenerations	48
6D. Halphen Surfaces	49
6E. Morrison’s degenerations	50
7. Cusp Correspondence	50

7A.	Cusp Diagrams	50
7B.	Coble cusps	52
7C.	Enriques cusps	53
7D.	Sterk and K3 cusps	55
7E.	Enriques to Coble correspondence	56
8.	Stable Limits	59
8A.	Geometric preliminaries	59
8B.	Integral affine structures	60
8C.	KSBA stable limits	66
9.	The Moduli Space of Degree-2 Enriques Surfaces	68
9A.	The moduli space $F_{\text{En},2}$: main results	68
9B.	KSBA and semitoroidal compactifications of $F_{\text{En},2}$	71
9C.	The classifying map and the Enriques involution	72
10.	Open Problems	73
10A.	Open problems and technical gaps	73
11.	Appendix	76
11A.	Formulae	76
11B.	Diagrams	76
12.	Extracted References	79
12A.	Extracted references	79
	References	84

1. INTRODUCTION

1A. Overview.

Remark 1A.1: A Coble surface is a smooth projective rational surface S with $|-K_S| = \emptyset$ but $|-2K_S| \neq \emptyset$. Such surfaces arise from the work of [Cob19] and [Cob29] on Cremona transformations of $\mathbf{P}^2$ preserving an irreducible rational sextic C with ten nodes – the blowup S along these nodes yields a Coble surface. [Cob19] shows that the Cremona class of such a curve C can be written as a union of finitely many projective equivalence classes, and that if C is sufficiently general then $\text{Aut}(S) \cong W(E_{10})$, the Weyl group of an infinite root system of type E_{10} . Moreover, he shows that modulo $\text{Aut}(S)$, there are only finitely many smooth rational negative curves on S .

Coble surfaces occur as degenerations of Enriques surfaces and were ultimately classified in [DZ99]. As such, they are intimately tied to the theory of algebraic K3 surfaces equipped with a nonsymplectic involution, which were classified by [Nik79]. For a reduced plane sextic C , the double cover of S branched along the proper transform of C is a K3 surface X which can be realized as a degeneration of the universal double cover of an Enriques surface, where X acquires an A_1 singularity fixed by the Enriques involution. The resulting quotient has a quartic singularity whose resolution is an irreducible smooth rational curve $\tilde{C}$ satisfying $\tilde{C}^2 = -4$, and thus by [Nue15, §3] is a Coble surface.

We will be interested in Coble surfaces in the strong sense, where $|-2K_S| = \{C_1 + \cdots + C_n\}$ is a single reduced divisor comprised of n disjoint smooth rational curves referred to as *boundary components* of S . It is known that $1 \leq n \leq 10$, and for each such n , there is a moduli space $F_{\text{Co},n}$ of Coble surfaces with n boundary components. When $n = 1$, the moduli space $F_{\text{Co}} := F_{\text{Co},1}$ of Coble surfaces can be described as a boundary divisor $\mathcal{H}_{-2}$ in the 10-dimensional moduli space F_{En} of unpolarized Enriques surfaces, and thus F_{Co} is 9-dimensional. Moreover, F_{Co} was shown to be rational in [DK13] via a comparison to a moduli space of cuspidal plane quintics. A natural question is whether or not F_{Co} admits a geometrically meaningful, modular compactification $\overline{F_{\text{Co}}}$, and if so, if the boundary $\partial \overline{F_{\text{Co}}}$ can be described and classified. Toward this end, we turn to the stable pair compactifications of Kollár, Shepherd-Barron, and Alexeev [KS88; Ale96; Kol23].

Remark 1A.2: The search for modular compactifications of moduli spaces is a central problem in algebraic geometry. The prototypical example stems from the work of [DM69; KM76; Knu83] on the Deligne-Mumford-Knudsen compactification $\overline{\mathcal{M}}_{g,n}$ of the moduli space of stable pointed curves $\mathcal{M}_{g,n}$. By [Mum65; Nam76; AN99; Ale02], a similarly modular compactification $\overline{\mathcal{A}}_g$ of the moduli space $\mathcal{A}_g$ of principally polarized abelian varieties via stable pairs exists, the normalization of which coincides with a particular choice of toroidal [Ash+10] compactification by the work of [KS88] and [Ale96]. By [PS71], the coarse moduli space F_{2d} of primitively polarized K3 surfaces of degree $2d$ is isomorphic to an arithmetic quotient $D_{L_{2d}}/\Gamma_{L_{2d}}$ of a Type IV Hermitian symmetric domain associated to a lattice L_{2d} , where $D_{L_{2d}}$ is a period domain classifying Hodge structures of K3 type and $\Gamma_{L_{2d}}$ is an arithmetic subgroup of the orthogonal group $O(L_{2d})$. By [BB66], such domains admit a Baily-Borel compactification which is a projective variety. For F_{2d} and related lattice-polarized moduli spaces F_S of K3 surfaces, the boundary of the compactification consists of a configuration of 0-cusps (points) and 1-cusps (curves), see [Sca87]. See fig. 1. By [Ash+10], there additionally exist infinitely many toroidal compactifications described by a choice of fans at each cuspidal point of the Baily-Borel compactification. The work of [Loo85] on semitoroidal compactifications simultaneously generalizes the Baily-Borel and toroidal compactifications, retaining the advantage that the boundary can be understood in terms of semifans and studied using toric geometry.

Semitoroidal compactifications are advantageous due to their explicit and combinatorial nature, but do not a priori carry a clear modular interpretation along the boundary strata. Alternatively, letting (S, R) be a pair consisting of a surface and a suitably chosen divisor, the work of [KS88; Ale96; Kol23] yields a compactification obtained by taking the closure of pairs $(S, \varepsilon R)$ in the space of KSBA stable pairs. Although KSBA compactifications admit strong modular interpretations, their boundary strata are generally quite difficult to describe. A recent strategy employed in [ABE22; AET23; AE22; Ale+25] is to simultaneously leverage the advantages of both semitoric and KSBA compactifications by finding comparison morphisms between them. This allows the modular boundary of the KSBA compactification to be studied and classified using the combinatorics of toric geometry, lattice theory, and critically, advances in integral affine geometry and mirror symmetry.

The moduli space F_{Co} of Coble surfaces with $n = 1$ boundary components admits a Hodge-theoretic period domain description of the form $D(T_{\text{Co}})/\Gamma_{\text{Co}}$ where T_{Co} is a fixed lattice and $\Gamma_{\text{Co}} := O^+(T_{\text{Co}})^*$ is the stable

orthogonal group preserving the period domain component (an arithmetic subgroup of finite index in $O(T_{C_0})$). Its Baily-Borel compactification $\overline{F}_{C_0}^{\text{BB}}$ contains only one 0-cusp p_0 , and thus the combinatorial data of a semitoroidal compactification is determined by a single Γ_{C_0} -invariant semifan associated to a lattice at p_0 . A canonical choice one can take is the Coxeter fan, formed by a fundamental domain of the action of the lattice's Weyl group, along with its reflections; see fig. 2. Perhaps more naturally, one can also search for a semifan $\mathcal{F}$ such that the resulting compactification $\overline{F}_{C_0}^{\mathcal{F}}$ is isomorphic to the KSBA compactification $\overline{F}_{C_0}$ for a suitably chosen divisor, and indeed this is what we do in this paper. To this end, we prove the following:

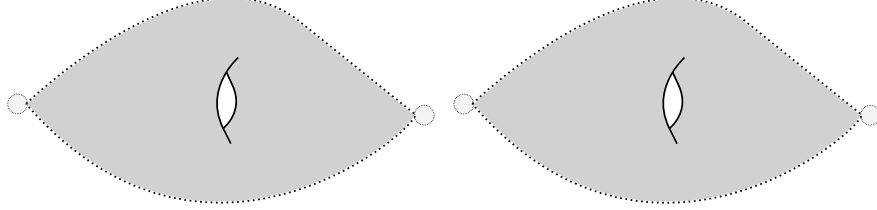


FIGURE 1. Schematic of a Baily-Borel compactification: the shaded arithmetic quotient together with the boundary strata adjoined at its cusps.

This schematic is carried over from the earlier draft, where it was drawn but never captioned; the caption above describes only what is drawn. Confirm what the two components and the interior lens are intended to denote.

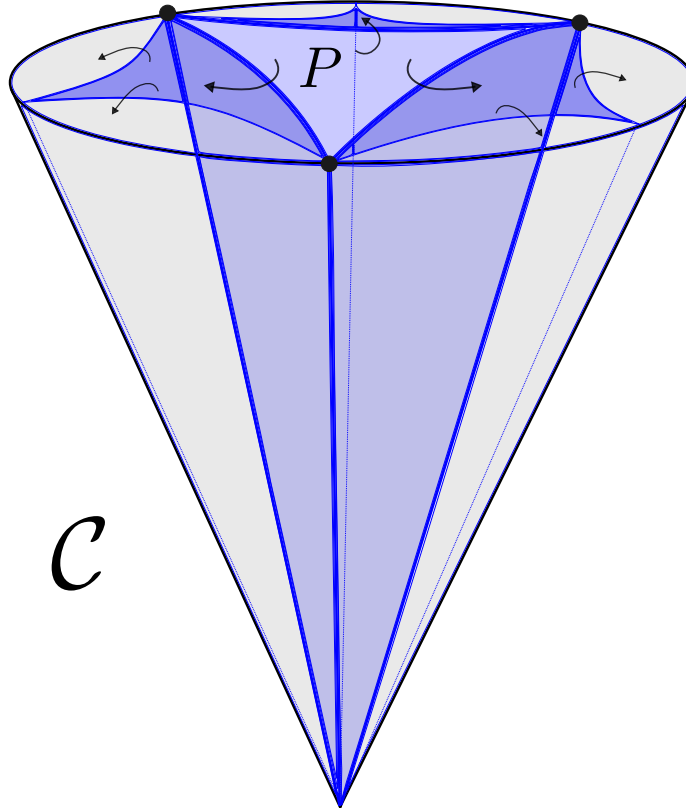


FIGURE 2. A fundamental polyhedron P for the Weyl group of a hyperbolic lattice acting on the positive cone $\mathcal{C}$, together with its images under the reflections in the walls of P ; the Coxeter fan is the resulting decomposition of $\mathcal{C}$. Such a polyhedron is computed by Vinberg's algorithm [Vin75, §1].

Theorem 1A.3.

There is a semifan $\mathcal{F}$ such that there exists a morphism

$$\Psi : (\overline{F_{\text{Co}}})^\nu \rightarrow \overline{F_{\text{Co}}}^{\mathcal{F}}$$

from the normalization of the KSBA compactification to the semitoroidal compactification associated with $\mathcal{F}$. The Coxeter fan of T_{Co} is a refinement of $\mathcal{F}$, and stable Coble surfaces in the boundary of $\overline{F_{\text{Co}}}$ admit explicit descriptions in terms of surfaces associated to sub-Dynkin diagrams of Coxeter diagrams.

Remark 1A.4: This result is made possible by recent advances in [AE22; AEH24] on compactifications of K3 surfaces with nonsymplectic automorphisms, along with the theory of recognizable divisors developed in [AE23]. We also critically leverage the related stable pair compactification of the moduli spaces of Enriques surfaces studied in [Ale+25]. In particular, we use the folding theory of Coxeter diagrams, their associated integral affine structures, and the theory of $ADE + BC$ surfaces in order to explicitly describe stable degenerations of Coble surfaces.

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2. SURFACES AND MODULI

2A. Surfaces: K3, Enriques, and rational.

Remark 2A.1 (Orientation): The moduli theory of Coble surfaces is governed by the geometry of a small stock of ambient surfaces — K3 surfaces, Enriques surfaces, the log del Pezzo ADE and BC surfaces, the quadric $\mathbf{P}^1 \times \mathbf{P}^1$ and its toric quotient W — together with the moduli space of elliptic K3 surfaces. This section collects their definitions and basic invariants for reference. The associated lattices (K3 lattice, Enriques lattice, root lattices) are treated in the Lattice Theory sections; where a root lattice appears we use the algebraic-geometry sign convention, so that A_n, D_n, E_n are negative definite with roots of norm -2 (see definitions 4B.2 and 4B.6). All surfaces are taken over $k = \mathbf{C}$.

2A.1. K3 surfaces.

Definition 2A.2 (K3 surface)

A **K3 surface** is a smooth, projective, geometrically integral surface X over $k = \mathbf{C}$ such that its canonical bundle is trivial,

$$\omega_X \cong \mathcal{O}_X \quad (\text{equivalently } K_X \sim 0),$$

and its irregularity vanishes,

$$q(X) := h^{0,1}(X) = h^1(X, \mathcal{O}_X) = 0.$$

Remark 2A.3 (The K3 lattice): The integral cohomology $H^2(X; \mathbf{Z})$ of a K3 surface is modeled by the K3 lattice

$$L_{\text{K3}} = U^3 \oplus E_8^2 = \Pi_{3,19},$$

the orthogonal direct sum of three hyperbolic planes and two copies of the negative-definite E_8 lattice of definition 4B.6; it is even, unimodular, of signature $(3, 19)$. In the positive-definite (Euclidean) normalization for E_8 this is written $U^3 \oplus E_8(-1)^2$, the two descriptions denoting the same lattice. A **marked K3 surface** is one equipped with an isometry

$$H^2(X; \mathbf{Z}) \xrightarrow{\sim} L_{\text{K3}}.$$

Remark 2A.4 (Hodge diamond): The Hodge numbers of a K3 surface are

$$h^{1,1} = 20, \quad h^{2,0} = h^{0,2} = 1,$$

with all other Hodge numbers vanishing except $h^{0,0} = h^{2,2} = 1$; the Hodge diamond is

$$\begin{array}{ccccc} & & 1 & & \\ & 0 & & 0 & \\ 1 & & 20 & & 1 \\ & 0 & & 0 & \\ & & 1 & & \end{array}$$

In particular the second Betti number is $b_2(X) = 1 + 20 + 1 = 22$, matching the rank of L_{K3} .

2A.2. Enriques surfaces.

Definition 2A.5 (Enriques surface)

An **Enriques surface** is a non-rational algebraic surface Z of Kodaira dimension $\kappa(Z) = 0$ for which

$$h^1(\mathcal{O}_Z) = h^2(\mathcal{O}_Z) = 0$$

and the canonical class K_Z is a nontrivial 2-torsion element of $\text{Pic}(Z)$.

Equivalently, Enriques surfaces are the quotients $Z = X/\iota$ of K3 surfaces X (definition 2A.2) by fixed-point-free involutions ι — called **Enriques involutions** — and satisfy $2K_Z \sim 0$ and $q(Z) = 0$.

Remark 2A.6 (*Basic properties*):

- Enriques surfaces are among the first examples of non-rational surfaces with geometric genus $p_g = 0$.
- They are minimal surfaces.
- The universal cover of Z is the K3 surface X , and $\pi_1(Z) \cong \mathbf{Z}/2\mathbf{Z}$ is generated by the covering involution ι , which acts freely and properly discontinuously on X .

Remark 2A.7 (*Hodge diamond and cohomology*): The Hodge numbers of an Enriques surface are

$$h^{1,1}(Z) = 10, \quad h^{2,0}(Z) = h^{0,2}(Z) = 0,$$

confirming $p_g = 0$, with all other Hodge numbers vanishing except $h^{0,0} = h^{2,2} = 1$. The second integral cohomology carries a 2-torsion summand,

$$\text{NS}(Z) \cong H^2(Z; \mathbf{Z}) \cong \mathbf{Z}^{10} \oplus \mathbf{Z}/2\mathbf{Z},$$

and its free part is modeled by the **Enriques lattice** $E_{10} = U \oplus E_8 \xrightarrow{\sim} \text{II}_{1,9}$, even unimodular of signature $(1, 9)$ (definition 4E.6). Its 2-twist $E_{10}(2) = U(2) \oplus E_8(2)$ arises as the invariant lattice of the covering involution on the K3 cover; the Enriques period domain is governed instead by the transcendental lattice $U \oplus U(2) \oplus E_8(2)$ of signature $(2, 10)$.

2A.3. ADE and BC surfaces.

Definition 2A.8 (ADE surface)

An **ADE surface** (Y, C) is a log canonical log del Pezzo surface of index two whose boundary C has dual intersection graph a simply-laced Dynkin diagram $(A_n, D_n, \text{ or } E_n)$. These surfaces have canonical toric models determined by lattice polytopes in $\mathbf{Z}^2$ [AT17].

Definition 2A.9 (Priming)

A **priming** is a weighted blowup at a boundary node (a corner blowup) that reduces the lattice length of a long edge. Allowed primings are restricted so as to preserve log canonicity and toricity [AT17].

Definition 2A.10 (Folding)

A **folding** passes from a simply-laced ADE diagram to a non-simply-laced BC diagram by taking the quotient by a diagram automorphism τ [AT17; AET23]. For instance A_{2n-1} folds to C_n .

Folding identifies symmetric nodes, producing double bonds and merged nodes in the diagram. Geometrically, the surface is replaced by a quotient Y/τ : boundary components may become non-reduced, acquire multiplicities, or exhibit half-integral coefficients (half-divisors). Passing to the canonical double cover branched over the half-integral loci restores the integral ADE boundary.

Remark 2A.11: The precise relationship between these foldings, the fixed-point subalgebras, and the orbit-sum construction (which can introduce a B/C Langlands duality) is discussed in definition 5B.3 and the Foldings section.

2A.4. The base surface $\mathbf{P}^1 \times \mathbf{P}^1$.

Definition 2A.12 (The base surface $Y = \mathbf{P}^1 \times \mathbf{P}^1$)

Let $Y = \mathbf{P}^1 \times \mathbf{P}^1$ over $\mathbf{C}$. The Picard group $\text{Pic}(Y) \cong \mathbf{Z}^2$ is freely generated by the pullbacks of hyperplanes ℓ_1, ℓ_2 from the two factors. The intersection numbers are

$$\ell_1^2 = \ell_2^2 = 0, \quad \ell_1 \cdot \ell_2 = 1,$$

yielding the lattice isometry

$$(\text{Pic}(Y), \cdot) \xrightarrow{\sim} \left(\mathbf{Z}^2, \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \right) = U.$$

The canonical divisor is $K_Y = -2\ell_1 - 2\ell_2$, of self-intersection

$$K_Y^2 = (-2\ell_1 - 2\ell_2)^2 = 8(\ell_1 \cdot \ell_2) = 8,$$

so that Y is a del Pezzo surface of degree 8.

2A.5. The toric surface W .

Remark 2A.13 (The toric surface W): Let $Y = \mathbf{P}^1 \times \mathbf{P}^1$ and let $\tau: (x, y) \mapsto (-x, -y)$ be the coordinate involution, which has 4 fixed points. The quotient $W = Y/\tau$ is a singular del Pezzo surface of degree 4, with four A_1 singularities. Both Y and W are toric, sharing the same polytope

$$Q_{4,4} := [0, 4]^2,$$

but with different lattices:

$$M_Y = \mathbf{Z}^2, \quad M_W = \mathbf{Z}_{\text{ev}}^2 := \left\{ (i, j) \in \mathbf{Z}^2 \mid i + j \in 2\mathbf{Z} \right\}.$$

2A.6. Moduli of elliptic $K3$ surfaces.

Remark 2A.14 (The moduli space F_{ell}): The moduli space F_{ell} parametrizes elliptic $K3$ surfaces with a chosen section, which fixes a hyperbolic summand $U_1 \subseteq L_{K3}$. The period map is defined on the orthogonal complement

$$T_{\text{ell}} := U_1^\perp \cap L_{K3} \cong U^2 \oplus E_8^2,$$

of signature $(2, 18)$.

Remark 2A.15 (KSBA and semitoroidal compactifications): Alexeev–Brunyate–Engel construct two KSBA (stable pair) compactifications of F_{ell} , each isomorphic to a semitoroidal compactification of the period domain [ABE22]:

1. **Ramification divisor compactification $\overline{F}^{\text{ram}}$.** The polarization is $[R] = 3(s + 2f)$, the ramification divisor of the double cover of $\mathbf{P}(1, 1, 4)$. The associated fan is a coarsening of the Coxeter fan.
2. **Rational curve divisor compactification $\overline{F}^{\text{rc}}$.** The polarization is $R = s + m \sum_{i=1}^{24} f_i$. The associated fan is a refinement of the Coxeter fan.

Geometric models for the boundary strata are constructed using **integral-affine spheres with 24 singularities** (IAS²); see the integral affine structures material (definition 8B.2). The Baily–Borel boundary contains a unique 0-cusp η and two 1-cusps, yielding the boundary lattices E_8^2 and D_{16}^+ .

2B. $K3$ double covers and their involutions. This section records the general machinery of the $K3$ double cover of a rational surface, the distinguished geometric involutions it carries, and the lattice involutions they induce on H^2 together with their invariant and coinvariant sublattices. The Coble-specific instance of the double-cover construction — where the base is a terminal Coble surface and the branch locus is the anticanonical curve — is treated in the section on $K3$ covers of Coble surfaces; here we develop the construction for a general base and the involution-theoretic apparatus common to the del Pezzo, Enriques, and Nikulin quotients. The general invariant/coinvariant lattice formalism L^G, L_G is developed in definition 4F.2 and proposition 4F.3; we cross-reference it rather than restating it.

2B.1. The double cover construction.

Remark 2B.1 (The K3 double cover construction): A standard method to produce Calabi-Yau varieties involves taking a double cover $\pi : X \rightarrow Y$ branched over a reduced divisor B . If $B \in |-2K_Y|$, one takes $\mathcal{L} = -K_Y$. The cover X is defined via the $\mathcal{O}_Y$ -algebra

$$\mathcal{A} = \mathcal{O}_Y \oplus \mathcal{L}^{-1}.$$

Adjunction yields

$$K_X = \pi^*(K_Y + \mathcal{L}),$$

and substituting $\mathcal{L} = -K_Y$ gives $K_X = \mathcal{O}_X$, so that X has trivial canonical bundle.

By taking $B \subset Y = \mathbf{P}^1 \times \mathbf{P}^1$ to be a smooth τ -invariant curve in $|-2K_Y| = |\mathcal{O}_Y(4, 4)|$, we obtain a 10-dimensional family of K3 surfaces, after quotienting by the toric automorphisms $D_4 \rtimes (\mathbf{C}^\times)^2$.

Remark 2B.2: The specialization of remark 2B.1 to a terminal Coble surface of K3 type is carried out in the section on K3 covers of Coble surfaces: there one takes $\mathcal{L} = \mathcal{O}_S(-K_S)$ for S the (rational) base and a section $s \in H^0(\mathcal{L}^{\otimes 2})$ cutting out the anticanonical curve C , and the resulting double cover $f : X \rightarrow S$ is a smooth K3 surface with $\text{Pic}(X)$ a 2-elementary lattice of invariants $(r, a, \delta)_1 = (10 + n, 12 - n, \delta)_1$ [DK25, Prop. 9.1.1; CDL25, Def. 5.4.3].

2B.2. Geometric involutions on the double cover.

Remark 2B.3 (Geometric involutions on the K3 double cover): For the K3 double cover $X \rightarrow Y = \mathbf{P}^1 \times \mathbf{P}^1$, there are three distinguished involutions acting on the coordinates (x, y, z) :

- $\iota_{\text{dP}}(x, y, z) = (x, y, -z)$: the **del Pezzo** (deck) involution, the generator of the deck transformations of the double cover. Its quotient is a K3 quotient and $\iota_{\text{dP}}^* \omega_X = -\omega_X$.
- $\iota_{\text{En}}(x, y, z) = (-x, -y, -z)$: the **Enriques** involution, with smooth Enriques quotient Z , satisfying $\iota_{\text{En}}^* \omega_X = -\omega_X$.
- $\iota_{\text{Nik}}(x, y, z) = (-x, -y, z)$: the **Nikulin** involution, with singular K3 quotient Z' carrying $\geq 8A_1$ singularities, satisfying $\iota_{\text{Nik}}^* \omega_X = +\omega_X$.

These generate a faithful representation of the Klein four-group $\mathbf{Z}_2^2 \hookrightarrow \text{Aut}(X)$. The del Pezzo and Enriques involutions act on the holomorphic 2-form by -1 (they are nonsymplectic), whereas the Nikulin involution acts by $+1$ (it is symplectic); consistently with the group law, $\iota_{\text{En}} \circ \iota_{\text{dP}} = \iota_{\text{Nik}}$.

2B.3. Induced lattice involutions on H^2 .

Definition 2B.4 (Lattice involutions for K3 covers)

Fix a basis of the K3 lattice L_{K3} corresponding to the decomposition $U^3 \oplus E_8^2$, and write a general vector as $(u_1, u_2, u_3, \alpha_1, \alpha_2)$ with $u_i \in U$ and $\alpha_j \in E_8$. The three geometric involutions ι_{dP} , ι_{En} , and ι_{Nik} of remark 2B.3 induce isometries of L_{K3} , denoted I_{dP} , I_{En} , and I_{Nik} respectively, given in this basis by

$$\begin{aligned} I_{\text{dP}}(u_1, u_2, u_3, \alpha_1, \alpha_2) &= (-u_1, u_3, u_2, -\alpha_1, -\alpha_2), \\ I_{\text{En}}(u_1, u_2, u_3, \alpha_1, \alpha_2) &= (-u_1, u_3, u_2, \alpha_2, \alpha_1), \\ I_{\text{Nik}}(u_1, u_2, u_3, \alpha_1, \alpha_2) &= (u_1, u_2, u_3, -\alpha_2, -\alpha_1). \end{aligned}$$

Remark 2B.5 (Properties of the lattice involutions): The lattice involutions satisfy the following.

- The group $\langle I_{\text{dP}}, I_{\text{En}}, I_{\text{Nik}} \rangle$ is isomorphic to $\mathbf{Z}_2^2$; in particular these involutions mutually commute, and $I_{\text{En}} \circ I_{\text{dP}} = I_{\text{Nik}}$.
- For each involution $I_\star$, the **invariant sublattice** is denoted $S_\star := L_{K3}^{I_\star=1}$ and the **coinvariant sublattice** is $T_\star := L_{K3}^{I_\star=-1}$, in the sense of definition 4F.2: $S_\star$ is the $+1$ -eigenlattice and $T_\star$ the -1 -eigenlattice of $I_\star$, and by proposition 4F.3 these are orthogonal and rationally span $L_{K3\mathbf{Q}}$.
- The transcendental lattices T_Z of the Enriques surfaces Z primitively embed into these invariant sublattices.

The isometry classes of the invariant and coinvariant sublattices of the three involutions, together with their 2-elementary invariants $(r, a, \delta)_n$ (where $n = n_+$ is the number of positive eigenvalues) and discriminant groups A_L , are collected in table 1. For the del Pezzo and Enriques involutions these lattices also appear in the Lattice Summary alongside the chain of primitive embeddings of lemma 4G.2. The Enriques (co)invariant lattices are built from the twist $E_{10}(2) = U(2) \oplus E_8(2)$ ($S_{\text{En}} = E_{10}(2)$, $T_{\text{En}} = U \oplus E_{10}(2)$), whereas the del

olution τ of $\mathbf{P}^1 \times \mathbf{P}^1$ used to the invariance on the branch is not specified source note; ident is the involution at lifts to the ic involutions of 2B.3).

Pezzo lattices $S_{\text{dP}} = U(2)$, $T_{\text{dP}} = U \oplus U(2) \oplus E_8^2$ retain an untwisted E_8^2 . The 2-elementarity of both $S_\star$ and $T_\star$ is the content of proposition 4F.6, and their classification is Nikulin's [Nik80].

TABLE 1. Invariant sublattices $S_\star$ and coinvariant sublattices $T_\star$ of the del Pezzo, Enriques, and Nikulin involutions on $L_{K3} = U^3 \oplus E_8^2$. For the Nikulin involution the invariant and coinvariant lattices are written L_{Nik}^+ and L_{Nik}^- ; each complementary pair has ranks summing to $22 = \text{rank } L_{K3}$.

L	Isometry class	$\text{rank}_{\mathbf{Z}}(L)$	$\text{sig}(L)$	$(r, a, \delta)_n$	A_L
S_{dP}	$U(2)$	2	(1, 1)	$(2, 2, 0)_1$	$\mathbf{Z}_2^2$
T_{dP}	$U \oplus U(2) \oplus E_8^2$	20	(2, 18)	$(20, 2, 0)_2$	$\mathbf{Z}_2^2$
S_{En}	$U(2) \oplus E_8(2)$	10	(1, 9)	$(10, 10, 0)_1$	$\mathbf{Z}_2^{10}$
T_{En}	$U \oplus U(2) \oplus E_8(2)$	12	(2, 10)	$(12, 10, 0)_2$	$\mathbf{Z}_2^{10}$
L_{Nik}^+	$U^3 \oplus E_8(2)$	14	(3, 11)	$(14, 8, 0)_3$	$\mathbf{Z}_2^8$
L_{Nik}^-	$E_8(2)$	8	(0, 8)	$(8, 8, 0)_0$	$\mathbf{Z}_2^8$

2C. Numerical polarizations and Noether–Lefschetz loci.

2C.1. Numerical polarizations.

Definition 2C.1 (Numerical polarization)

A **numerical polarization** $[h]$ on an algebraic surface Z — for instance an Enriques surface — is the numerical class of $h := c_1(\mathcal{L})$ for an ample line bundle $\mathcal{L} \in \text{Pic}(Z)$, often written $[\mathcal{L}]$.

Remark 2C.2 (*Numerical classes on an Enriques surface*): For an Enriques surface Z the first Chern class induces an isomorphism

$$c_1: \text{Pic}(Z) \xrightarrow{\sim} H^2(Z; \mathbf{Z}),$$

so that the free part $H^2(Z; \mathbf{Z})_f$ is identified with the group of numerical divisor classes $\text{Num}(Z)$. Under the intersection pairing this free part is the even unimodular lattice of signature (1, 9), i.e. the Enriques lattice E_{10} of definition 4E.6 [CDL25]; the numerical polarization $[h]$ is thus an ample class in $\text{Num}(Z) \xrightarrow{\sim} E_{10}$.

Remark 2C.3 (*Degree of a numerical polarization*): The **degree** of a numerical polarization $[h]$ is its self-intersection h^2 computed in $\text{Num}(Z)$. We are primarily interested in the **degree-2** case $h^2 = 2$; the corresponding moduli space of degree-2 numerically polarized Enriques surfaces is the space $F_{\text{En},2}$ appearing below.

2C.2. The Noether–Lefschetz locus for Enriques surfaces.

Remark 2C.4 (*The canonical cover and its involutions*): The canonical double cover $\pi: X \rightarrow Z$ of a degree-2 polarized Enriques surface yields a K3 surface X carrying two commuting involutions: the fixed-point-free **Enriques involution** ι_{En} (the deck transformation of the canonical cover) and the **del Pezzo involution** ι_{dP} [Ale+25].

Definition 2C.5 (Noether–Lefschetz locus $\text{NL}_{S_{\text{En}}}$)

The moduli space $F_{\text{En},2}$ of degree-2 numerically polarized Enriques surfaces embeds into the K3 moduli space $F_{(2,2,0)}$ via a canonical map j . Its image is the **Noether–Lefschetz locus** $\text{NL}_{S_{\text{En}}}$ defined by the primitive embedding of the **invariant** (algebraic) lattice of ι_{En} ,

$$S_{\text{En}} = E_{10}(2) = U(2) \oplus E_8(2),$$

the rank-10, 2-elementary, signature-(1, 9) lattice of type (10, 10, 0) (see definition 4E.6 and the Special Lattices section). The appearance of these extra invariant classes in NS is what cuts out the locus.

Remark 2C.6 (*KSBA limits and non-normality*): The Zariski closure

$$B := \overline{j(F_{\text{En},2})} \subseteq \overline{F_{(2,2,0)}}$$

parameterizes all KSBA limits of these double covers. The space fails to be normal along the boundary, due to branching and boundary divisor intersections.

The two source no (Numerical polariz Noether–Lefschetz locus for Enriques S faces) carried no citations. The attributions here ($\mathcal{L}$, N marking $\rightarrow$ CDL2 canonical cover, it involutions, the m $j: F_{\text{En},2} \rightarrow F_{(2,2,0)}$ the locus $\text{NL}_{S_{\text{En}}}$, the KSBA-limit cl $\rightarrow$ AEGS25) are s as the standard so for the concepts th name; the author confirm the inten mary references.

3. COBLE SURFACES

3A. Basics of Coble surfaces.

Remark 3A.1 (On defining Cobles): For the general theory of Coble surfaces, we refer to [CDL25; DK25], along with [DM20; DZ99; DK13; CD89; CD12; Dol17]. Following [DM20, §5.1], a **Coble surface** is a smooth projective rational surface with $|-K_S| = \emptyset$ but $|-2K_S| \neq \emptyset$. Such surfaces were first studied in [Cob19], [Cob29], and were ultimately classified in [DZ99]. We say S is **terminal of K3 type** if $|-2K_S|$ contains a smooth¹ divisor $C = C_1 + \cdots + C_n$, the disjoint union of n reduced smooth rational curves C_i satisfying $C_i^2 = -4$ ², and thus $C_i C_j = -4\delta_{ij}$. We refer to C as the *anti-bicanonical curve* of S , and note that $K_S \cdot C_i = -2$. The C_i are referred to as the *boundary components* of S . One can show $n = -K_S^2$ and $n \leq 10$, c.f. [DK25, Cor. 9.1.5].

Remark 3A.2 (Cobles as blowups): Such a Coble surface S can be shown to be a *basic rational surface* [Dol17, §5.1], i.e. it admits a birational morphism to $\pi : S \rightarrow \mathbf{P}^2$ which decomposes as blowups of N points.³ The number of points N depends on n ; more precisely, one can show $N = 9 + n$. The image of C is a divisor in $|-2K_{\mathbf{P}^2}| = |\mathcal{O}_{\mathbf{P}^2}(6)|$ and is thus always a sextic curve, generically a nodal plane sextic where the images of the C_i are its irreducible components. We primarily restrict to the case $n = 1$, whence $K_S^2 = -1$ and S is obtained as the blowup of a special set of ten points in $\{p_1, \dots, p_{10}\} \subset \mathbf{P}^2$.

In this case, $\tilde{C} := \pi(C)$ is an irreducible rational sextic plane curve, and the ten points p_i are precisely its 10 A_1 singularities, some of which may be infinitely near. This is the classical example studied in [Cob19] and [Cob29] – the blowup S of $\mathbf{P}^2$ at the ten A_1 singularities of an irreducible plane sextic curve with at worst A_1 singularities, corresponding to the $n = 1, N = 10$ case above. [Cob19] shows, among other things, that there are finitely many $\text{Aut}(S)$ -orbits of smooth rational curves of negative self-intersection.

Note that conversely, starting with such a plane sextic $\tilde{C}$, one can show that the proper transform C of $\tilde{C}$ is in $|-2K_S|$. Blowing up the singularities of C yields a Coble surface with *smooth* anti-bicanonical divisor; such Coble surfaces are not the image of any birational but not biregular morphism from another Coble surface and are said to be **terminal**. We say S is **minimal** if the blowdown of any (-1) -curve on S is no longer a Coble surface, or equivalently if S does not admit a birational but not biregular morphism onto another Coble surface.

Remark 3A.3 (On the importance of Cobles): Following [CD12, §1.4], a point set $\mathcal{P} = \{p_1, \dots, p_n\}$ of $\mathbf{P}^2$ is called **Cremona special** if $n \geq 9$ and if the surface S obtained by blowing up $\mathcal{P}$ is such that $\text{Aut}^*(S)$, the image of $\text{Aut}(S)$ in $\text{O}(\text{Pic}(S))$, has finite index in the infinite Weyl group W_S . The surface S is then also called Cremona special. This condition of finiteness roughly expresses that $\text{Aut}^*(X)$ is as large as possible. A point set is called *unnodal* if its blowup does not contain any (-2) -curves. In [CD12], it is proved that a Cremona special complex rational surface is either an unnodal Halphen surface (with $n = 9$) or an unnodal Coble surface (with $n = 10$), that unnodal point sets define a Zariski open subset in the moduli space of point sets defining Halphen and Coble surfaces, and that Cremona special Halphen and Coble surfaces are precisely the unnodal ones.

3B. K3 covers.

Remark 3B.1 (On relation to K3s): Let $\mathcal{L} := \mathcal{O}_S(-K_S) \in \text{Pic}(S)$; By [DK25, Prop. 9.1.1], taking a section $s \in H^0(\mathcal{L}^{\otimes 2})$ with $Z(s) = C$ yields a branched double cover $f : X \rightarrow S$ where X is a smooth K3 surface. Let σ be the involution generating the deck transformations of this cover. Then the fixed locus $\text{Fix}(\sigma)$ is a union of n smooth rational curves which are precisely $f^{-1}(C_i)$ where C_i are the irreducible components of C in S . By [CDL25, Def. 5.4.3], the preimages $f^{-1}(C_i)$ are disjoint (-2) -curves and $\text{Pic}(X)$ is a 2-elementary lattice with invariants of the form

$$(r, a, \delta)_1 = (10 + n, 12 - n, \delta)_1.$$

By [CDL25, Def. 5.4.3, Eqn. 5.3.1], the ramification divisor R is explicitly of one of the following forms:

¹More generally, by [DK25], this can be relaxed to a reduced simple normal crossings divisor.

²This follows from an application of the genus formula:

$$p_a(C) = 1 + \frac{1}{2}(C^2 + C \cdot K_S) = 1 + K_S^2 = 0,$$

and so C is a smooth rational curve satisfying $C^2 = 4K_S^2 = -4$ since $K_S^2 = -1$. This is [CD12, §3.1].

³By the proof of [DK25, Prop. 9.1.3].

1. $R = \emptyset$ if $(r, a, \delta) = (10, 10, 0)$,
2. R is a sum of two elliptic curves if $(r, a, \delta) = (10, 8, 0)$,
3. R is the sum of a single rational curve and $n - 1$ other disjoint (-2) -curves otherwise.

It is also known that $\delta = 1$ unless $n = 8$, c.f. [CDL25, Table 5.1]. Thus if S is a terminal Coble surface of K3 type with $n = 1$, the ramification locus of the K3 cover is a single smooth rational curve, and we obtain a lattice with invariants

$$S_{\text{Co}} := (11, 11, 1)_1 \cong \langle -2 \rangle \oplus E_{10}(2)$$

with orthogonal complement

$$T_{\text{Co}} := S_{\text{Co}}^{\perp L_{\text{K3}}} = (11, 11, 1)_2 \cong I_{2,9}(2) \cong \langle 2 \rangle \oplus E_{10}(2).$$

The lattices S_{Co} and T_{Co} will be of fundamental importance in constructing the Hodge-theoretic period domain for Coble surfaces, yielding a coarse space for the corresponding moduli space; the identifications above are derived in the section on period domains.

Proposition 3B.2.

Let $f : X \rightarrow S$ be the K3 double cover of a terminal Coble surface of K3 type with $n = 1$, and let σ be the covering involution. Write $\text{Pic}(S) = \langle H, E_1, \dots, E_{10} \rangle \cong I_{1,10}$, where H is the pullback of a line and the E_i are the exceptional curves of $S = \text{Bl}_{p_1, \dots, p_{10}} \mathbf{P}^2$, so that $H^2 = 1$, $E_i^2 = -1$, and $H \cdot E_i = E_i \cdot E_j = 0$ for $i \neq j$. Then $f^* : \text{Pic}(S) \rightarrow \text{Pic}(X)$ is injective, its image

$$S_{\text{Co}} := f^* \text{Pic}(S) = \langle f^* H, f^* E_1, \dots, f^* E_{10} \rangle$$

is the invariant lattice $H^2(X, \mathbb{Z})^\sigma$, and

$$S_{\text{Co}} \cong \langle 2 \rangle \oplus \langle -2 \rangle^{10} = I_{1,10}(2) = (11, 11, 1)_1.$$

Proof.

Injectivity and the twist by 2. If $f^* D = 0$ then $2D = f_* f^* D = 0$, and $\text{Pic}(S) \cong I_{1,10}$ is torsion-free, so $D = 0$; thus f^* is injective. As f has degree 2, one has $f_* f^* = 2$, and the projection formula gives, for all $D, D' \in \text{Pic}(S)$,

$$((f^* D \cdot f^* D')) = D \cdot f_* f^* D' = D \cdot 2D' = 2((D \cdot D')).$$

Applied to the orthogonal basis $H, E_1, \dots, E_{10}$, this shows that $f^* H, f^* E_1, \dots, f^* E_{10}$ have Gram matrix $\text{diag}(2, -2, \dots, -2)$, so that $S_{\text{Co}} \cong \langle 2 \rangle \oplus \langle -2 \rangle^{10} = I_{1,10}(2)$, of signature $(1, 10)$. This recovers, by the projection formula, the twist-by-2 described geometrically in remark 3B.3.

Rank of the invariant lattice via Lefschetz. The lattice S_{Co} is σ -invariant, so $S_{\text{Co}} \subseteq H^2(X, \mathbb{Z})^\sigma$. The fixed locus $\text{Fix}(\sigma) = R$ is a single smooth rational curve (the case $n = 1$), so $\chi(R) = 2$. On the K3 surface X , both H^0 and H^4 are σ -invariant of rank 1 and $H^1 = H^3 = 0$, so the topological Lefschetz fixed-point formula reads

$$\chi(R) = \sum_i (-1)^i \text{tr}((\sigma^* \mid H^i(X, \mathbb{Z}))) = 2 + \text{tr}((\sigma^* \mid H^2(X, \mathbb{Z}))).$$

Since $\chi(R) = 2$, we obtain $\text{tr}(\sigma^* \mid H^2(X, \mathbb{Z})) = 0$. As $H^2(X, \mathbb{Z})$ has rank 22 and σ^* is an involution, the invariant lattice has rank $\frac{1}{2}(22 + 0) = 11$ and the coinvariant lattice rank $\frac{1}{2}(22 - 0) = 11$, so that $11 + 11 = 22$; cf. proposition 4F.3. Hence S_{Co} , of rank 11, is a finite-index sublattice of $H^2(X, \mathbb{Z})^\sigma$.

The Nikulin invariants and equality. The lattice $S_{\text{Co}} \cong \langle 2 \rangle \oplus \langle -2 \rangle^{10}$ has

$$|\det S_{\text{Co}}| = 2 \cdot 2^{10} = 2^{11}, \quad A_{S_{\text{Co}}} \cong (\mathbb{Z}/2\mathbb{Z})^{11}, \quad q_{S_{\text{Co}}} \cong \langle \frac{1}{2} \rangle \oplus \langle -\frac{1}{2} \rangle^{10},$$

so S_{Co} is 2-elementary of rank $r = 11$ and length $a = 11$. Because $q_{S_{\text{Co}}}$ takes the value $\frac{1}{2} \notin \mathbb{Z}$, one has $\delta = 1$, giving $(r, a, \delta) = (11, 11, 1)$. For a nonsymplectic involution whose fixed locus is a single rational curve, the fixed-locus formula assigns the invariant lattice these same invariants $(r, a, \delta) = (11, 11, 1)$ [CDL25, Def. 5.4.3, Eqn. 5.3.1], so $|\det H^2(X, \mathbb{Z})^\sigma| = 2^{11}$. Since $S_{\text{Co}} \subseteq H^2(X, \mathbb{Z})^\sigma$ have equal rank 11 and equal absolute determinant 2^{11} , the index $[H^2(X, \mathbb{Z})^\sigma : S_{\text{Co}}]$ is 1: the two coincide, and by proposition 4F.4 the invariant lattice – hence S_{Co} – is primitive in $H^2(X, \mathbb{Z})$. That the invariants

$(11, 11, 1)_1$ determine the isometry class $\langle 2 \rangle \oplus \langle -2 \rangle^{10} \cong \langle -2 \rangle \oplus E_{10}(2)$ is Nikulin's classification of indefinite even 2-elementary lattices [Nik80]. This derivation makes explicit the invariants stated by citation in remark 3B.1. ■

Remark 3B.3: Following [CD12], we note that this computation is a special case of a general construction. Let S be any basic rational surface and write S as the blowup of $\mathbf{P}^2$ at N points $p_1, \dots, p_N$ with $N \geq 9$. It is a fact that $\text{Pic}(S) \cong I_{1,N}$, since one can construct a **geometric basis** in the following way: let e_0 be the class of the total transform of a hyperplane class in $\mathbf{P}^2$ and for $1 \leq i \leq N$, let e_i be the class of the total transform of the exceptional divisor over p_i . Then $\text{Pic}(S) = \langle e_0, e_1, \dots, e_N \rangle$ and $\rho(X) = N + 1$; one verifies that $e_0^2 = 1$, and for $i \geq 1$, that $e_i^2 = -1$. Moreover $e_i e_j = 0$ for $i \neq j$, making this an orthogonal basis with respect to the intersection pairing, yielding $I_{1,N}$. In the case of Coble surfaces, the effect of taking the K3 double cover is to twist this lattice by 2, yielding $\text{Pic}(X) = I_{1,N}(2)$, generated by preimages of the e_i . We remark that

$$K_S = -3e_0 + e_1 + \dots + e_N.$$

3C. Calculations.

Lemma 3C.1.

Let $C \subset \mathbf{P}^2$ be an irreducible sextic curve with at worst A_1 singularities. Then C is a rational curve if and only if it has exactly ten A_1 singularities.

Proof.

One has the genus-degree formula for singular plane curves:

$$p_a(C) = \frac{1}{2}(d-1)(d-2) - \sum_p \delta_p$$

where the sum is over the singular locus of C . Since C has at worst A_1 singularities, each singularity contributes $\delta_p = 1$. Noting that $d = 6$, we obtain the geometric genus $g = 10 - k$, where k is the number of A_1 singularities. Thus $g = 0$ if and only if $k = 10$. ■

Remark 3C.2(Severi varieties and degenerate sextics): We note that an arbitrary rational irreducible sextic need not have exactly ten singularities, nor must its singularities be A_1 singularities. The condition $g = 0$ merely requires that the sum of the δ -invariants of its singularities equals 10. The locus of rational A_1 -singular sextics forms a Severi variety $V_{6,10}$ [Ble+12] whose generic element has precisely 10 A_1 singularities. However, $V_{6,10}$ contains lower-dimensional strata parametrizing rational sextics with fewer singularities of higher multiplicity. For instance, there exist rational A_2 -singular sextics lacking 10 A_1 singularities entirely [GK20], provided their singularities satisfy the bounded condition $\sum \delta_p = 10$. The classical unnodal Coble surface construction specifically requires a sextic from the generic, 10- A_1 locus.

Remark 3C.3(Canonical class of Cobles): Let $W \subset \mathbf{P}^2$ be a generic irreducible rational sextic with ten A_1 singularities. The classical Coble surfaces S , first constructed in [Cob19] in a study of Cremona transformations of $\mathbf{P}^2$ which preserve such a rational sextic W . These surfaces can be realized as the blowup of $\mathbf{P}^2$ along the divisor $D = p_1 + \dots + p_{10}$ comprised of the ten A_1 singularities of W . In this construction, letting C be the proper transform of W , one can show that C is a rational curve and $|-2K_S| = \{C\}$ by adjunction. Moreover, the square of the canonical class can be computed as follows:

$$\begin{aligned}
K_S^2 &= (\pi^* K_{\mathbf{P}^2} + \sum_{k=1}^{10} E_k)^2 \\
&= (\pi^* K_{\mathbf{P}^2})^2 + \left(\sum_{k=1}^{10} E_k\right)^2 + 2\left(\sum_{k=1}^{10} \pi^* K_{\mathbf{P}^2} \cdot E_k\right) \\
&= (-3H)^2 + \sum_{k=1}^{10} E_k^2 \\
&= 9 + 10 \cdot (-1) \\
&= -1.
\end{aligned}$$

Thus S is of K3 type, and the 2-to-1 cover $X \rightarrow S$ branched along C is a K3 surface with $\rho(Y) = 11$, where a primitive sublattice $I_{1,10}(2)$ of $\text{Pic}(X)$ is generated by the pullbacks of a hyperplane class E_0 and ten exceptional curves E_i . We note that the group of deck transformations of $X \rightarrow S$ is generated by a nonsymplectic involution ι which preserves $\text{NS}(X)$. Thus if S is of K3 type, then the universal double cover $f : X \rightarrow S$ is a singular normal K3 surface X at worst A_1 singularities which is naturally equipped with a nonsymplectic involution ι . Moreover, by the proof of [AD18, Lem. 2.2], a Hurwitz-type formula yields

$$0 = 2K_X = 2(f^*(K_S) + \sum_{k=1}^{10} E_k) = f^*(2K_S + C)$$

yielding $C \sim -2K_S$.

Remark 3C.4 (Invariants of antibicanonical curves): By [CD12, §3.1], if S is Coble surface of K3 type with $n = 1$ boundary components and $C \in |-2K_S|$ is an irreducible curve, then

$$p_a(C) = 1 + \frac{1}{2}(C^2 + C \cdot K_S) = 1 + K_S^2 = 0,$$

noting that $K_S^2 = -1$. This forces C to be a smooth rational curve with $C^2 = 4K_S^2 = -4$.

4. LATTICES AND MODULI

4A. Lattice Theory.

Remark 4A.1: We refer to:

- [Vin85]
- [Vin75]

4A.1. Basic Theory.

Remark 4A.2: The study of semitoroidal compactifications of moduli spaces of Coble surfaces largely reduces to lattice theory, of which we will now recall the essential notions.

Remark 4A.3 (Basic invariants): By a **lattice**, we mean a free $\mathbf{Z}$ -module L of finite rank equipped with a nondegenerate⁴ symmetric integral bilinear form $\beta_L : L \otimes_{\mathbf{Z}} L \rightarrow \mathbf{Z}$. We abbreviate $vw := \beta_L(v, w)$ and $v^2 := \beta_L(v, v)$ and refer to the latter as the **norm** of v . We write $L_R := L \otimes_{\mathbf{Z}} R$ and β_{L_R} for $R = \mathbf{Q}, \mathbf{R}, \mathbf{C}$ for the $\mathbf{Z}$ -linear extensions of (L, β_L) to the rational, real, and complex numbers respectively.

A submodule $M \subseteq L$ is a **sublattice** if the restricted bilinear form $\beta_L|_M$ endows M with the structure of a lattice. A vector $v \in L$ is **isotropic** if $v^2 = 0$, i.e. it is norm zero, and more generally a sublattice $M \subseteq L$ is isotropic if $\beta_L|_M \equiv 0$. A lattice is said to be **even** if $x^2 \in 2\mathbf{Z}$ for all $x \in L$, and **odd** otherwise. A nondegenerate symmetric bilinear form can be linearly extended to $L_{\mathbf{R}}$ and by Sylvester's theorem, diagonalized with only 1 or -1 on the diagonal. We write n_+ and n_- respectively for the number of ± 1 entries on the diagonal. The **signature** of L is the pair (n_+, n_-) , and the **index** is $n_+ - n_-$. We say L is **definite** if either n_+ or n_- is zero, and **indefinite** otherwise. More precisely, if $n_- = 0$ we say L is **positive definite**, and similarly if $n_+ = 0$, we say L is **negative definite**. A negative definite lattice generated by elements of norm -2 is said to be a **root lattice**, and such elements are referred to as **roots**.

⁴A bilinear form is *nondegenerate* if for any x , $\beta_L(x, L) = 0$ implies $x = 0$.

The **rank** r of a lattice is its rank as a free $\mathbf{Z}$ -module and is given by $r = n_+ + n_- = \dim_{\mathbf{Q}} L_{\mathbf{Q}}$. An indefinite lattice of signature $(1, r-1)$ is said to be **hyperbolic**. Fixing a generating set e_i of L , we define the **Gram matrix** of L as the matrix $G_L := (\beta_L(e_i, e_j))_{ij}$, and the **discriminant** as $\text{disc } L := \det G_L$. The discriminant is independent of the choice of generating set.

Remark 4A.4(Discriminant forms): The dual lattice to L is denoted $L^\vee := \text{Hom}_{\mathbf{Z}}(L, \mathbf{Z})$, and there is an morphism

$$\begin{aligned} \iota : L &\hookrightarrow L^\vee \\ x &\mapsto \beta_L(x, \cdot) \end{aligned}$$

which, if L is nondegenerate, is an injection with finite index image. The **discriminant group** is $A_L := \text{coker } \iota \cong L^\vee / L$; this is a finite order group of order $|\text{disc } L|$. We say L is **unimodular** if any of the following equivalent conditions hold:

1. A_L is the trivial group,
2. ι is an isomorphism and $L \cong L^\vee$,
3. $|\text{disc } L| = 1$.

If $A_L \cong (\mathbf{Z}/p\mathbf{Z})^a$ for some a , we say L is **p -elementary**; in our applications we will often have $p = 2$. For even lattices, the form β_L descends to a well-defined quadratic form

$$\begin{aligned} q_L : A_L &\rightarrow \mathbf{Q}/2\mathbf{Z} \\ x + L &\mapsto \beta_{L_{\mathbf{Q}}}(x, x) \bmod 2\mathbf{Z} \end{aligned}$$

We call the pair (A_L, q_L) the **discriminant quadratic form** of L . A **morphism** between two lattices is a morphism of $\mathbf{Z}$ -modules $\eta : L \rightarrow L'$ respecting the bilinear forms in the sense that $\beta_L(x, y) = \beta_{L'}(\eta(x), \eta(y))$, and is a **primitive embedding** if η is injective and $\text{coker } \eta$ is torsionfree. An **isometry** of lattices is an isomorphism, defined in the obvious way. We write $\text{O}(L)$ for the group of lattice automorphisms of L , denoted the **orthogonal group** of L , and similarly $\text{O}(q_L)$ for the $\mathbf{Z}$ -module automorphisms of the discriminant group A_L which preserve the quadratic form. There is a natural group homomorphism $\text{O}(L) \rightarrow \text{O}(q_L)$, the kernel is denoted $\tilde{\text{O}}(L)$, equivalently written $\text{O}(L)^*$; we call it the **stable orthogonal group**. When L has signature $(2, n)$, the associated period domain Ω_L has two connected components interchanged by complex conjugation, and we write $\text{O}^+(L)$ for the index-two subgroup of $\text{O}(L)$ preserving each component (equivalently, preserving the orientation of the positive-definite part, i.e. those isometries of real spinor norm $+1$). We write $\text{O}^+(L)^* := \text{O}^+(L) \cap \tilde{\text{O}}(L)$ for the intersection, which is the arithmetic group acting on a single component D_L .

Remark 4A.5(Orthogonal complements): Given two lattices L_1, L_2 we write $L_1 \oplus L_2$ for the **orthogonal direct sum**, which is the direct sum of the underlying modules with bilinear form defined by

$$\beta_{L_1 \oplus L_2}(v_1 + v_2, w_1 + w_2) := \beta_{L_1}(v_1, w_1) + \beta_{L_2}(v_2, w_2).$$

We write $L^{\oplus n}$ for the direct sum of n copies of L . Let $\eta : M \hookrightarrow L$ be a primitive embedding of lattices, for example the inclusion of a sublattice. We write

$$M^{\perp L} := \left\{ x \in L \mid \beta_L(x, M) = 0 \right\}.$$

If the ambient lattice L is understood, we often abuse notation and simply write $M^\perp$ without reference to L . Note that $M^{\perp L} \oplus M \subseteq L$ may not be saturated, and is generally a finite index sublattice of L . We note that $M^{\perp L} \cap M \neq \{0\}$ in general.⁵ We also note that for any lattices L_i ,

$$A_{L_1 \oplus \dots \oplus L_n} = A_{L_1} \oplus \dots \oplus A_{L_n}$$

Remark 4A.6(2-elementary lattices): Let L be a 2-elementary lattice. The **divisibility** of a vector $v \in L$, denoted $\text{div}_L(v)$, is defined by $\beta_L(v, L) = \text{div}_L(v)\mathbf{Z}$, i.e. the positive integral generator of the image of the map $\beta_L(v, \cdot) : L \rightarrow \mathbf{Z}$. For 2-elementary lattices, one always has $\text{div}_L(v) \in \{1, 2\}$. We set $v^* := v / \text{div}_L(v) \in A_L$. Letting $q_L : A_L \rightarrow \frac{1}{2}\mathbf{Z}/\mathbf{Z}$ be the induced quadratic form on A_L , we say v^* is **characteristic** if $q_L(x) = \beta_L(v^*, x) \bmod \mathbf{Z}$ for all $x \in A_L$, and is **ordinary** otherwise. We say that a primitive isotropic vector $e \in L$ is

⁵For example, let $L = U = \langle e, f \rangle$ and $M = \langle e \rangle$. Then $M^{\perp L} = M$.

1. **odd** if $\text{div}_L(e) = 1$,
2. **even ordinary** if $\text{div}_L(e) = 2$ and e^* is ordinary, or
3. **even characteristic** if $\text{div}_L(e) = 2$ and e^* is characteristic.

The 2-elementary hyperbolic lattices admitting a primitive embedding into L_{K3} were classified by Nikulin in [Nik80, §3.6.2]. An indefinite 2-elementary lattice is determined up to isometry by a triple of invariants (r, a, δ) . Here, $r := \text{rank}_{\mathbf{Z}}(L)$ is the rank, $a = \text{rank}_{\mathbb{F}_2} A_L$ is the exponent appearing in $A_L = (\mathbf{Z}/2\mathbf{Z})^a$, and $\delta \in \{0, 1\}$ is the **coparity**: we set $\delta = 0$ if $q_L(A_L) \subseteq \mathbf{Z}$, so $q_L(x) \equiv 0 \pmod{\mathbf{Z}}$ for all $x \in A_L$, and $\delta = 1$ otherwise. We accordingly specify such lattices using the notation $(r, a, \delta)_{n+}$.

Remark 4A.7 (Twists of a lattice): If L is a lattice with bilinear form β_L , define $L(n)$ to be the twist of L by n , which has the same underlying $\mathbf{Z}$ -module but is equipped with the scaled bilinear form

$$\beta_{L(n)}(v, w) := n \cdot \beta_L(v, w).$$

Remark 4A.8 (The lattice $\langle n \rangle$): The lattice $\langle n \rangle$ is defined as the rank 1 lattice $\mathbf{Z}$ with one generator v satisfying $\beta_{\langle n \rangle}(v, v) = n$. The Gram matrix is the 1×1 matrix $G_{\langle n \rangle} = [n]$, and the associated quadratic form is $q_{\langle n \rangle}(x) = nx^2$.

Remark 4A.9 (The hyperbolic lattice): In rank 2, there are two unimodular hyperbolic lattices: the odd $I_{1,1} := \langle 1 \rangle \oplus \langle -1 \rangle$, and the even $U := \Pi_{1,1}$. We refer to the latter as the **hyperbolic lattice**, which can be realized as $U := \mathbf{Z}e \oplus \mathbf{Z}f$ with $e^2 = f^2 = 0$ and $ef = 1$, and thus the following Gram matrix:

$$G_U = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}.$$

Remark 4A.10 (ADE lattices): Any Dynkin diagram of type A_n, D_n, E_6, E_7, E_8 corresponds to a root lattice of the respective type. By convention, we take the negative definite twists of these lattices. Of particular importance to us is the E_8 lattice associated to the following Dynkin diagram:

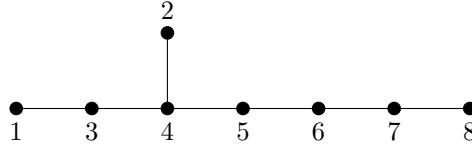


FIGURE 3. The Dynkin diagram E_8 .

Remark 4A.11 (The lattice $I_{p,q}$): For any pair of non-negative integers (p, q) , there exists an odd indefinite unimodular lattice determined up to isomorphism by its rank and signature:

$$I_{p,q} := \langle 1 \rangle^{\oplus p} \oplus \langle -1 \rangle^{\oplus q}.$$

Remark 4A.12 (The lattice $\Pi_{p,q}$): Let L be an even indefinite unimodular lattice of signature (p, q) . Then $p - q \equiv 0 \pmod{8}$, and L is uniquely determined up to isomorphism by its rank and signature:

$$\Pi_{p,q} := \begin{cases} U^{\oplus p} \oplus E_8^{\oplus \frac{q-p}{8}}, & p < q \\ U^{\oplus q} \oplus E_8(-1)^{\oplus \frac{p-q}{8}}, & p > q. \end{cases}$$

4B. Root systems and root lattices.

Remark 4B.1 (Orientation and sign convention): The lattices governing the moduli theory of Coble surfaces are assembled from the classical root lattices, whose combinatorics we collect here for reference. Recall from the Lattice Theory section that we work with the algebraic-geometry sign convention: a **root lattice** is *negative definite*, generated by vectors of norm -2 , which we call **roots**.

It is however most transparent to construct the classical root lattices inside a *positive-definite* Euclidean space, where the roots have norm $+2$; the lattice used elsewhere in the monograph is then the negative-definite twist $L(-1)$. Throughout this section we describe the positive-definite Euclidean model and understand A_n, D_n, E_n to denote its negative twist. All of the combinatorial invariants recorded below—root counts, simple systems, Weyl groups, and discriminant groups—are invariant under twisting, so no ambiguity results.

4B.1. *General root lattices.***Definition 4B.2**

Let L be a lattice (in the Euclidean model, positive definite). The set of **roots** of L is

$$R(L) := \left\{ v \in L \mid v^2 = 2 \right\}.$$

A **simple root system** $\Phi(L) \subseteq R(L)$ is a subset of roots $\{\alpha_i\}$ such that every root $v \in R(L)$ is either a nonnegative integer combination or a nonpositive integer combination of the α_i — the **positive** and **negative** roots respectively; that is,

$$R(L) = \mathbf{Z}_{\geq 0} \Phi(L) \sqcup \mathbf{Z}_{\leq 0} \Phi(L),$$

so every $v \in R(L)$ can be written as $v = \sum_{\alpha_i \in \Phi(L)} c_i \alpha_i$ with all $c_i \in \mathbf{Z}_{\geq 0}$ or all $c_i \in \mathbf{Z}_{\leq 0}$. The elements of $\Phi(L)$ are the **simple roots**. Any simple root system spans a sublattice $\mathbf{Z}\Phi(L) \hookrightarrow L$, and we say L is a **root lattice** if this inclusion has index 1, i.e. $L = \mathbf{Z}\Phi(L)$.

Remark 4B.3 (The Weyl group): For a root $\alpha \in R(L)$, the associated **reflection** is the isometry

$$\begin{aligned} s_\alpha : L_{\mathbf{Q}} &\rightarrow L_{\mathbf{Q}} \\ x &\mapsto x - \frac{2\beta_L(x, \alpha)}{\beta_L(\alpha, \alpha)} \alpha = x - \beta_L(x, \alpha) \alpha, \end{aligned}$$

the second equality holding because $\alpha^2 = 2$. It fixes the hyperplane $H_\alpha := \alpha^\perp$ pointwise and sends $\alpha \mapsto -\alpha$. The **Weyl group** of L is the subgroup

$$W(L) := \left\langle s_{\alpha_i} \mid \alpha_i \in \Phi(L) \right\rangle \leq O(L)$$

generated by the reflections in the simple roots.

For a root lattice we may identify the Gram matrix G_L with $G_{\mathbf{Z}\Phi(L)}$, the matrix of pairings $\beta_L(\alpha_i, \alpha_j)$ between the simple roots; its diagonal entries are $\alpha_i^2 = 2$ and its off-diagonal entries record the intersections of simple roots.

4B.2. *The ADE lattices.* The simply-laced root lattices A_n, D_n, E_n have a single root length and are the ones that occur in the lattice theory of $K3$ and Coble surfaces.

Definition 4B.4

Let $\mathbf{Z}^{n+1}$ denote the standard diagonal lattice $\langle 1 \rangle^{\oplus(n+1)}$ with orthonormal generators $\{e_1, \dots, e_{n+1}\}$, and set $v_{n+1} := \sum_{i=1}^{n+1} e_i$. The root lattice A_n is realized as the primitive sublattice

$$A_n \cong v_{n+1}^\perp \mathbf{Z}^{n+1} = \left\{ x \in \mathbf{Z}^{n+1} \mid \sum_{i=1}^{n+1} x_i = 0 \right\}.$$

Its roots are the difference vectors

$$R(A_n) = \left\{ r_{ij} := e_i - e_j \mid 1 \leq i \neq j \leq n+1 \right\},$$

of which there are $n(n+1)$. A simple system is given by the n vectors

$$\alpha_i := e_i - e_{i+1}, \quad 1 \leq i \leq n.$$

Definition 4B.5

The root lattice D_n is the sublattice of $\mathbf{Z}^n$ cut out by the even-coordinate-sum condition, i.e. the kernel of the reduction $\varepsilon_2 : \mathbf{Z}^n \rightarrow \mathbf{Z}/2\mathbf{Z}, x \mapsto \sum_i x_i \pmod{2}$:

$$D_n \cong \mathbf{Z}_{\text{ev}}^n := \left\{ x = (x_1, \dots, x_n) \in \mathbf{Z}^n \mid \sum_{i=1}^n x_i \equiv 0 \pmod{2} \right\} \hookrightarrow \mathbf{Z}^n.$$

Its roots are

$$R(D_n) = \left\{ \pm e_i \pm e_j \mid 1 \leq i < j \leq n \right\},$$

of which there are $2n(n-1)$. A simple system is given by

$$\alpha_i := e_i - e_{i+1} \quad (1 \leq i \leq n-1), \quad \alpha_n := e_{n-1} + e_n.$$

Definition 4B.6

The root lattice E_8 is realized inside $\mathbf{Q}^8$ as

$$E_8 \cong \left\{ x \in \mathbf{Z}^8 \cup (\mathbf{Z}^8 + \tfrac{1}{2}v_8) \mid \sum_{i=1}^8 x_i \equiv 0 \pmod{2} \right\}, \quad v_8 = (1, \dots, 1),$$

so that E_8 consists of all integer vectors and all half-integer vectors whose coordinate sum is even. The lattices E_7 and E_6 are obtained as orthogonal complements inside E_8 :

$$E_7 := L_1^\perp E_8 \quad \text{for any } L_1 \leq E_8 \text{ isometric to } A_1, \quad E_6 := L_2^\perp E_8 \quad \text{for any } L_2 \leq E_8 \text{ isometric to } A_2.$$

(Here $A_1 = \langle 2 \rangle$ is generated by a single root and A_2 by two roots meeting at the usual angle; the isometry type of the complement is independent of the chosen sublattice.)

Example 4B.7: The lattice E_8 contains exactly

$$2^4 \cdot 3 \cdot 5 = 240$$

roots, of two kinds:

1. the 112 difference vectors $\pm e_i \pm e_j$ for $1 \leq i < j \leq 8$ (these are the roots of the D_8 sublattice), and
2. the 128 half-sum vectors $\frac{1}{2}(\pm e_1 \pm \dots \pm e_8)$ with an even number of minus signs.

Indeed $\binom{8}{2} \cdot 4 = 112$ and $2^{8-1} = 128$, giving 240 in total. There are 8 simple roots comprising $\Phi(E_8)$, and the Weyl group $W(E_8)$ acts transitively on the root set $R(E_8)$.

Theorem 4B.8.

E_8 is the unique even, unimodular, positive-definite lattice of rank 8. In particular it is self-dual, $E_8 \cong E_8^\vee$.

Remark 4B.9: The characterization in theorem 4B.8 is a classical fact of lattice theory [CS10]. The construction of definition 4B.6 exhibits E_8 concretely as the overlattice D_8^+ of D_8 (see definition 4B.12 and theorem 4B.13), which makes both the evenness and the unimodularity transparent.

4B.3. Non-simply-laced root systems. The following root systems carry two root lengths and so are not root lattices in the sense of definition 4B.2; we record them for completeness in the Euclidean model.

Definition 4B.10

The root system $R(B_n) \subseteq \mathbf{Z}^n$ consists of the **short roots** $r_i := \pm e_i$ for $1 \leq i \leq n$ together with the **long roots** $r_{ij} := \pm e_i \pm e_j$ for $1 \leq i < j \leq n$; there are $2n + 2n(n-1) = 2n^2$ roots in total ($2n$ short, $2n(n-1)$ long). A simple system is

$$\alpha_i := e_i - e_{i+1} \quad (1 \leq i \leq n-1), \quad \alpha_n := e_n.$$

The root system $R(C_n)$ is dual to B_n : it consists of the **long roots** $2r_i = \pm 2e_i$ together with the **short roots** $r_{ij} = \pm e_i \pm e_j$, again $2n^2$ roots in total ($2n$ long, $2n(n-1)$ short). A simple system is

$$\alpha_i := e_i - e_{i+1} \quad (1 \leq i \leq n-1), \quad \alpha_n := 2e_n.$$

Definition 4B.11

The root system F_4 is realized in $\mathbf{Z}^4$ with root set $R(F_4)$ comprising the 24 long roots $\pm e_i \pm e_j$ for $1 \leq i < j \leq 4$ together with the 24 short roots consisting of the 8 vectors $\pm e_i$ ($1 \leq i \leq 4$) and the 16 half-sums $\frac{1}{2}(\pm e_1 \pm e_2 \pm e_3 \pm e_4)$ ranging over all 2^4 sign patterns, giving 48 roots in all. A system of simple roots is

$$\alpha_1 = e_2 - e_3, \quad \alpha_2 = e_3 - e_4, \quad \alpha_3 = e_4, \quad \alpha_4 = \tfrac{1}{2}(e_1 - e_2 - e_3 - e_4).$$

The root system G_2 embeds into the trace-zero sublattice $\mathbf{Z}_\Delta^3 := \left\{ x \in \mathbf{Z}^3 \mid \sum_i x_i = 0 \right\} \subseteq \mathbf{Z}^3$. Its roots are the vectors $\pm(e_i - e_j)$ and $\pm(2e_i - e_j - e_k)$, twelve in total. A simple system is

$$\alpha_1 = e_1 - e_2, \quad \alpha_2 = -e_1 + 2e_2 - e_3.$$

4B.4. *The plus construction and $D_8^+ \cong E_8$.* The realization of E_8 in definition 4B.6 is an instance of a general overlattice construction for the D_n family.

Definition 4B.12

Define two vectors in $\mathbf{Q}^n$ by

$$v_+ = \frac{1}{2}(1, 1, 1, \dots, 1), \quad v_- = \frac{1}{2}(-1, 1, 1, \dots, 1).$$

The **plus** and **minus** overlattices of D_n are

$$D_n^+ := D_n \cup (v_+ + D_n), \quad D_n^- := D_n \cup (v_- + D_n),$$

each obtained by adjoining one half-integer coset to D_n .

Theorem 4B.13.

Let $D_n^+ = D_n \cup (v_+ + D_n)$ with $v_+ = \frac{1}{2}(1, \dots, 1)$ be the overlattice of definition 4B.12.

1. D_n^+ is a lattice if and only if n is even. For n odd, $2v_+ = (1, \dots, 1) \notin D_n$ — equivalently $[v_+]$ has order 4 in $A_{D_n} \cong \mathbf{Z}/4\mathbf{Z}$ — so the coset union is not closed under addition and is not a lattice.
2. For n even, D_n^+ contains D_n as a sublattice of index 2 and has determinant 1.
3. D_n^+ is an integral lattice (hence unimodular) if and only if $4 \mid n$, and is even if and only if $8 \mid n$.
For $n \equiv 2 \pmod{4}$ the lattice D_n^+ has determinant 1 but is non-integral, since $v_+^2 = n/4 \notin \mathbf{Z}$.
4. In particular $D_8^+ \cong E_8$ is the even unimodular lattice of rank 8.

Remark 4B.14: The properties in theorem 4B.13 are standard for the D_n family [CS10]. Because D_8^+ is unimodular and contains D_8 with index 2, the even-coordinate-sum sublattice D_8 has discriminant group of order 4; the half-integer coset $v_+ + D_8$ supplies precisely the 128 half-sum roots of E_8 recorded in example 4B.7, while the 112 roots of D_8 supply the difference vectors. This recovers the count $112 + 128 = 240$ and identifies E_8 with D_8^+ .

4C. Discriminant forms and the genus.

Remark 4C.1: The Lattice Theory section introduced the dual lattice $L^\vee$, the discriminant group $A_L := L^\vee/L$, and the discriminant quadratic form $q_L : A_L \rightarrow \mathbf{Q}/2\mathbf{Z}$ of an even lattice. We now develop the theory these constructions specialize: the general correspondence between symmetric bilinear forms and quadratic forms, its torsion-valued counterpart, the resulting structure on A_L , the properties of the dual lattice, and the classification of lattices up to genus. Throughout, (L, β_L) denotes a lattice in the sense of the Lattice Theory section, with $L_{\mathbf{Q}} := L \otimes_{\mathbf{Z}} \mathbf{Q}$ and $\beta_{L_{\mathbf{Q}}}$ the $\mathbf{Q}$ -linear extension of β_L .

4C.1. Quadratic forms and the polarization identity.

Definition 4C.2

A **quadratic form** on a $\mathbf{Z}$ -module L is a map of sets $q : L \rightarrow \mathbf{Q}$ such that $q(\lambda v) = \lambda^2 q(v)$ for all $v \in L$ and all $\lambda \in \mathbf{Z}$, and whose **polar form** β_q is a symmetric bilinear form on L :

$$\begin{aligned} \beta_q : L \otimes_{\mathbf{Z}} L &\rightarrow \mathbf{Q} \\ (v, w) &\mapsto \beta_q(v, w) := q(v + w) - q(v) - q(w). \end{aligned}$$

We say q is **integral** if $q(L) \subseteq \mathbf{Z}$, and we call the pair (L, q) a **quadratic $\mathbf{Z}$ -module**.

Remark 4C.3: Setting $w = v$ in the polar form and using homogeneity gives

$$\beta_q(v, v) = q(2v) - 2q(v) = 4q(v) - 2q(v) = 2q(v),$$

so that $q(v) = \frac{1}{2}\beta_q(v, v)$. The polar form is thus always **even**, meaning $\beta_q(v, v) \in 2\mathbf{Z}$ whenever q is integral. This identity is the source of the bijection recorded below.

4C.2. The correspondence between bilinear and quadratic forms.

Lemma 4C.4.

Every $\mathbf{Q}$ -valued symmetric bilinear module (L, β) determines a $\mathbf{Q}$ -valued quadratic module (L, q_β) by

$$q_\beta(v) := \beta(v, v),$$

and q_β depends only on the symmetric part of β . Conversely, every $\mathbf{Q}$ -valued quadratic module (L, q) determines a symmetric bilinear module (L, β_q) via its polar form of definition 4C.2.

Lemma 4C.5.

There is a bijection between even symmetric integral bilinear forms on L and integral quadratic forms on L :

$$\left\{ \beta \in \text{Sym}_{\mathbf{Z}}^2(L^\vee) \mid \beta \text{ is even} \right\} \longleftrightarrow \text{Quad}_{\mathbf{Z}}(L),$$

under which a bilinear form β is sent to $q(v) := \frac{1}{2}\beta(v, v)$, and a quadratic form q is sent to its polar form β_q . Note that this forward map $\beta \mapsto \frac{1}{2}\beta(v, v)$ differs by the factor $\frac{1}{2}$ from the map $\beta \mapsto q_\beta(v) = \beta(v, v)$ of lemma 4C.4, so this bijection is a distinct construction rather than a restriction of that lemma; only the backward (polar-form) direction is shared. Concretely, the polar form of any integral quadratic form is an even symmetric integral bilinear form, and conversely every even symmetric integral bilinear form β is the polar form of the integral quadratic form $q(v) := \frac{1}{2}\beta(v, v)$.

Proof.

If q is integral then $\beta_q(v, v) = 2q(v) \in 2\mathbf{Z}$, so β_q is even and integral. Conversely, if β is even and integral then $q(v) := \frac{1}{2}\beta(v, v)$ takes values in $\mathbf{Z}$, and its polar form recovers β :

$$\beta_q(v, w) = q(v + w) - q(v) - q(w) = \frac{1}{2}(\beta(v + w, v + w) - \beta(v, v) - \beta(w, w)) = \beta(v, w),$$

using symmetry of β . The two assignments are mutually inverse. ■

4C.3. *Torsion bilinear and quadratic forms.***Definition 4C.6**

A **torsion bilinear form** is a pair (G, β) where G is a finitely generated torsion $\mathbf{Z}$ -module and

$$\beta : G \otimes_{\mathbf{Z}} G \rightarrow \mathbf{Q}/\mathbf{Z}$$

is a symmetric bilinear form. A **torsion quadratic form** is a pair (G, q) where G is a finitely generated torsion $\mathbf{Z}$ -module and $q : G \rightarrow \mathbf{Q}/\mathbf{Z}$ is a quadratic form, i.e. $q(\lambda x) = \lambda^2 q(x)$ for all $x \in G$ and $\lambda \in \mathbf{Z}$, whose polar form is a torsion bilinear form.

Remark 4C.7: The discriminant group A_L of the Lattice Theory section is a finite, hence finitely generated torsion, $\mathbf{Z}$ -module, so its associated forms are instances of this notion. We record the discriminant data as two separate forms: the discriminant quadratic form $q_L : A_L \rightarrow \mathbf{Q}/2\mathbf{Z}$ (matching the $\mathbf{Q}/2\mathbf{Z}$ normalization of the Lattice Theory section) and the discriminant bilinear form $b_L : A_L \times A_L \rightarrow \mathbf{Q}/\mathbf{Z}$. These share the numerator $\beta(\tilde{x}, \tilde{y})$ read modulo different lattices; the diagonal $q_L(\bar{x}) = \beta(\tilde{x}, \tilde{x}) \bmod 2\mathbf{Z}$ reduces to $b_L(\bar{x}, \bar{x}) = \beta(\tilde{x}, \tilde{x}) \bmod \mathbf{Z}$ under the natural surjection $\mathbf{Q}/2\mathbf{Z} \twoheadrightarrow \mathbf{Q}/\mathbf{Z}$, not under any multiplication-by-2 isomorphism. ✍

4C.4. *The discriminant bilinear and quadratic forms.***Definition 4C.8**

Let (L, β_L) be a nondegenerate even lattice with discriminant group $A_L = L^\vee/L$, and let β also denote the $\mathbf{Q}$ -valued extension of the form to $L^\vee$. The **discriminant bilinear form** of L is the torsion bilinear form

$$b_L : A_L \times A_L \rightarrow \mathbf{Q}/\mathbf{Z}$$

$$(\bar{x}, \bar{y}) \mapsto \beta(x, y) \bmod \mathbf{Z},$$

computed on any lifts $x, y \in L^\vee$ of $\bar{x}, \bar{y}$. Its associated **discriminant quadratic form** is

$$q_L(\bar{x}) := \beta(x, x) \bmod 2\mathbf{Z} \in \mathbf{Q}/2\mathbf{Z},$$

for any lift x of $\bar{x}$; this is the $\mathbf{Q}/2\mathbf{Z}$ -valued form q_L recalled from the Lattice Theory section. Reducing modulo $\mathbf{Z}$ recovers the diagonal of b_L , i.e. $q_L(\bar{x}) \bmod \mathbf{Z} = b_L(\bar{x}, \bar{x})$, via the surjection $\mathbf{Q}/2\mathbf{Z} \twoheadrightarrow \mathbf{Q}/\mathbf{Z}$. The **orthogonal group** $O(A_L)$ is the group of automorphisms of A_L preserving q_L . The **length** $\ell(L)$ of L is the minimal number of generators of the abelian group A_L .

Remark 4C.9: Both b_L and q_L are well defined: replacing a lift x by $x + m$ with $m \in L$ changes $\beta(x, y)$ by $\beta(m, y) \in \mathbf{Z}$ (so b_L is well defined modulo $\mathbf{Z}$) and changes $\beta(x, x)$ by $\beta(2x, m) + \beta(m, m) \in 2\mathbf{Z}$ (so q_L is well defined modulo $2\mathbf{Z}$), since $\beta(L^\vee, L) \subseteq \mathbf{Z}$ and L is even. These forms are Nikulin's discriminant forms [Nik80].

Proposition 4C.10.

The discriminant forms b_L and q_L of a nondegenerate lattice L are themselves nondegenerate, meaning that $b_L(\bar{x}, \cdot) = 0$ in $\text{Hom}(A_L, \mathbf{Q}/\mathbf{Z})$ implies $\bar{x} = 0$. For any $\bar{x}, \bar{y} \in A_L$ the $\mathbf{Q}/\mathbf{Z}$ -valued bilinear form is recovered from the $\mathbf{Q}/2\mathbf{Z}$ -valued quadratic form by

$$b_L(\bar{x}, \bar{y}) = \frac{1}{2} (q_L(\bar{x} + \bar{y}) - q_L(\bar{x}) - q_L(\bar{y})),$$

where the bracketed difference lies in $\mathbf{Q}/2\mathbf{Z}$ and equals $2\beta(\tilde{x}, \tilde{y}) \bmod 2\mathbf{Z}$; halving this even representative yields a well-defined element $\beta(\tilde{x}, \tilde{y}) \bmod \mathbf{Z}$ of $\mathbf{Q}/\mathbf{Z}$.

Proof.

Nondegeneracy is the statement that the induced map $A_L \rightarrow \text{Hom}(A_L, \mathbf{Q}/\mathbf{Z})$ is an isomorphism; this holds because A_L is finite and the pairing b_L is the pairing induced by the perfect pairing $L^\vee/L \times L^\vee/L \rightarrow \mathbf{Q}/\mathbf{Z}$ coming from a nondegenerate β_L . The polarization identity is the reduction modulo $\mathbf{Z}$ of the identity $\beta(x, y) = \frac{1}{2}(\beta(x + y, x + y) - \beta(x, x) - \beta(y, y))$ on lifts. ■

4C.5. Properties of the dual lattice.

Proposition 4C.11.

Let L and M be nondegenerate lattices. The dual lattice $L^\vee = \text{Hom}_{\mathbf{Z}}(L, \mathbf{Z})$ satisfies the following.

1. Duality commutes with orthogonal direct sums: $(L \oplus M)^\vee = L^\vee \oplus M^\vee$.
2. If L has Gram matrix G_β in a basis B_L , then the dual basis is $B_{L^\vee} = (B_L^t)^{-1}$, and the Gram matrix of the dual form is $G_{\beta^\vee} = G_\beta^{-1}$.
3. The discriminant of the dual satisfies $\text{disc}(L^\vee) = 1/\text{disc}(L)$.
4. The dual of a twist is $(L(m))^\vee = L^\vee(1/m)$, where $L(m)$ is the twist of L by m from the Lattice Theory section.

Remark 4C.12: Property (2) is the source of (3), since $\text{disc}(L^\vee) = \det(G_\beta^{-1}) = 1/\det(G_\beta) = 1/\text{disc}(L)$. Property (1) is compatible with the direct-sum decomposition $A_{L \oplus M} = A_L \oplus A_M$ of discriminant groups recalled in the Lattice Theory section.

4C.6. Geometric identification of the dual lattice.

Theorem 4C.13.

For a nondegenerate integral lattice (L, β_L) , the dual lattice is identified with a $\mathbf{Z}$ -submodule of $L_{\mathbf{Q}} = L \otimes_{\mathbf{Z}} \mathbf{Q}$ via

$$L^\vee \cong \left\{ v \in L_{\mathbf{Q}} \mid \beta_{L_{\mathbf{Q}}}(v, L) \subseteq \mathbf{Z} \right\},$$

where a functional $\varphi \in L^\vee$ corresponds to the unique vector $v_\varphi \in L_{\mathbf{Q}}$ such that $\varphi(w) = \beta_{L_{\mathbf{Q}}}(v_\varphi, w)$ for all $w \in L$. Under this identification one has the chain of inclusions

$$L \subseteq L^\vee \subseteq L_{\mathbf{Q}}.$$

Proof.

Nondegeneracy of β_L makes the $\mathbf{Q}$ -linear extension $L_{\mathbf{Q}} \rightarrow \text{Hom}_{\mathbf{Q}}(L_{\mathbf{Q}}, \mathbf{Q}), v \mapsto \beta_{L_{\mathbf{Q}}}(v, \cdot)$, an isomorphism,

so each $\varphi \in L^\vee \subseteq \text{Hom}_{\mathbf{Q}}(L_{\mathbf{Q}}, \mathbf{Q})$ has a unique preimage $v_\varphi \in L_{\mathbf{Q}}$. The condition $\varphi(L) \subseteq \mathbf{Z}$ translates to $\beta_{L_{\mathbf{Q}}}(v_\varphi, L) \subseteq \mathbf{Z}$, giving the stated image. The inclusion $L \subseteq L^\vee$ is the map ι of the Lattice Theory section, and $L^\vee \subseteq L_{\mathbf{Q}}$ holds because the pairing takes rational values. ■

4C.7. The genus, class group, and class number.

Definition 4C.14

Two lattices L_1, L_2 belong to the same **genus** if $L_{1, \mathbf{Z}_p} \cong L_{2, \mathbf{Z}_p}$ for every prime p , where $L_{i, \mathbf{Z}_p} := L_i \otimes_{\mathbf{Z}} \mathbf{Z}_p$, and $L_{1, \mathbf{R}} \cong L_{2, \mathbf{R}}$. Lattices in the same genus share the same rank, signature, and determinant, but need not be isometric over $\mathbf{Z}$. The **class group** $\text{cl}(L)$ is the set of isometry classes of lattices in the genus of L , and the **class number** is the cardinality $|\text{cl}(L)|$.

Remark 4C.15: For indefinite even lattices L of rank ≥ 3 the class number is 1, so that the genus determines the isometry class; this is Eichler's theorem on the spinor genus of indefinite forms (see [CS10] for a general reference). For definite lattices the situation is reversed: class number 1 is comparatively rare. 

Proposition 4C.16.

If $\text{rank}(L) > 16 + \ell(L)$, where $\ell(L)$ is the length of definition 4C.8, then the class number satisfies $|\text{cl}(L)| \geq 2$.

Proof.

This is the bound of [Sca87]. ■

4D. Embeddings, overlattices, and gluing.

Remark 4D.1: The basic vocabulary of lattices, primitive embeddings, orthogonal complements, discriminant groups, and the unimodular lattices $\text{I}_{p,q}$ and $\text{II}_{p,q}$ was fixed in section 4A. We now develop the finer theory that underlies the lattice computations of this monograph: the several equivalent characterizations of a primitive sublattice, the classification of primitive embeddings up to isometry, Nikulin's correspondence between overlattices and isotropic subgroups of the discriminant form, the splitting of a unimodular sublattice off its ambient lattice, the classification of unimodular lattices, the finiteness of the set of primitive embeddings into an even unimodular lattice, and the behaviour of the discriminant group under scaling. Throughout, S , T , and L denote nondegenerate lattices in the sense of section 4A, β denotes the ambient bilinear form when no confusion can arise, and $A_L := L^\vee/L$ is the discriminant group with its quadratic form q_L . 

4D.1. Primitive and saturated sublattices.

Definition 4D.2 (Saturation)

Let $S \subseteq L$ be a sublattice. The **saturation** of S in L is the sublattice

$$\text{Sat}_L(S) := \left\{ v \in L \mid nv \in S \text{ for some } n \in \mathbf{Z} \setminus \{0\} \right\}.$$

Equivalently, $\text{Sat}_L(S) = S_{\mathbf{Q}} \cap L$, where $S_{\mathbf{Q}} := S \otimes_{\mathbf{Z}} \mathbf{Q}$ is regarded inside $L_{\mathbf{Q}}$. We say S is **saturated** in L if $S = \text{Sat}_L(S)$.

Proposition 4D.3 (Characterization of primitive sublattices).

Let $S \subseteq L$ be a sublattice. The following conditions are equivalent.

1. The inclusion $S \hookrightarrow L$ is a primitive embedding, i.e. $\text{coker}(S \hookrightarrow L)$ is torsionfree.
2. S is saturated in L : $S = \text{Sat}_L(S)$.
3. $S_{\mathbf{Q}} \cap L = S$, where the intersection is taken in $L_{\mathbf{Q}}$.
4. S is a direct summand of L as a $\mathbf{Z}$ -module, i.e. $L \cong S \oplus T$ for some submodule $T \subseteq L$.
5. Every $\mathbf{Z}$ -basis of S extends to a $\mathbf{Z}$ -basis of L .
6. Every $\mathbf{Z}$ -linear functional $S \rightarrow \mathbf{Z}$ is the restriction of a $\mathbf{Z}$ -linear functional $L \rightarrow \mathbf{Z}$.

7. $S = (S^{\perp L})^{\perp L}$, the double orthogonal complement of S in L .
 A sublattice satisfying these conditions is called a **primitive sublattice**.

Proof.

We show (1) $\iff$ (2), (2) $\iff$ (3), (2) $\iff$ (4) $\iff$ (5) $\iff$ (6), and (2) $\iff$ (7).

(1) $\iff$ (2): By definition $\text{coker}(S \hookrightarrow L) = L/S$, and L/S is torsionfree if and only if no nonzero element of L/S is torsion. An element $v + S$ is torsion in L/S precisely when $nv \in S$ for some nonzero $n \in \mathbf{Z}$, i.e. when $v \in \text{Sat}_L(S)$. Thus L/S is torsionfree if and only if $\text{Sat}_L(S) = S$.

(2) $\iff$ (3): This is the second description of the saturation recorded in definition 4D.2. If $v \in S_{\mathbf{Q}} \cap L$, then $v = \sum_i (a_i/n) s_i$ with $s_i \in S$, $a_i \in \mathbf{Z}$, and $n \in \mathbf{Z} \setminus \{0\}$, so $nv \in S$ and $v \in \text{Sat}_L(S)$; conversely if $nv \in S$ then $v = (nv)/n \in S_{\mathbf{Q}}$, so $v \in S_{\mathbf{Q}} \cap L$. Hence $\text{Sat}_L(S) = S_{\mathbf{Q}} \cap L$, and (2) and (3) are the same equation.

(2) $\implies$ (4): The quotient L/S is a finitely generated $\mathbf{Z}$ -module, hence isomorphic to $\mathbf{Z}^k \oplus F$ with F finite. The torsion subgroup F is exactly $\text{Sat}_L(S)/S$, which vanishes by (2), so $L/S \cong \mathbf{Z}^k$ is free. A short exact sequence $0 \rightarrow S \rightarrow L \rightarrow L/S \rightarrow 0$ with L/S free splits, yielding $L \cong S \oplus T$ with $T \cong L/S$.

(4) $\implies$ (5): If $L \cong S \oplus T$, then the union of a $\mathbf{Z}$ -basis of S and a $\mathbf{Z}$ -basis of T is a $\mathbf{Z}$ -basis of L , so any basis of S extends.

(5) $\implies$ (6): Fix a basis $e_1, \dots, e_r$ of S , extended to a basis $e_1, \dots, e_r, e_{r+1}, \dots, e_n$ of L . Given $\varphi: S \rightarrow \mathbf{Z}$, define $\tilde{\varphi}: L \rightarrow \mathbf{Z}$ by $\tilde{\varphi}(e_i) = \varphi(e_i)$ for $i \leq r$ and $\tilde{\varphi}(e_j) = 0$ for $j > r$; then $\tilde{\varphi}$ restricts to φ on S .

(6) $\implies$ (2): Suppose $v \in \text{Sat}_L(S)$ with $nv \in S$, $n \neq 0$, but $v \notin S$. Choose a $\mathbf{Z}$ -basis $s_1, \dots, s_r$ of S and write $nv = \sum_i a_i s_i$ with $a_i \in \mathbf{Z}$. Since $v \notin S$, some coefficient a_i is not divisible by n ; fix such an i and let $\varphi: S \rightarrow \mathbf{Z}$ be the coordinate functional dual to s_i , so $\varphi(nv) = a_i$. By (6) there is $\tilde{\varphi}: L \rightarrow \mathbf{Z}$ restricting to φ , and then $n\tilde{\varphi}(v) = \tilde{\varphi}(nv) = \varphi(nv) = a_i$, forcing $n \mid a_i$, a contradiction. Hence $\text{Sat}_L(S) = S$.

(2) $\implies$ (7): In general $S \subseteq (S^{\perp L})^{\perp L}$ and the right-hand side is saturated (it is cut out by the vanishing of rational linear conditions, hence equals its own $\mathbf{Q}$ -span intersected with L). Because L is nondegenerate, S and $(S^{\perp L})^{\perp L}$ have the same $\mathbf{Q}$ -span, so $(S^{\perp L})^{\perp L} = S_{\mathbf{Q}} \cap L$. If S is saturated, this equals S .

(7) $\implies$ (2): If $S = (S^{\perp L})^{\perp L}$, then S equals a double orthogonal complement, which as just noted equals $S_{\mathbf{Q}} \cap L$; by (3) this is condition (2). ■

4D.2. Equivalence of embeddings.

Definition 4D.4 (Equivalence of primitive embeddings)

Two primitive embeddings $\iota_1: S \hookrightarrow L_1$ and $\iota_2: S \hookrightarrow L_2$ are **equivalent** if there is an isometry $f \in \text{Isom}(L_1, L_2)$ with $f \circ \iota_1 = \iota_2$. When $L_1 = L_2 = L$, two primitive embeddings $\iota_1, \iota_2: S \hookrightarrow L$ are **equivalent** if they are related in this way by an element $f \in \text{O}(L)$. The set of equivalence classes of primitive embeddings of S into a fixed lattice L is denoted $\text{Emb}(S, L)$.

Remark 4D.5: Since an equivalence identifies $\iota_1(S)$ with $\iota_2(S)$ as sublattices of L , describing $\text{Emb}(S, L)$ amounts to classifying the $\text{O}(L)$ -orbits of primitive sublattices of L isometric to S , together with the choice of isometry onto each such sublattice. This is the object controlled by the gluing theory of the next subsection and made finite, in the even unimodular case, by proposition 4D.16.

4D.3. Overlattices and gluing.

Definition 4D.6 (Overlattice)

An **overlattice** of a lattice S is a lattice L containing S as a finite-index sublattice, with $\beta_L|_S = \beta_S$. Equivalently, L is a lattice with $S \subseteq L \subseteq S^{\vee}$, where the inclusions use the canonical map $S \hookrightarrow S^{\vee}$ of section 4A and its dual; the finite quotient L/S is then a subgroup of $A_S = S^{\vee}/S$.

Theorem 4D.7 (*Nikulin's gluing correspondence*).

Let S be an even lattice. There is a bijection

$$\{\text{even overlattices } L \text{ of } S\} \longleftrightarrow \{\text{isotropic subgroups } H \leq A_S\}$$

between the set of even overlattices of S and the set of subgroups $H \leq A_S$ on which the discriminant quadratic form q_S vanishes identically. The bijection sends an overlattice to the isotropic subgroup it cuts out, and an isotropic subgroup to the overlattice it glues:

$$\begin{aligned} L &\longmapsto H_L := L/S \subseteq A_S, \\ H &\longmapsto \eta^{-1}(H) \subseteq S^\vee, \end{aligned}$$

where $\eta: S^\vee \rightarrow A_S$ is the quotient map. Under this correspondence,

$$[L : S] = |H|, \quad A_L \cong H^\perp/H, \quad \text{disc } L = \frac{\text{disc } S}{|H|^2},$$

where $H^\perp \leq A_S$ is the orthogonal complement of H with respect to q_S . Two even overlattices L, L' of S are isometric by an isometry restricting to an element of $O(S)$ if and only if H_L and $H_{L'}$ are conjugate under the image of $O(S)$ in $O(q_S)$.

Proof.

Any lattice L intermediate between S and $S^\vee$ contains S with finite index, and the pairing β_S extends to L with integer values precisely when the image $H_L = L/S \subseteq A_S$ is isotropic for q_S : for $x + S, y + S \in H_L$ one has $\beta_{S_{\mathbf{Q}}}(x, y) \in \mathbf{Z}$ if and only if the associated bilinear form on A_S vanishes on H_L , and evenness of L requires in addition $q_S(x + S) = 0$ for all $x + S \in H_L$. Conversely, given an isotropic $H \leq A_S$, the preimage $\eta^{-1}(H) \subseteq S^\vee$ is an even overlattice of S with $\eta^{-1}(H)/S = H$; the two constructions are mutually inverse.

For the numerical statements, $[L : S] = |L/S| = |H|$. Since disc scales by the square of the index under passage to a finite-index sublattice, $\text{disc } S = [L : S]^2 \text{disc } L = |H|^2 \text{disc } L$, giving the displayed formula. Finally, the discriminant form of L is computed on $L^\vee/L$; one has $L^\vee = \eta^{-1}(H^\perp)$ inside $S_{\mathbf{Q}}$, whence $A_L = L^\vee/L \cong H^\perp/H$.

The orbit statement is immediate from the definitions: an isometry of S extends to an isometry of $S^\vee$ and hence acts on A_S through the natural map $O(S) \rightarrow O(q_S)$, carrying the overlattice attached to H to the one attached to its image. ■

Remark 4D.8: The correspondence of theorem 4D.7 is the engine behind the classification of primitive embeddings. A primitive embedding $S \hookrightarrow L$ with orthogonal complement $T := S^{\perp L}$ realizes L as an even overlattice of the orthogonal direct sum $S \oplus T$, whose discriminant group is $A_S \oplus A_T$ by the additivity recorded in section 4A. The corresponding isotropic subgroup $H \leq A_S \oplus A_T$ is the graph of an isometry $\gamma: H_S \xrightarrow{\sim} H_T$ between subgroups $H_S \leq A_S$ and $H_T \leq A_T$, anti-isometric for the two discriminant forms; the embedding is thus determined by the gluing data (H_S, H_T, γ) . Comparing discriminants across the overlattice L of $S \oplus T$ gives the **discriminant formula**

$$|\text{disc } T| = \frac{|\text{disc } L| \cdot |H|^2}{|\text{disc } S|}, \quad H := H_L,$$

recovering, when L is unimodular, the statement that $|\text{disc } T| = |\text{disc } S|$ and that $H_S = A_S, H_T = A_T$ so that $\gamma: A_S \xrightarrow{\sim} A_T$ is an isometry onto A_T equipped with the negated form; see proposition 4D.16. The framework and these formulas are due to Nikulin [Nik80, §1.4–1.5]. ✍

4D.4. Splitting of unimodular sublattices.

Definition 4D.9 (Splitting)

Let $S \hookrightarrow L$ be a primitive embedding with orthogonal complement $T := S^{\perp L}$. We say S **splits** L if $L = S \oplus T$; equivalently, if L is the trivial index-1 overlattice of $S \oplus T$, i.e. the isotropic subgroup $H_L \leq A_{S \oplus T}$ of theorem 4D.7 is the zero group.

Proposition 4D.10 (*Unimodular sublattices split*).

Let S be a unimodular lattice admitting a primitive embedding into a nondegenerate lattice L , and let

$T := S^{\perp L}$ be its orthogonal complement. Then S splits L :

$$L \cong S \oplus T.$$

Moreover, if L is unimodular then T is unimodular as well.

Proof.

First observe that $S \cap T = \{0\}$: any $x \in S \cap T$ satisfies $\beta_L(x, S) = 0$, and since S is nondegenerate this forces $x = 0$. Thus the sum $S + T$ inside L is direct, giving an inclusion $S \oplus T \subseteq L$.

We show this inclusion is an equality. Let $x \in L$ be arbitrary and consider the functional $\beta_L(x, \cdot)|_S : S \rightarrow \mathbf{Z}$. Since S is unimodular, the canonical map $S \rightarrow S^\vee$ is an isomorphism, so there exists $s \in S$ with $\beta_L(x, y) = \beta_S(s, y)$ for all $y \in S$; that is, $\beta_L(x - s, S) = 0$, so $x - s \in T$. Hence $x = s + (x - s) \in S \oplus T$, and $L = S \oplus T$, which is the first claim.

For the second claim, $L = S \oplus T$ gives $\text{disc } L = \text{disc } S \cdot \text{disc } T$. When L is unimodular, $\text{disc } L = \pm 1$, and since S is unimodular with $\text{disc } S = \pm 1$ we obtain $\text{disc } T = \pm 1$, so T is unimodular. ■

Lemma 4D.11 (Divisibility of isotropic vectors in unimodular lattices).

Let L be a nondegenerate unimodular lattice and let $v \in L$ be a primitive isotropic vector. Then there exists $w \in L$ with $\beta_L(v, w) = 1$. In particular $\text{div}_L(v) = 1$ for every primitive vector $v \in L$.

Proof.

Since L is unimodular, the canonical map $L \rightarrow L^\vee$ is an isomorphism, so every functional in $L^\vee = \text{Hom}_{\mathbf{Z}}(L, \mathbf{Z})$ has the form $\beta_L(u, \cdot)$ for a unique $u \in L$. Because v is primitive, by proposition 4D.3 it extends to a $\mathbf{Z}$ -basis $e_1 = v, e_2, \dots, e_n$ of L . Let $\varphi : L \rightarrow \mathbf{Z}$ be the dual-basis functional with $\varphi(v) = 1$ and $\varphi(e_j) = 0$ for $j > 1$. Writing $\varphi = \beta_L(w, \cdot)$ for the corresponding $w \in L$ and using symmetry gives $\beta_L(v, w) = \beta_L(w, v) = \varphi(v) = 1$. For the divisibility statement, the image $\beta_L(v, L) = \text{div}_L(v)\mathbf{Z}$ contains $\beta_L(v, w) = 1$, so $\text{div}_L(v) = 1$. ■

Corollary 4D.12 (Hyperbolic splitting of unimodular lattices).

Let L be a nondegenerate unimodular lattice containing a primitive isotropic vector. Then L splits off a rank-2 unimodular hyperbolic plane:

$$L \cong P \oplus P^{\perp L}, \quad P \cong \begin{cases} U, & L \text{ even,} \\ I_{1,1}, & L \text{ odd.} \end{cases}$$

Proof.

Let $e \in L$ be a primitive isotropic vector. By lemma 4D.11 choose $w \in L$ with $\beta_L(e, w) = 1$, and set $k := w^2$. The sublattice $P := \langle e, w \rangle$ has Gram matrix $\begin{bmatrix} 0 & 1 \\ 1 & k \end{bmatrix}$ of determinant -1 , hence P is a rank-2 unimodular sublattice, and it is primitive since a unimodular sublattice is saturated by proposition 4D.3.

The isometry type of P is governed by the parity of $k = w^2$: the Gram matrix $\begin{bmatrix} 0 & 1 \\ 1 & k \end{bmatrix}$ gives $P \cong U$ when k is even and $P \cong I_{1,1}$ (the odd rank-2 unimodular hyperbolic lattice) when k is odd. We adjust w within its coset to realize the parity dictated by L ; note that adding to w any vector of $\langle e \rangle^{\perp L}$ preserves $\beta_L(e, w) = 1$.

If L is even then $k = w^2$ is even; replacing w by $w - \frac{k}{2}e$ (and $e \in \langle e \rangle^{\perp L}$ since $e^2 = 0$) leaves $\beta_L(e, w) = 1$ unchanged and makes $w^2 = 0$, so $P \cong U$.

If L is odd we arrange w^2 to be odd, so that $P \cong I_{1,1}$. Every $x \in L$ satisfies $x - \beta_L(e, x)w \in \langle e \rangle^{\perp L}$, so $L = \langle e \rangle^{\perp L} + \mathbf{Z}w$. Were $\langle e \rangle^{\perp L}$ to consist entirely of even-norm vectors and w^2 even, every $x = y + mw$ ($y \in \langle e \rangle^{\perp L}$) would have $x^2 = y^2 + 2m\beta_L(y, w) + m^2w^2$ even, forcing L even — contrary to hypothesis.

Hence either w^2 is already odd, or $\langle e \rangle^{\perp L}$ contains a vector u of odd norm; in the latter case replace w by $w + u$, which preserves $\beta_L(e, w) = 1$ and gives $(w + u)^2 = w^2 + 2\beta_L(w, u) + u^2$ odd. Either way $k = w^2$ is odd and $P \cong I_{1,1}$.

Since P is unimodular, proposition 4D.10 gives $L \cong P \oplus P^{\perp L}$. ■

4D.5. Classification of unimodular lattices.

Theorem 4D.13 (Classification of indefinite unimodular lattices).

Any indefinite unimodular lattice is determined up to isometry by its rank, index, and parity. Explicitly, an indefinite unimodular lattice of signature (p, q) is isometric to $I_{p,q} = \langle 1 \rangle^{\oplus p} \oplus \langle -1 \rangle^{\oplus q}$ if it is odd, and to $II_{p,q}$ (which requires $p - q \equiv 0 \pmod{8}$) if it is even. The same uniqueness by rank, index, and parity holds for definite unimodular lattices of rank at most 8.

Theorem 4D.14 (Classification of small unimodular lattices).

Let L be any unimodular lattice, definite or indefinite, with $\text{rank}_{\mathbf{Z}} L \leq 4$. Then either

1. L is odd and $L \cong I_{p,q}$ for some (p, q) with $p + q = \text{rank}_{\mathbf{Z}} L$, or
2. L is even and $L \cong U$ or $L \cong U^{\oplus 2}$.

Remark 4D.15: The indefinite classification is Serre's theorem on integral quadratic forms [Ser73, Ch. V]; the uniqueness statement for definite unimodular lattices of small rank, and the rank ≤ 4 enumeration, are classical and may be found in Milnor–Husemoller [MH73, Ch. II] and in Conway–Sloane [CS10, Ch. 15]. The even case is especially rigid: in the range $\text{rank}_{\mathbf{Z}} L \leq 4$ the only even unimodular lattices are U (rank 2) and $U^{\oplus 2}$ (rank 4), there being no even unimodular lattice of odd rank and none of rank 0 other than the zero lattice. The constraint $p - q \equiv 0 \pmod{8}$ for $II_{p,q}$ reflects the fact that the signature of an even unimodular lattice is divisible by 8. ✍

4D.6. Finiteness of embeddings.

Proposition 4D.16 (Finiteness of embeddings into even unimodular lattices).

If S and L are even lattices and L is unimodular, then $\text{Emb}(S, L)$ is a finite set.

Proof.

By proposition 4D.10 and remark 4D.8, a primitive embedding $S \hookrightarrow L$ into an even unimodular L is equivalent to the data of its orthogonal complement $T := S^{\perp L}$ together with an isometry

$$\gamma: A_S \xrightarrow{\sim} A_T(-1),$$

because unimodularity of L forces the gluing subgroup to be the whole graph of γ (equivalently $H_S = A_S$, $H_T = A_T$). Two such embeddings are equivalent in $\text{Emb}(S, L)$ if and only if the corresponding γ differ by the action of $O(q_T)$; the equivalence class therefore depends only on the isometry class of T and on a $O(q_T)$ -orbit of isometries $A_S \xrightarrow{\sim} A_T(-1)$.

Each ingredient is finite. The discriminant groups A_S and A_T are finite abelian groups, so the set $\text{Isom}(A_S, A_T(-1))$ of discriminant-form isometries and the group $O(q_T)$ are finite. The isometry class of T is constrained to a fixed genus in the sense of definition 4C.14 (its signature is $\text{sign } L - \text{sign } S$ and its discriminant form is $-q_S$), and a genus of lattices contains only finitely many isometry classes; equivalently, the class group $\text{cl}(T)$ of definition 4C.14 is finite. A finite union of finite sets of $O(q_T)$ -orbits is finite, so $\text{Emb}(S, L)$ is finite. ■

4D.7. The exact sequence for scaled lattices.

Proposition 4D.17 (Discriminant group of a scaled lattice).

Let L be a lattice and m a positive integer, and let $L(m)$ be the twist of L by m (section 4A). There is

a short exact sequence of finite abelian groups

$$0 \rightarrow L/mL \rightarrow A_{L(m)} \rightarrow A_L \rightarrow 0.$$

In particular, if L is unimodular then $A_L = 0$ and $A_{L(m)} \cong L/mL \cong (\mathbf{Z}/m\mathbf{Z})^{\text{rank } L}$.

Proof.

The underlying module of $L(m)$ is L , with form $\beta_{L(m)} = m\beta_L$. Consequently the canonical map $\iota_{L(m)}: L(m) \rightarrow L(m)^\vee$ is m times the canonical map $\iota_L: L \rightarrow L^\vee$ after the identification $L(m)^\vee \cong L^\vee$ of underlying modules. Thus $L(m)^\vee \cong L^\vee$ and the image of $L(m)$ inside $L(m)^\vee$ is $m\iota_L(L)$. The discriminant group is

$$A_{L(m)} = L(m)^\vee / L(m) \cong L^\vee / mL.$$

The inclusion $mL \subseteq L \subseteq L^\vee$ then yields the short exact sequence

$$0 \rightarrow L/mL \rightarrow L^\vee / mL \rightarrow L^\vee / L \rightarrow 0,$$

which is the claimed sequence $0 \rightarrow L/mL \rightarrow A_{L(m)} \rightarrow A_L \rightarrow 0$. If L is unimodular then ι_L is an isomorphism, so $A_L = 0$ and $A_{L(m)} \cong L/mL \cong (\mathbf{Z}/m\mathbf{Z})^{\text{rank } L}$. ■

Proposition 4D.18 (The orthogonal group exact sequence).

Let L be a lattice. Every isometry $f \in \text{O}(L)$ extends by functoriality to an isometry of $L^\vee$ and hence induces an automorphism of the discriminant form A_L , defining a group homomorphism

$$\psi: \text{O}(L) \rightarrow \text{O}(q_L).$$

Writing $\tilde{\text{O}}(L) := \ker \psi$ for the **stable orthogonal group** of L (section 4A) and $\text{O}^*(q_L) := \text{coker } \psi$, there is an exact sequence

$$0 \rightarrow \tilde{\text{O}}(L) \rightarrow \text{O}(L) \xrightarrow{\psi} \text{O}(q_L) \rightarrow \text{O}^*(q_L) \rightarrow 0.$$

Remark 4D.19: The cokernel $\text{O}^*(q_L)$ measures the obstruction to lifting an automorphism of the discriminant form q_L to an isometry of L . By Nikulin's surjectivity criterion, ψ is surjective, i.e. $\text{O}^*(q_L) = 0$, whenever L is indefinite and

$$\ell(A_L) + 2 \leq \text{rank } L,$$

where $\ell(A_L)$ is the length of definition 4C.8, i.e. the minimal number of generators of A_L [Nik80, Cor. 1.5.2, Thm. 1.14.2]. For a unimodular lattice such as U or E_8 the discriminant group is trivial, so $\text{O}(q_L) = 0$ and $\tilde{\text{O}}(L) = \text{O}(L)$; the same triviality underlies the surjectivity statement invoked for L_{K3} in lemma 4G.2.

4E. Special lattices and transformations.

Remark 4E.1: The lattice-theoretic study of moduli of Coble and Enriques surfaces is carried out on a small stock of recurring lattices and on the group-theoretic transformations that move vectors within them. This section collects those lattices together with their discriminant data, and records the Eichler–Siegel transvections that generate the arithmetic groups acting on the associated period domains. Throughout we use the notions of the Lattice Theory section: the dual lattice $L^\vee$, the discriminant group $A_L = L^\vee / L$, the discriminant quadratic form q_L , the twist $L(n)$, the rank-one lattice $\langle n \rangle$, the hyperbolic lattice U , the negative-definite root lattices A_n, D_n, E_n , and the 2-elementary invariants (r, a, δ) . The discriminant computations below are instances of the general exact sequence for scaled lattices, proposition 4D.17.

4E.1. The hyperbolic plane and its orthogonal group.

Remark 4E.2 (The hyperbolic plane): Recall that the **hyperbolic plane** $U = \text{II}_{1,1}$ is the unique even unimodular lattice of signature $(1, 1)$, realized as $U = \mathbf{Z}e \oplus \mathbf{Z}f$ with $e^2 = f^2 = 0$ and $ef = 1$, and Gram matrix

$$G_U = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}.$$

Since U is unimodular, $U^\vee = U$ and the discriminant group A_U is trivial. The set of primitive isotropic vectors of U is exactly

$$\{\pm e, \pm f\}.$$

Proposition 4E.3.

The orthogonal group of U is a Klein four-group,

$$\mathrm{O}(U) = \left\{ \pm \mathrm{id}, \pm \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \right\} \cong C_2 \times C_2,$$

generated by the two involutions $\sigma : (e, f) \mapsto (-e, -f)$ and $\tau : (e, f) \mapsto (f, e)$. Over the rationals the group $\mathrm{O}(U_{\mathbf{Q}})$ is much larger: it contains the two one-parameter families of isometries

$$i_a = \begin{bmatrix} a & 0 \\ 0 & a^{-1} \end{bmatrix}, \quad j_b = \begin{bmatrix} 0 & b \\ b^{-1} & 0 \end{bmatrix}, \quad a, b \in \mathbf{Q}^\times,$$

written in the basis (e, f) .

Proof.

An isometry of U permutes the four primitive isotropic vectors $\{\pm e, \pm f\}$ and is determined by its action on the ordered isotropic frame (e, f) . Preservation of $ef = 1$ forces the image frame to be one of $(e, f), (-e, -f), (f, e), (-f, -e)$, giving the four listed matrices. Over $\mathbf{Q}$ the vectors ae and $a^{-1}f$ are again isotropic with $(ae)(a^{-1}f) = ef = 1$, so $i_a \in \mathrm{O}(U_{\mathbf{Q}})$, and likewise $j_b(e) = b^{-1}f$, $j_b(f) = be$ satisfy $(b^{-1}f)(be) = fe = 1$, so $j_b \in \mathrm{O}(U_{\mathbf{Q}})$. ■

4E.2. *The scaled hyperbolic plane $U(2)$.***Proposition 4E.4.**

The scaled lattice $U(2)$ has Gram matrix

$$G_{U(2)} = \begin{bmatrix} 0 & 2 \\ 2 & 0 \end{bmatrix},$$

discriminant $\mathrm{disc}U(2) = -4$, and dual lattice

$$U(2)^\vee = \tfrac{1}{2}U.$$

Its discriminant group is

$$A_{U(2)} \cong U/2U \cong (\mathbf{Z}/2\mathbf{Z})^2,$$

a 2-elementary group of rank 2.

Proof.

The dual form has Gram matrix $G_{U(2)}^{-1} = \tfrac{1}{2}G_U$, so under the geometric identification of $U(2)^\vee$ as a submodule of $U_{\mathbf{Q}}$ (theorem 4C.13) the dual lattice is $\tfrac{1}{2}U$, that is, the module $\tfrac{1}{2}\mathbf{Z}e \oplus \tfrac{1}{2}\mathbf{Z}f$ with the form of U . Because U is unimodular of rank 2, the scaled-lattice sequence $0 \rightarrow U/2U \rightarrow A_{U(2)} \rightarrow A_U \rightarrow 0$ of proposition 4D.17 has $A_U = 0$, giving $A_{U(2)} \cong U/2U \cong (\mathbf{Z}/2\mathbf{Z})^2$. Concretely $A_{U(2)} = (\tfrac{1}{2}U)/U$ is generated by the images of $\tfrac{1}{2}e$ and $\tfrac{1}{2}f$. ■

4E.3. *The lattice $E_8(2)$.***Proposition 4E.5.**

The scaled root lattice $E_8(2)$, with E_8 the negative-definite E_8 lattice of definition 4B.6, satisfies

$$\mathrm{disc}E_8(2) = 2^8, \quad E_8(2)^\vee \cong \tfrac{1}{2}E_8, \quad A_{E_8(2)} \cong E_8/2E_8 \cong (\mathbf{Z}/2\mathbf{Z})^8,$$

so that $E_8(2)$ is 2-elementary of rank 8 with $A_{E_8(2)}$ of rank 8.

Proof.

For any lattice L of rank r and any positive integer m one has $\mathrm{disc}L(m) = m^r \mathrm{disc}L$, since the Gram matrix of $L(m)$ is m times that of L . Taking $L = E_8$, which is unimodular of rank 8 with $\mathrm{disc}E_8 = 1$,

gives $\text{disc} E_8(2) = 2^8$. Since E_8 is unimodular, $E_8^\vee = E_8$, and the dual form of $E_8(2)$ has Gram matrix $G_{E_8(2)}^{-1} = \frac{1}{2} G_{E_8}^{-1}$, whence $E_8(2)^\vee \cong \frac{1}{2} E_8$. The scaled-lattice sequence of proposition 4D.17 then yields $A_{E_8(2)} \cong E_8/2E_8 \cong (\mathbf{Z}/2\mathbf{Z})^8$. ■

4E.4. The Enriques lattice E_{10} and its twist.

Definition 4E.6 (The Enriques lattice)

The **Enriques lattice** is

$$E_{10} := U \oplus E_8 \xrightarrow{\sim} \Pi_{1,9},$$

the orthogonal direct sum of the hyperbolic plane and the negative-definite E_8 lattice. It is even, unimodular, of signature $(1, 9)$, and is the unique even unimodular lattice of that signature up to isometry.

Remark 4E.7 (Role in Enriques moduli): For an Enriques surface Z the second integral cohomology carries a torsion summand,

$$\text{NS}(Z) \cong H^2(Z; \mathbf{Z}) \cong \mathbf{Z}^{10} \oplus \mathbf{Z}/2\mathbf{Z},$$

and the intersection pairing endows the free part $H^2(Z; \mathbf{Z})_f$ with the structure of an even unimodular lattice of signature $(1, 9)$. There is an isometry $H^2(Z; \mathbf{Z})_f \xrightarrow{\sim} E_{10}$, and a **marking** of Z is a choice of such isometry [CDL25]. The lattice E_{10} is thus the numerical incarnation of the Enriques surface, while its twist governs the period map.

Remark 4E.8 (The twisted Enriques lattice): The relevant period lattice is the twist

$$E_{10}(2) = U(2) \oplus E_8(2),$$

which is even of signature $(1, 9)$ and, being the twist by 2 of a rank-10 unimodular lattice, is 2-elementary with $A_{E_{10}(2)} \cong E_{10}/2E_{10} \cong (\mathbf{Z}/2\mathbf{Z})^{10}$ by proposition 4D.17. In the invariant notation of the Lattice Theory section it has 2-elementary type $(r, a, \delta) = (10, 10, 0)$. This is the lattice denoted $S_{\text{En}} = E_{10}(2)$ in the lattice summary, and it recurs throughout the Coble tables as the base summand $E_{10}(2)$ appearing alongside $U(2)$ and the root lattices; compare table 2.

4E.5. Degree $2d$ K3 lattices.

Definition 4E.9 (Degree $2d$ K3 lattices)

For a positive integer d and an integer $m \geq 0$, define the lattices

$$L_{\text{K3}, 2d}^{(m)} := \langle -2d \rangle \oplus U^2 \oplus E_8^m, \quad L_{\text{K3}, 2d} := L_{\text{K3}, 2d}^{(2)}.$$

The distinguished case $m = 2$ gives a lattice of signature $(2, 19)$, which we call the **degree $2d$ K3 lattice**.

Remark 4E.10: The K3 lattice is $L_{\text{K3}} = U^3 \oplus E_8^2 = \Pi_{3,19}$, of signature $(3, 19)$. Take a primitive vector $h = e + df$ in one hyperbolic summand $U = \langle e, f \rangle$, so that $h^2 = 2d$; it spans a sublattice isometric to $\langle 2d \rangle$. Its orthogonal complement within that U is spanned by $e - df$, which has $(e - df)^2 = -2d$, so $h^\perp U = \langle e - df \rangle \cong \langle -2d \rangle$. Taking the orthogonal complement of h in all of $L_{\text{K3}} = U^3 \oplus E_8^2$ then realizes the orthogonal complement of a degree- $2d$ polarization as

$$h^\perp L_{\text{K3}} \cong \langle -2d \rangle \oplus U^2 \oplus E_8^2 = L_{\text{K3}, 2d}.$$

The lattice $L_{\text{K3}, 2d}$ therefore models the transcendental-type lattice governing the moduli space F_{2d} of K3 surfaces carrying a polarization of degree $2d$; its period domain has signature $(2, 19)$ [Sca87]. The auxiliary lattices $L_{\text{K3}, 2d}^{(m)}$ with other values of m arise when the number of E_8 summands is varied.

4E.6. Nikulin's 2-adic lattices V_k and U_k .

Definition 4E.11 (Nikulin's lattices V_k and U_k)

Let $\mathbf{Z}_2$ denote the ring of 2-adic integers. For an integer $k \geq 0$ define the rank-two $\mathbf{Z}_2$ -lattices

$$V_k := \left(\mathbf{Z}_2^2, \begin{bmatrix} 2^{k+1} & 2^k \\ 2^k & 2^{k+1} \end{bmatrix} \right), \quad U_k := \left(\mathbf{Z}_2^2, \begin{bmatrix} 0 & 2^k \\ 2^k & 0 \end{bmatrix} \right).$$

We abbreviate $V := V_1$. Each of V_k and U_k is even, nondegenerate, and of 2-adic rank 2.

Remark 4E.12: The lattice U_k is the 2-adic hyperbolic plane scaled by 2^k : one has $U_0 \cong U \otimes_{\mathbf{Z}} \mathbf{Z}_2$ and $U_1 \cong U(2) \otimes_{\mathbf{Z}} \mathbf{Z}_2$. The lattice V_k has determinant $\det \begin{bmatrix} 2^{k+1} & 2^k \\ 2^k & 2^{k+1} \end{bmatrix} = 2^{2k+2} - 2^{2k} = 3 \cdot 2^{2k}$, which is positive with positive diagonal entries, so its associated real form is positive definite of signature $(2, 0)$. We caution that the symbol U_k collides with the global notation U for the hyperbolic plane over $\mathbf{Z}$; in the 2-adic setting U without a subscript is never used, and U_k always denotes the 2-adic form above. These lattices are the local building blocks out of which the indecomposable factors of general even 2-elementary lattices are assembled, in the sense of the decomposition below.

4E.7. Eichler–Siegel transvections.

Definition 4E.13 (Eichler–Siegel transvection)

Let L be a lattice, let $e \in L_{\mathbf{Q}}$ be an isotropic vector, and let $a \in \langle e \rangle^{\perp L_{\mathbf{Q}}}$ be a vector orthogonal to e , i.e. $ae = 0$. The **Eichler–Siegel transvection** associated to the pair (e, a) is the isometry $E_{e,a} \in \mathrm{O}(L_{\mathbf{Q}})$ defined by

$$E_{e,a}(v) = v + (ev)a - \left(av + \frac{1}{2}(aa)(ev) \right) e.$$

If a is itself isotropic this simplifies to

$$E_{e,a}(v) = v + (ev)a - (av)e.$$

The construction is due to Eichler [Eic74].

Definition 4E.14 (The Eichler transvection group)

The **Eichler transvection group** $E(L) \leq \mathrm{O}(L)$ is generated by all transvections $E_{e,a}$ for which $e \in L$ is a primitive isotropic vector of divisibility $\mathrm{div}_L(e) = 1$ and $a \in \langle e \rangle^{\perp L}$. For lattices containing two orthogonal hyperbolic planes it acts transitively on primitive vectors of fixed norm and divisibility, and coincides with the stable orthogonal group up to finite index; see [GHS08].

Theorem 4E.15 (*Eichler transvections on $U \oplus U$*).

There is an isometry

$$\phi: U \oplus U \xrightarrow{\sim} (\mathrm{Mat}_{2 \times 2}(\mathbf{Z}), q = \det),$$

identifying the rank-four even lattice $U \oplus U$ with the lattice of integer 2×2 matrices equipped with the determinant as its quadratic form (so that the associated bilinear form is $2 \det$). Under ϕ the group generated by the Eichler–Siegel transvections of $U \oplus U$ corresponds to the image of $\mathrm{SL}_2(\mathbf{Z}) \times \mathrm{SL}_2(\mathbf{Z})$ acting on a matrix $X \in \mathrm{Mat}_{2 \times 2}(\mathbf{Z})$ by

$$(A, B) \cdot X := AXB^{-1}.$$

Remark 4E.16: The quadratic form $q(X) = \det X$ on $\mathrm{Mat}_{2 \times 2}(\mathbf{Z})$ is integer-valued, so the lattice is even, and its associated bilinear form is $b(X, X) = 2 \det X$; it has signature $(2, 2)$, matching that of $U \oplus U$, and indeed $(\mathrm{Mat}_{2 \times 2}(\mathbf{Z}), \det) \cong U \oplus U$ is even unimodular. This realizes the exceptional isogeny between $\mathrm{O}(U \oplus U)$ and a quotient of $\mathrm{SL}_2 \times \mathrm{SL}_2$. Since $\det(AXB^{-1}) = \det X$ for $A, B \in \mathrm{SL}_2$, each pair (A, B) acts by an isometry, and the transvections $E_{e,a}$ generate exactly the image of $\mathrm{SL}_2(\mathbf{Z}) \times \mathrm{SL}_2(\mathbf{Z})$; a proof is given in [GHS08].

4E.8. Decomposability of even 2-elementary lattices.

Theorem 4E.17 (*Decomposition of 2-elementary discriminant forms*).

Let L be an even 2-elementary lattice. Then the discriminant quadratic form on A_L is an orthogonal

direct sum of the four elementary discriminant forms p, q, u, v , subject only to the relations

$$\begin{aligned} u^{\oplus 2} &= v^{\oplus 2}, & p^{\oplus 4} &= q^{\oplus 4}, \\ u \oplus p &= (p \oplus q) \oplus p, & u \oplus q &= (p \oplus q) \oplus q, \\ v \oplus p &= q^{\oplus 3}, & v \oplus q &= p^{\oplus 3}. \end{aligned}$$

Here p and q are the two rank-one forms on $\mathbf{Z}/2\mathbf{Z}$, realized as the discriminant forms of $\langle 2 \rangle$ and $\langle -2 \rangle$, while u and v are the two rank-two forms on $(\mathbf{Z}/2\mathbf{Z})^2$, realized as the discriminant forms of $U(2)$ and of $V = V_1$ respectively.

Remark 4E.18: The relations record the coincidences among finite 2-adic quadratic forms established by Nikulin [Nik80]: for instance $u^{\oplus 2} = v^{\oplus 2}$ expresses that two orthogonal copies of the even rank-two form agree with two copies of the odd one, and $v \oplus p = q^{\oplus 3}$ resolves the odd rank-two form against the rank-one forms. Together they present the semigroup of 2-elementary discriminant forms under orthogonal sum, so that every A_L has a normal form in the generators p, q, u, v .

Theorem 4E.19 (Building blocks of K3-embeddable 2-elementary lattices).

The even 2-elementary lattices admitting a primitive embedding into the K3 lattice L_{K3} decompose as orthogonal direct sums of basic lattices, such as

$$A_1, \quad D_4, \quad D_6, \quad D_8, \quad E_7, \quad E_8, \quad E_8(2), \quad \langle 2 \rangle, \quad U, \quad U(2).$$

This standard building-block list is not asserted to be exhaustive.

Remark 4E.20: This building-block list is drawn from Nikulin's study of the even 2-elementary hyperbolic lattices embedding into L_{K3} [Nik80], the same theory that underlies the invariant triple (r, a, δ) of the Lattice Theory section. The Coble lattices of table 2 are exactly of this form: each 2-elementary lattice M appearing there is assembled from these blocks — for example $M = E_8(2) \oplus U \oplus A_1^{\oplus 2}$ at $n = 2$ and $M = E_8 \oplus D_8 \oplus U(2)$ at $n = 8$ — so the table is a concrete census of the decomposition in the range relevant to Coble surfaces.

4F. Invariant and coinvariant lattices.

Remark 4F.1: We record the lattice-theoretic constructions attached to a group of isometries of a lattice, and specialize them to the case of an involution. This apparatus underpins the lattice theory of the K3-cover / Enriques-cover picture used throughout: the 2-elementary lattices S_{Co} and T_{Co} arise as the invariant and coinvariant lattices of the nonsymplectic involution on the K3 cover of a Coble surface, and it is these lattices that enter the chain of primitive embeddings of lemma 4G.2.

Definition 4F.2

Let $G \hookrightarrow O(L)$ be a faithful representation realizing a group G acting on a lattice L by isometries. The **invariant sublattice** and the **coinvariant sublattice** of the action are defined as

$$L^G := \left\{ v \in L \mid g.v = v \text{ for all } g \in G \right\}, \quad L_G := (L^G)^{\perp L}.$$

Thus L^G is the sublattice of vectors fixed by every element of G , and L_G is its orthogonal complement in L .

Proposition 4F.3.

Suppose $G = \langle I \rangle$ is generated by an involution $I \in O(L)$, so that $I^2 = \text{id}$. Then, for the induced action of I on $L_{\mathbf{Q}}$, the rational spans $(L^G)_{\mathbf{Q}}$ and $(L_G)_{\mathbf{Q}}$ are precisely the eigenspaces of I for the eigenvalues $+1$ and -1 respectively, and

$$L_{\mathbf{Q}} = (L^G)_{\mathbf{Q}} \perp (L_G)_{\mathbf{Q}}.$$

Proof.

Since $I^2 = \text{id}$, the induced isometry of $L_{\mathbf{Q}}$ is diagonalizable with all eigenvalues in $\{+1, -1\}$, so $L_{\mathbf{Q}} = V_+ \oplus V_-$ where $V_{\pm}$ denotes the ± 1 eigenspace. The fixed space V_+ is by definition $(L^G)_{\mathbf{Q}}$. For $v \in V_+$ and $w \in V_-$, using that I is an isometry,

$$\beta_{L_{\mathbf{Q}}}(v, w) = \beta_{L_{\mathbf{Q}}}(Iv, Iw) = \beta_{L_{\mathbf{Q}}}(v, -w) = -\beta_{L_{\mathbf{Q}}}(v, w),$$

so $\beta_{L_{\mathbf{Q}}}(v, w) = 0$. Hence $V_- \subseteq V_+^\perp$, and comparing dimensions gives $V_- = V_+^\perp = (L_G)_{\mathbf{Q}}$. ■

Proposition 4F.4.

Let $G \hookrightarrow \mathrm{O}(L)$ be a faithful representation of a finite cyclic group G . Then L^G and L_G are primitive sublattices of L , and the orthogonal direct sum $L^G \oplus L_G$ is a finite-index sublattice of L [Nik80].

Proposition 4F.5.

Let L be an even nondegenerate lattice and let $G = \langle I \rangle \cong \mathbf{Z}/2\mathbf{Z}$ be generated by an involution $I \in \mathrm{O}(L)$. The invariant and coinvariant sublattices L^G and L_G are primitive and mutually orthogonal. Moreover, each $v \in L$ decomposes orthogonally, up to a factor of 2, as

$$2 \cdot v = v_+ + v_-, \quad v_+ := v + I(v) \in L^G, \quad v_- := v - I(v) \in L_G.$$

Proof.

Orthogonality and primitivity are the case $G = \langle I \rangle$ of proposition 4F.4, and $L_G = (L^G)^\perp$ is orthogonal to L^G by definition. For the decomposition, fix $v \in L$ and set $v_+ := v + I(v)$ and $v_- := v - I(v)$, both of which lie in L . Applying I and using $I^2 = \mathrm{id}$,

$$I(v_+) = I(v) + I^2(v) = I(v) + v = v_+, \quad I(v_-) = I(v) - v = -v_-,$$

so v_+ is fixed by I and hence $v_+ \in L^G$, while v_- lies in the -1 eigenspace of I and hence, by proposition 4F.3, in $(L_G)_{\mathbf{Q}} \cap L = L_G$. Adding the two expressions gives $v_+ + v_- = 2v$, and the summands are orthogonal since $v_+ \in L^G$ and $v_- \in L_G = (L^G)^\perp$. ■

Proposition 4F.6.

Let L be a 2-elementary lattice and let $I \in \mathrm{O}^*(L)$ be an involution lying in the stable orthogonal group $\mathrm{O}^*(L) = \tilde{\mathrm{O}}(L)$. Then both the invariant lattice L^I and the coinvariant lattice L_I are again 2-elementary.

4G. Lattice Summary. We summarize the lattices that will be relevant to our discussion:

$$\begin{aligned} L_{K3} &= (22, 0, 0)_3 = U^3 \oplus E_8^2 = \mathrm{II}_{3,19} & E_{10} &= (10, 0, 0)_1 = U \oplus E_8 = \mathrm{I}_{1,9} \\ S_{\mathrm{En}} &= (10, 10, 0)_1 = E_{10}(2) & T_{\mathrm{En}} &= (12, 10, 0)_2 = U \oplus E_{10}(2) \\ S_{\mathrm{Co}} &= (11, 11, 1)_1 = \langle -2 \rangle \oplus E_{10}(2) & T_{\mathrm{Co}} &= (11, 11, 1)_2 = \langle 2 \rangle \oplus E_{10}(2) \\ S_{\mathrm{dP}} &= (2, 2, 0)_1 = U(2) & T_{\mathrm{dP}} &= (20, 2, 0)_2 = U \oplus U(2) \oplus E_8^2 \end{aligned}$$

Lemma 4G.1.

Writing

$$\begin{aligned} T_{\mathrm{Co}} &= \langle 2 \rangle \oplus E_{10}(2) = \langle h, e', f', \alpha_1, \dots, \alpha_8 \rangle \\ T_{\mathrm{En}} &= U \oplus E_{10}(2) = \langle \tilde{e}, \tilde{f}, \tilde{e}', \tilde{f}', \tilde{\alpha}_1, \dots, \tilde{\alpha}_8 \rangle, \end{aligned}$$

there is an embedding of lattices $T_{\mathrm{Co}} \hookrightarrow T_{\mathrm{En}}$:

$$\begin{aligned} \eta : \langle 2 \rangle \oplus E_{10}(2) &\rightarrow U \oplus E_{10}(2) \\ (h, x) &\mapsto (\tilde{e} + \tilde{f}, x) \end{aligned}$$

which sends the generator h of $\langle 2 \rangle$ to $\tilde{e} + \tilde{f} \in U$ and is the identity on the $E_{10}(2)$ summand. Since $\mathrm{coker} \eta$ is torsionfree, η is a primitive embedding.

Lemma 4G.2.

There is a sequence of primitive embeddings

$$T_{\text{Co}} \hookrightarrow T_{\text{En}} \hookrightarrow T_{\text{dP}} \hookrightarrow L_{K3}$$

which is unique up to $O(L_{K3})$. In particular, this yields an embedding

$$\begin{aligned} \langle 2 \rangle \oplus U(2) \oplus E_8(2) &\hookrightarrow U \oplus U(2) \oplus E_8^2 \\ (h, x, y) &\mapsto (\tilde{e} + \tilde{f}, x, y, y) \end{aligned}$$

and thus an embedding $F_{\text{Co}} \hookrightarrow F_{(2,2,0)}$.

Proof.

By [Ale+25, Lem. 2.4], it suffices to show uniqueness of $S_{\text{En}} \hookrightarrow S_{\text{Co}}$, i.e.

$$E_{10}(2) \hookrightarrow \langle -2 \rangle \oplus E_{10}(2),$$

or equivalently by untwisting,

$$E_{10} \hookrightarrow \langle -1 \rangle \oplus E_{10}.$$

This embedding is unique since one can write the codomain as $E_{10}^\perp \oplus E_{10}$. Similarly, by [Nik80, Cor. 1.5.2, Thm. 3.6.3], the homomorphism $O(L_{K3}) \rightarrow O(T_{\text{Co}})$ is surjective. ■

Lemma 4G.3.

The embeddings of lattices $\eta : T_{\text{Co}} \hookrightarrow T_{\text{En}}$ (resp. $T_{\text{Co}} \hookrightarrow T_{\text{dP}}$) induce locally closed embedding $F_{\text{Co}} \hookrightarrow F_{\text{En}}$ (resp. $F_{\text{Co}} \hookrightarrow F_{(2,2,0)}$) which extend to a morphisms on the Baily-Borel compactifications.

Proof.

This follows from [KK72, §5, Thm.2]. ■

Theorem 4G.4.

F_{Co} is the normalization of a closed subvariety of F_{En} .

Proof.

It suffices to show that $D(T_{\text{Co}})/O^+(T_{\text{Co}})^* \rightarrow D(T_{\text{En}})/O^+(T_{\text{En}})^*$ is a finite morphism which is generically injective. The lattice embedding $T_{\text{Co}} \hookrightarrow T_{\text{En}}$ induces an injective morphism $D(T_{\text{Co}}) \hookrightarrow D(T_{\text{En}})$. It remains to show that the stabilizer of T_{Co} in $O(T_{\text{En}})$ is precisely $O(T_{\text{Co}})$ and the morphism is finite.

This morphism is finite because...

The stabilizer statement follows from... ■

I don't know how to prove this. Maybe one should embed into T_{dP} instead to get the stabilizer statement? Finiteness is still unclear. Maybe one can use finite $\iff$ proper and finite fibers, using Stacks tag 02LS. This can be checked Zariski locally?

Maybe this can be proved using Zariski's main theorem: a birational morphism to a normal variety with finite fibers is an isomorphism onto an open subset. Is this morphism birational? What are the fibers, and how can we tell if they are finite?

4H. Coble lattices. The following can be found in an unpublished note of Dolgachev.⁶ Write $|-2K_S| = \{C\}$ where $C = C_1 + \dots + C_n$ has n irreducible components. By adjunction, $C_i \cong \mathbf{P}^1$ and $C_i^2 = -4$, so $K_S^2 = -n$. Let Σ be a general Coble set of points in $\mathbf{P}^2$. There are 10 irreducible families of K3s obtained as the double cover of S branched over C , where the fixed point locus of the deck transformation consists of n smooth rational curves. Generally the rank of $\text{Pic}(X)$ is $r = 10 + n$ and $\ell = 12 - n$, and $K_S^2 = 9 - |\Sigma| = -n$.

⁶<https://homepage.mi-ras.ru/~prokhorov/conf/isk20/Dolgachev.pdf>

TABLE 2. Table of Coble lattices

n	$ \Sigma $	K_V^2	$M = (r, l, \delta)$	2-elementary lattice M	$N = M^\perp$
1	10	-1	(11, 11, 1)	$E_{10}(2) \oplus A_1$	$I_{2,9}(2)$
2	11	-2	(12, 10, 1)	$E_8(2) \oplus U \oplus A_1^{\oplus 2}$	$I_{2,8}(2)$
3	12	-3	(13, 9, 1)	$D_4^{\oplus 2} \oplus A_1^{\oplus 3} \oplus U(2)$	$I_{2,7}(2)$
4	13	-4	(14, 8, 1)	$D_4^{\oplus 2} \oplus A_1^{\oplus 4} \oplus U(2)$	$I_{2,6}(2)$
5	14	-5	(15, 7, 1)	$E_8 \oplus A_1^{\oplus 5} \oplus U$	$I_{2,5}(2)$
6	15	-6	(16, 6, 1)	$E_{10} \oplus A_1^{\oplus 6}$	$I_{2,4}(2)$
7	16	-7	(17, 5, 1)	$E_8 \oplus D_6 \oplus A_1 \oplus U(2)$	$I_{2,3}(2)$
8	17	-8	(18, 4, 0)	$E_8 \oplus D_8 \oplus U(2)$	$U(2)^{\oplus 2}$
8	17	-8	(18, 4, 1)	$E_{10} \oplus D_6 \oplus A_1^{\oplus 2}$	$I_{2,2}(2)$
9	18	-9	(19, 3, 1)	$E_{10} \oplus D_8 \oplus A_1$	$I_{2,1}(2)$
10	19	-10	(20, 2, 1)	$E_{10} \oplus D_{10}$	$I_{2,0}(2)$

This table is reproduced in [CDL25, Table 5.1, p.553]. The fixed point locus is described by [CDL25, Eqn. 5.3.1]:

$$X^g = \begin{cases} \emptyset & \text{if } (r, l, \delta) = (10, 10, 0), \\ C_1^{(1)} + C_2^{(1)} & \text{if } (r, l, \delta) = (10, 8, 0), \\ C^{(g)} + \sum_{i=1}^k R_i & \text{otherwise} \end{cases}$$

where $C^{(g)}$ denotes a curve of genus $g \geq 0$, R_i are disjoint (-2) -curves, and

$$g = \frac{1}{2}(22 - r - l), \quad k = \frac{1}{2}(r - l).$$

The ramification divisor for the involution on the K3 cover is described by $g = 0$ and $k = n - 1$ by [CDL25, Def. 5.4.3].

4I. Nodal Enriques Surfaces. The lattices for S_{Nod} comes from [DK13, Prop. 3.1]. There is a primitive embedding

$$\begin{aligned} S_{\text{En}} &\hookrightarrow S_{\text{Nod}} \\ U(2) \oplus E_8(2) &\hookrightarrow U \oplus \langle -4 \rangle \oplus E_8(2) \\ ((e, f), x) &\mapsto ((2\tilde{f} + \tilde{g} + h, \tilde{g}), x) \end{aligned}$$

where $U(2) = \langle e, f \rangle$, $U = \langle \tilde{e}, \tilde{f} \rangle$, and $\langle -4 \rangle = \langle h \rangle$; this map is the identity on the $E_8(2)$ component. This formula is given on [CDL25, p. 561]. There it is claimed that this induces a map $F_{S_{\text{Nod}}} \hookrightarrow F_{S_{\text{En}}}$ where the former coincides with the moduli space of nodal Enriques surfaces. Geometrically, a nodal Enriques surface is one that contains a smooth rational curve, i.e. a (-2) -curve [CDL25]; the embedding above is thus the lattice-theoretic avatar of this locus inside $F_{S_{\text{En}}}$.

$$S_{\text{Nod}} = (11, 10, 1)_1 = \langle -4 \rangle \oplus E_{10}(2) \quad T_{\text{Nod}} = (11, 8, 1)_2 = \langle 4 \rangle \oplus U \oplus E_8(2)$$

4J. Moduli.

Remark 4J.1 (Coble moduli as a divisor in Enriques moduli): For each $1 \leq n \leq 10$, there is a moduli space of Coble surfaces with n boundary components. As noted, we will primarily be interested in the $n = 1$ case. In this case, we have two distinct constructions of a moduli space of such Coble surfaces. Coble surfaces appear as degenerations of Enriques surfaces, and in fact they form a boundary divisor $\mathcal{H}_{-2}$ in the 10-dimensional moduli space F_{En} of Enriques surfaces. Thus they form a 9-dimensional moduli space. These are realized by allowing the K3 cover of an Enriques surface to acquire an A_1 singularity fixed by the Enriques involution; the resulting quotient has a quartic singularity whose resolution is an irreducible smooth rational curve C satisfying $C^2 = -4$, and by [Nue15, p. 8] is thus a Coble surface S .

Remark 4J.2 (Hodge/lattice theoretic moduli): Separately, one can construct a Hodge-theoretic period domain directly using lattice theory.⁷ Write $|-2K_S| = \{C\}$ where $C = C_1 + \cdots + C_n$ has n irreducible components. By adjunction and the genus formula, $C_i \cong \mathbf{P}^1$ and $C_i^2 = -4$, so $K_S^2 = -n$. Let Σ be a sufficiently general Coble point set in $\mathbf{P}^2$. There are 10 irreducible families of K3s obtained as the double cover of S branched over C , where the fixed locus of the deck transformation is comprised of n smooth rational curves. Generally the rank of $\text{Pic}(X)$ is $r = 10 + n$ and $\ell = 12 - n$, and $K_S^2 = 9 - |\Sigma|$.

See table 2.

The case of interest to us is $n = 1$, and thus the lattice $M = (11, 11, 1) = A_1 \oplus E_{10}(2)$ and its complement N in the K3 lattice will be used to construct a period domain quotiented by an appropriate arithmetic subgroup, producing the coarse moduli space of interest.

Remark 4J.3 (On nonsymplectic involutions): As described in [DZ99], the double cover $\pi : X \rightarrow S$ realizes X as a degeneration of a K3 cover of an Enriques surface, which thus describes a family of K3s equipped with a nonsymplectic involution as studied in [AE22] and a corresponding family of Enriques surfaces as studied in [Ale+25]. It seems that these degenerations correspond to the (weakly projective) **flowerpot** degenerations of [Mor81], and I conjecture that the corresponding Kulikov models correspond to integral-affine discs (as opposed to spheres or real projective spaces).

For introduction, need to say more about how the nonsymplectic involution machinery is being used in this particular case?

4K. Moduli Spaces Summary. We summarize the relevant moduli spaces:

$$\begin{aligned} F_{\text{En}} &= \mathbb{D}(T_{\text{En}})/\text{O}^+(T_{\text{En}})^* & F_{\text{En},2} &= \mathbb{D}(T_{\text{En}})/\Gamma_{\text{En},2} \\ F_{\text{Co}} &= \mathbb{D}(T_{\text{Co}})/\text{O}^+(T_{\text{Co}})^* & F_{\text{Co},2} &= \mathbb{D}(T_{\text{Co}})/\Gamma_{\text{Co},2} \end{aligned}$$

where

$$\begin{aligned} \Gamma_{\text{En},2} &= \text{O}(T_{\text{En}}) \cap \text{O}(T_{\text{dP}}) \subseteq \text{O}(L_{\text{K3}}) \\ \Gamma_{\text{Co},2} &= \text{Stab}_{\text{O}(T_{\text{En}})}(T_{\text{Co}}) \subseteq \text{O}(T_{\text{En}}) \\ F_{\text{Nod}} &= \mathbb{D}(T_{\text{Nod}})/\text{O}^+(T_{\text{Nod}})^* \\ F_{\text{Nod},2} &= ??? \end{aligned}$$

where T_{dP} is described in [Ale+25]. Note that we implicitly use the embedding $\eta : T_{\text{Co}} \hookrightarrow T_{\text{En}}$ of lemma 4G.1.

Degree 2 polarized Coble surfaces do not seem to appear in previous literature, merely (unpolarized?) Cobles with n boundary components. So I can not yet determine a more explicit description of $\Gamma_{\text{Co},2}$.

We note, as in [DK13], that $T_{\text{Co}} \cong v^\perp T_{\text{En}}$ for some $v^2 = -2$, which implies that there is a birational isomorphism $F_{\text{Co}} \xrightarrow{\sim} \mathcal{H}_{-2}/\text{O}^+(T_{\text{En}})^*$.

$$\begin{array}{ccccccc} \mathcal{F}_{\text{Co},2} & \longrightarrow & \mathcal{F}_{\text{En},2} & & & & \\ \downarrow ? & & \downarrow & \searrow & & & \\ \mathcal{F}_{\text{Co}} & \longrightarrow & \mathcal{F}_{\text{En}} & \longrightarrow & \mathcal{F}_{2,2,0} & \hookrightarrow & \mathcal{F}_4 \end{array}$$

FIGURE 4. The tower of moduli spaces relating the degree-2 numerically polarized Coble and Enriques moduli $F_{\text{Co},2}, F_{\text{En},2}$, their unpolarized quotients $F_{\text{Co}}, F_{\text{En}}$, and the degree- $(2, 2, 0)$ K3 moduli space $F_{(2,2,0)} \hookrightarrow F_4$. The map $F_{\text{Co},2} \rightarrow F_{\text{Co}}$ marked ? is not yet determined (cf. the remark on $\Gamma_{\text{Co},2}$ above).

⁷An unpublished note of Dolgachev: <https://homepage.mi-ras.ru/~prokhorov/conf/isk20/Dolgachev.pdf>

4L. GIT Discussion.

Remark 4L.1 (GIT construction): Following [DK13], by varying the coefficients of p_i in the planar blowup construction, one can construct F_{Co} as a locally closed subvariety of $(\mathbf{P}^2)^{10}/\text{PGL}_3$, which is of dimension

$$\dim(\mathbf{P}^2)^{10}/\text{PGL}_3 = \dim(\mathbf{P}^2)^{10} - \dim \text{PGL}_3 = 2 \cdot 10 - (3^2 - 1) = 12.$$

A posteriori, the number of moduli for a Coble surface is 9, which means that there should be 3 conditions imposed upon the configuration of 10 points. These 3 conditions are precisely the *discriminant conditions* described in [Cob19, §2, Prop. (10)]. Letting D be the corresponding discriminant locus, we can identify F_{Co} as an open subset of $((\mathbf{P}^2)^{10} \setminus D)/\text{PGL}_3$ at the level of coarse moduli spaces.

4M. Horikawa model.

Remark 4M.1 (Horikawa's construction): Alternatively, Horikawa [Hor77] and more recently [Ale+25] consider the following: let $Y := \mathbf{P}^1 \times \mathbf{P}^1$ and define an involution $\tau(x, y) := (-x, -y)$. Letting $B \in |-2K_Y|^\tau$ be a τ -invariant anti-bicanonical curve in Y , if B passes through a τ -fixed point $x, y \in \{0, \pm\infty\}$, then the corresponding double branched cover branched over B is a nodal K3 surface X with A_1 singularities and covering involution ι_{dP} such that $Y = X/\langle \iota_{\text{dP}} \rangle$. Letting ι_{En} be a lift of τ , the quotient $Z := X/\langle \iota_{\text{En}} \rangle$ is a Coble surface. The case in which B does not pass through a τ -invariant point yields an Enriques surface Z , and an analysis of the corresponding moduli is carried out in [Ale+25].

It is well-known that there consequently exists a Coble surface S and a blowdown $S \rightarrow X/\iota$ along the strict transform of B . In this way, one realizes the moduli space F_{Co} of unpolarized Coble surfaces as a divisor in F_{En} , the 10-dimensional moduli space of unpolarized Enriques surfaces.

4N. Period Domains. By passing to the K3 cover, one can embed F_{Co} into an arithmetic quotient of a 9-dimensional Hermitian symmetric domain of type IV. Let $L_{\text{K3}} = U^3 \oplus E_8^2$ be the canonical K3 lattice. We recall that for any primitively embedded lattice $S \hookrightarrow L_{\text{K3}}$, letting $T := S^{\perp L_{\text{K3}}}$, there is a Hodge-theoretic description of the coarse moduli space F_S of S -polarized K3 surfaces given by

$$F_S := D_T/\text{O}^+(T)^*$$

where $\text{O}^+(T)^* := \text{O}^+(T) \cap \ker(\text{O}(T) \rightarrow \text{O}(q_T))$, the group $\text{O}^+(T)$ is the index-two subgroup of $\text{O}(T)$ preserving the component D_T , and D_T is a connected component of

$$\Omega_T := \left\{ [v] \in \mathbf{P}(T_{\mathbf{C}}) \mid v^2 = 0, v\bar{v} > 0 \right\}.$$

Letting $E_{10} := U \oplus E_8$, one can similarly consider the Enriques lattices $S_{\text{En}} := E_{10}(2)$ with $T_{\text{En}} = U \oplus E_{10}(2)$. Letting

$$\mathcal{H}_{-2d} = \bigcup_{\delta^2 = -2d} \delta^{\perp D_T},$$

be the union of the hyperplane sections cut out by the vectors δ of norm $-2d$, as in fig. 5, one can thus present

$$F_{\text{En}} = (D_{T_{\text{En}}} \setminus \mathcal{H}_{-2})/\text{O}^+(T_{\text{En}})^*$$

$$F_{\text{Co}} = \mathcal{H}_{-2}/\text{O}^+(T_{\text{En}})^*$$

$$F_{\text{Nod}} = (\mathcal{H}_{-4} \setminus \mathcal{H}_{-2})/\text{O}^+(T_{\text{En}})^*$$

where surfaces along the divisor $\mathcal{H}_{-2}$ in F_{En} correspond precisely to Coble surfaces and those along $\mathcal{H}_{-4} \setminus \mathcal{H}_{-2}$ correspond to Enriques surfaces with A_1 singularities.

cite: 2-elementary invariant and coinvariant lattices L^I, L_I of involution $I \in \text{O}^*(L)$ 2-elementary lattices

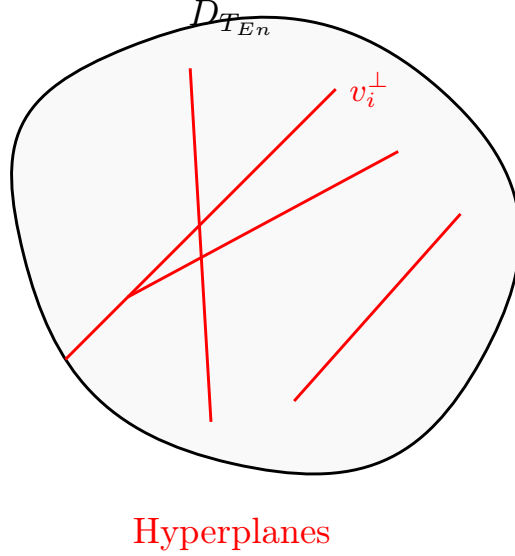


FIGURE 5. The period domain $D_{T_{E_n}}$ together with the hyperplanes $v_i^\perp$ cut out by vectors v_i of fixed negative norm. The Coble surfaces are precisely the periods lying on the hyperplanes of $\mathcal{H}_{-2}$.

Alternatively, one can construct the period domain for F_{C_0} directly. Following [DK13, Prop. 3.1] almost verbatim, let S be the blowup of $\mathbf{P}^2$ along 10 A_1 singularities of a rational sextic curve C . The double branched cover $f : X \rightarrow S$ branched along the proper transform of C contains the pullback of ten exceptional classes $E_1, \dots, E_{10}$ over the ten A_1 singularities, as well as E_0 , the pullback of a hyperplane class. Since $E_i^2 = -2$ for $1 \leq i \leq 10$, and one can show that $E_0^2 = 2$, the E_i generate a lattice isometric to

$$S_{C_0} := \langle 2 \rangle \oplus \langle -2 \rangle^{10} = I_{1,10}(2) = (11, 11, 1)_1 \cong \langle -2 \rangle \oplus E_{10}(2).$$

This follows from the fact that S_{C_0} is a 2-elementary lattice of signature $(1, 10)$ with discriminant group $(\mathbb{Z}/2\mathbb{Z})^{11}$ and thus uniquely determined up to isometry by its invariants (r, a, δ) . Similarly, its orthogonal complement T_{C_0} in the K3 lattice L_{K3} is a 2-elementary lattice of signature $(2, 9)$ and satisfies $q_{S_{C_0}} = -q_{T_{C_0}}$. The isomorphism class of $q_{T_{C_0}}$ determines T_{C_0} , and thus one has

$$T_{C_0} := S_{C_0}^\perp{}^{L_{K3}} = (11, 11, 1)_2 \cong \langle 2 \rangle \oplus E_{10}(2).$$

Alternatively, this follows immediately from the mirror move $S \rightsquigarrow T$ of [AE22, Thm. 5.10] applied to $S_{C_0} = (11, 11, 1)_1$. We obtain F_{C_0} as an open subset of the period domain $D_{T_{C_0}}/\mathcal{O}^+(T_{C_0})^*$, a normal quasiprojective variety of dimension 9, by [DK13, Prop. 3.2] and an application of the Torelli theorem for algebraic K3 surfaces from [PS71]. We note that [DK13] shows that F_{C_0} is rational by relating it to a codimension one subvariety of a moduli space of certain A_2 -singular quintics in $\mathbf{P}^2$.

40. KSBA Spaces.

Remark 40.1: By lemma 4G.3, there are morphisms $\overline{F}_{C_0}^{\text{BB}} \rightarrow \overline{F}_{E_n}^{\text{BB}}$ and $\overline{F}_{C_0}^{\text{BB}} \rightarrow \overline{F}_{(2,2,0)}^{\text{BB}}$ which induce correspondences between the boundary cusps.

Remark 40.2: We set up the moduli space of KSBA stable pairs for Coble surfaces, possibly using the ramification divisor of the K3 involution (which is in this case not fixed-point free). The above embeddings should allow us to take closures of stable pairs in already existing moduli spaces.

5. PERIOD DOMAINS AND COMPACTIFICATIONS

5A. Hyperbolic reflection groups and Vinberg's algorithm.

Remark 5A.1 (Orientation and sign convention): The Baily–Borel and semitoroidal compactifications of the period domains attached to Coble and Enriques surfaces are governed by the action of arithmetic reflection groups on hyperbolic space. This section collects the reflection-group and hyperbolic-polytope combinatorics used elsewhere in the monograph, culminating in Vinberg’s algorithm for a fundamental chamber and the Coxeter–Vinberg diagram that encodes it.

We work throughout with the algebraic-geometry sign convention (see the Lattice Theory section): roots are *negative-definite* directions, a **short root** having norm $v^2 = -2$ and a **long root** norm $v^2 = -4$. An indefinite lattice L of signature $(1, n)$ is called **hyperbolic**; its **positive cone**

$$C_L := \left\{ v \in L_{\mathbf{R}} \mid v^2 > 0 \right\}$$

has two connected components, and mirrors of roots are the hyperplanes orthogonal to negative-norm vectors. Fixing one component C_L^+ identifies $L_{\mathbf{R}}$ with a model of hyperbolic n -space, on which the reflection group acts.

5A.1. Reflections and the Weyl group.

Definition 5A.2

Let (L, β_L) be a nondegenerate integral lattice and $\alpha \in L$ an anisotropic vector (that is, $\beta_L(\alpha, \alpha) \neq 0$) satisfying the integrality condition

$$2\beta_L(\alpha, L) \subseteq \beta_L(\alpha, \alpha)\mathbf{Z}.$$

The **reflection in α** is the isometry

$$\begin{aligned} s_\alpha : L &\rightarrow L \\ x &\mapsto x - 2 \frac{\beta_L(\alpha, x)}{\beta_L(\alpha, \alpha)} \alpha. \end{aligned}$$

Each reflection has determinant -1 , fixes the orthogonal complement $\langle \alpha \rangle^{\perp L}$ pointwise, and sends $\alpha \mapsto -\alpha$. The integrality condition is exactly what guarantees $s_\alpha \in \mathrm{O}(L)$ rather than merely $s_\alpha \in \mathrm{O}(L_{\mathbf{Q}})$. The **Weyl group** of L is the subgroup

$$W(L) := \left\langle s_\alpha \mid \alpha \in L \text{ anisotropic, } s_\alpha \in \mathrm{O}(L) \right\rangle \leq \mathrm{O}(L)$$

generated by all such integral reflections.

5A.2. Roots and k -roots.

Definition 5A.3

Let L be a nondegenerate integral lattice and $v \in L$ a primitive vector. The reflection s_v is always an isometry of $L_{\mathbf{Q}}$, but is integral, $s_v \in \mathrm{O}(L)$, precisely when

$$2\beta_L(v, L) \subseteq v^2\mathbf{Z}.$$

For $k > 0$ the **k -Weyl group** $W^k(L)$ is the subgroup generated by the integral reflections s_v with $v^2 = -k$:

$$W^k(L) := \left\langle s_v \mid s_v \in \mathrm{O}(L), v^2 = -k \right\rangle.$$

A vector v is a **k -root** if $s_v \in W^k(L)$. The full Weyl group $W(L)$ of Definition 5A.2 is generated by integral reflections in vectors of any norm, so that $W(L) = \left\langle W^k(L) \mid k > 0 \right\rangle$.

Definition 5A.4

For a 2-elementary lattice L , the roots are precisely the vectors $v \in L$ of one of the two following types:

- a **short root**, with $v^2 = -2$; or
- a **long root**, with $v^2 = -4$ and $\mathrm{div}_L(v) = 2$.

Remark 5A.5: The two root types of Definition 5A.4 are exactly the primitive vectors of norm -2 or -4 satisfying the integrality condition of Definition 5A.3. For a short root, $2\beta_L(v, L) \subseteq 2\mathbf{Z} = v^2\mathbf{Z}$ holds automatically, and $s_v(x) = x + \beta_L(v, x)v$. For a norm -4 vector, integrality requires $\beta_L(v, L) \subseteq \mathbf{Z}$, which is precisely the condition $\mathrm{div}_L(v) = 2$; then $s_v(x) = x + \frac{1}{2}\beta_L(v, x)v$ lies in $\mathrm{O}(L)$.

5A.3. *Coroots and the weight space.***Definition 5A.6**

Let $R := \langle \Phi \rangle_{\mathbf{Z}}$ be a root lattice generated by a root system $\Phi = \{\alpha_1, \dots, \alpha_n\}$. The dual lattice $R^\vee$ is called the **weight lattice** of R , and the dual-basis elements ω_i determined by $(\omega_i, \alpha_j) = \delta_{ij}$ are the **fundamental weights**.

Remark 5A.7: For the unimodular root lattice E_8 one has $E_8^\vee \cong E_8$, and the fundamental weights $\omega_1, \dots, \omega_8$ — the dual basis to the simple roots — have coordinates in the simple-root basis given by the rows of the inverse Gram matrix. The **coroots** are $\alpha_i^\vee = 2\alpha_i/(\alpha_i, \alpha_i)$, which for the simply-laced E_8 coincide with the simple roots themselves — distinct from the ω_i .

5A.4. *Hyperbolic lattices and their polytopes.***Definition 5A.8**

Let L be a hyperbolic lattice and C_L^+ one connected component of its positive cone C_L . The intersection pairing induces a hyperbolic metric on $L_{\mathbf{R}}$, and the associated hyperbolic n -space $\mathbb{H}_L^n$ admits two standard models:

- the **unit hyperboloid model**

$$\mathbb{H}_L^n := \left\{ v \in L_{\mathbf{R}} \mid v^2 = 1, v \in C_L^+ \right\};$$

- the **projective (Klein) model**

$$\text{BB}_L^n := \mathbf{P}(C_L) \subset \mathbf{P}(L_{\mathbf{R}}),$$

the image of the positive cone in projective space.

To a vector v with $v^2 < 0$ we associate the hyperplane $H_v := v^\perp \cap \mathbb{H}_L^n$. Taking v, w to be unit normals of H_v, H_w , the relative position of two hyperplanes is governed by the pairing $v \cdot w := \beta_L(v, w)$:

- $|v \cdot w| < 1$: the hyperplanes meet in the interior of $\mathbb{H}_L^n$, at dihedral angle θ with $-v \cdot w = \cos \theta$;
- $|v \cdot w| = 1$: the hyperplanes are asymptotically parallel, meeting at an ideal point of the boundary $\partial \mathbb{H}_L^n$;
- $|v \cdot w| > 1$: the hyperplanes are ultra-parallel (divergent), at distance d with $-v \cdot w = \cosh d$.

5A.5. *Coxeter groups and Coxeter polytopes.***Definition 5A.9**

A **Coxeter group** is a group given by a presentation

$$S = \left\langle s_1, \dots, s_n \mid s_i^2 = (s_i s_j)^{m_{ij}} = e \right\rangle,$$

where $m_{ii} = 1$ and $m_{ij} = m_{ji} \geq 2$ for $i \neq j$.

Definition 5A.10

A **Coxeter polytope** P in $X = \mathbb{E}^n, S^n$, or $\mathbb{H}^n$ is a polytope such that:

1. each face lies in a hyperplane $H_i := \alpha_i^\perp$;
2. the dihedral angles between adjacent faces are π/m_{ij} for integers $m_{ij} \geq 2$; and
3. the reflection group $W = \langle s_{\alpha_i} \rangle$ acts properly discontinuously on X with fundamental domain P .

5A.6. *The discriminant locus and Weyl chambers.***Definition 5A.11**

The **discriminant locus** $\Delta(L)$ is the union of all mirrors, i.e. the hyperplanes orthogonal to the short roots $\Phi_2(L) = \left\{ v \in L \mid v^2 = -2 \right\}$:

$$\Delta(L) := \bigcup_{v \in \Phi_2(L)} v^\perp.$$

Definition 5A.12

The **Weyl chambers** of L are the connected components of

$$C_L^\circ := C_L \setminus \Delta(L),$$

where C_L is the cone of positive-norm vectors. A **simple system** is a set of roots whose mirrors form the bounding hyperplanes of a single Weyl chamber, i.e. a fundamental domain for the action of the 2-Weyl group $W^2(L)$ generated by the (-2) -reflections.

5A.7. *Coxeter–Vinberg diagrams.***Definition 5A.13**

The **Coxeter–Vinberg diagram** of a hyperbolic Coxeter polytope is a graph whose vertices correspond to the simple roots r_i (equivalently, to the walls H_i of the fundamental chamber), and whose edges encode the relative position of the mirrors H_i, H_j through the normalized pairing

$$g_{ij} := \frac{r_i \cdot r_j}{\sqrt{r_i^2 r_j^2}} = \cos\left(\frac{\pi}{m_{ij}}\right).$$

Edges are drawn according to the following conventions:

- **omitted** if $m_{ij} \leq 2$ (orthogonal mirrors);
- **multiple (or labeled) lines** if $3 \leq m_{ij} < \infty$, when $H_i \cap H_j$ lies in the interior of $\mathbb{H}^n$;
- **thick / bold** if $g_{ij} = 1$, when $H_i \cap H_j$ lies on the boundary $\partial\mathbb{H}^n$ (asymptotically parallel mirrors);
- **dotted** if $g_{ij} > 1$ (ultra-parallel mirrors).

For 2-elementary lattices, vertices are drawn **white** for short roots ($v^2 = -2$) and **black** for long roots ($v^2 = -4$).

5A.8. *Elliptic and parabolic subdiagrams.***Definition 5A.14**

Let G be a Coxeter–Vinberg diagram with vertices the simple roots v_i . A subdiagram $D \subseteq G$ is:

- **elliptic** if its Gram matrix $(v_i \cdot v_j)$ is positive definite; such a subdiagram defines a finite face of the Coxeter polytope;
- **parabolic** if each connected component is parabolic (its Gram matrix positive semidefinite of corank one) and it is not contained in any strictly larger elliptic subdiagram.

Definition 5A.15

A parabolic subdiagram is **maximal** if it is not properly contained in a larger parabolic subdiagram. Maximal parabolic subdiagrams correspond bijectively to the rank-2 isotropic subspaces $J \subset L$ containing a fixed isotropic line I — equivalently, to the 1-cusps adjacent to the 0-cusp $[I]$ in the Baily–Borel compactification.

5A.9. *Vinberg’s algorithm.***Theorem 5A.16 (Vinberg’s algorithm for a fundamental chamber).**

Let $\Gamma \subseteq O(L)$ be an arithmetic reflection subgroup of a hyperbolic lattice L . Vinberg’s algorithm constructs a fundamental chamber $P \subset \mathbb{H}^n$ for Γ as follows [Vin75].

1. **Control vector.** Choose a generic positive-norm vector $v_0 \in C_L^+$. Its stabilizer in Γ is the reflection group of the finite subdiagram of roots orthogonal to v_0 , which are used as the first walls.
2. **Enumeration by height.** Seek primitive vectors $v \in L$ with $v^2 < 0$, $s_v \in \Gamma$, and mirror H_v meeting the current cone $\tilde{P}$. Order the candidate roots by increasing **height**

$$h(v) := \frac{-2(v_0 \cdot v)}{v^2},$$

a quantity monotone in the hyperbolic distance from v_0 to H_v (for roots of a given norm), used to order the candidate roots and process them in successive batches of increasing height.

3. **Acceptance / pruning.** A candidate root v at the current height is kept only if its mirror bounds the chamber on the side containing v_0 ; equivalently, the inward normals of accepted walls pair non-negatively, so v is accepted iff $v \cdot r_i \geq 0$ for every previously accepted root r_i (a mirror with some $v \cdot r_i < 0$ would cut into the interior of $\tilde{P}$). Roots whose hyperplanes do not bound $\tilde{P}$ are discarded.
4. **Diagram.** The algorithm proceeds until $\tilde{P}$ becomes a convex polytope of finite volume. The Gram matrix of the accepted simple roots then determines the Coxeter–Vinberg diagram of P (Definition 5A.13).

The algorithm terminates in finitely many steps if and only if Γ has finite covolume in $O(L)$.

Remark 5A.17: The source note states step 3 as “discard those roots whose hyperplanes do not intersect $\tilde{P}$ transversely”; the explicit inequality $v \cdot r_i \geq 0$ recorded above is Vinberg’s acceptance criterion for that condition in the algebraic-geometry sign convention [Vin75].

5A.10. *Root-system conventions and tables.*

Remark 5A.18 (Bourbaki conventions and twists): The classical root systems are taken with the Bourbaki labeling, in the negative-definite twists appropriate to the algebraic-geometry convention:

$$A_n(-1), \quad B_n(-2), \quad C_n(-1), \quad D_n(-1), \quad E_n(-1) \quad (n \in \{6, 7, 8\}), \quad F_4(-2), \quad G_2(-1).$$

Both the simply-laced systems A_n, D_n, E_n and the non-simply-laced systems B_n, C_n, F_4, G_2 occur. The Coxeter diagrams and Gram matrices used throughout are derived from these conventions; the ranks, root counts, Weyl groups, discriminant groups, and explicit Euclidean basis constructions are collected in the root-lattice tables of the Root Systems section and the Diagrams appendix.

5B. Foldings of Dynkin diagrams.

Remark 5B.1 (Orientation): This section collects the folding constructions that produce non-simply-laced root systems from simply-laced ones by summing the roots in each orbit of a diagram automorphism, together with the specific root-folding criterion governing the involution $I = -I_{\text{En}}$ on T_{dP} and the *mirror moves* used to navigate the pyramid of 2-elementary lattices. As in the root systems material we work in the algebraic-geometry sign convention: root lattices are negative definite and roots have norm -2 (see definition 4B.2).

5B.1. Foldings of simply-laced diagrams.

Remark 5B.2 (Classical foldings of Dynkin diagrams): The process of folding by nontrivial diagram automorphisms produces non-simply-laced root systems from simply-laced diagrams. The roots in the same orbit under the folding group must be orthogonal.

Classical examples of foldings of simply-laced Dynkin diagrams include:

- $A_{2n-1} \rightarrow C_n$ via S_2 (horizontal reflection);
- $D_{n+1} \rightarrow B_n$ via S_2 (vertical reflection);
- $D_4 \rightarrow G_2$ via S_3 (rotation by $2\pi/3$);
- $E_6 \rightarrow F_4$ via S_2 (horizontal reflection).

These arise as the fixed-point subalgebras $\mathfrak{g}^\sigma$. The orbit-sum construction of Definition 5B.3 instead produces the **Langlands-dual** root system in types B/C ; accordingly the explicit foldings computed below yield B_n from A_{2n-1} and C_n from D_{n+1} (types F_4 and G_2 are self-dual and unaffected).

5B.2. Folded roots and folded root systems.

Definition 5B.3 (Folded root systems)

Let L be a lattice containing a root system Φ and $G \subset O(L)$ a finite group preserving Φ . Given a simple root $\alpha_i \in \Phi$, let $[\alpha_i] \subset \Phi$ denote its G -orbit.

The associated **folded root** and **folded root system** are defined by

$$\beta_{[\alpha_i]} := \sum_{\alpha_j \in [\alpha_i]} \alpha_j \in L^G, \quad \Phi^G := \left\{ \beta_{[\alpha_i]} \in L^G \mid [\alpha_i] \in \Phi/G \right\}.$$

Each folded root corresponds to a folded node in the Dynkin diagram. For an involution I , $\beta_{[\alpha_i]} = \alpha_i + I(\alpha_i) \in L^G$.

5B.3. Examples.

Example 5B.4 (Examples of classical foldings): Folding produces scaled root systems in the invariant lattice L^G . Explicit examples computed via definition 5B.3:

- $A_5 \rightarrow B_3(2)$: under horizontal reflection ($G = S_2$), $\Phi(A_5^G)$ has Gram matrix $G_{B_3(2)} = 2 \cdot G_{B_3} = \begin{pmatrix} 4 & -2 & 0 \\ -2 & 4 & -2 \\ 0 & -2 & 2 \end{pmatrix}$ (norms 4, 4, 2: two long, one short).
- $D_4 \rightarrow C_3$: under vertical reflection ($G = S_2$), $\Phi(D_4^G)$ has Gram matrix equal to G_{C_3} .
- $D_4 \rightarrow G_2$: under rotation by $2\pi/3$ ($G = \mathbf{Z}/3\mathbf{Z}$), $\Phi(D_4^G)$ has Gram matrix equal to G_{G_2} .
- $E_6 \rightarrow F_4(2)$: under horizontal reflection ($G = S_2$), $\Phi(E_6^G)$ has Gram matrix equal to $G_{F_4(2)}$.

Here the scaling notation $B_3(2)$ and $F_4(2)$ records that the invariant-lattice Gram matrix equals the corresponding root-system Gram matrix rescaled by 2. (These Gram matrices are written in the standard positive-definite normalization; negate for the AG convention of Definition 4B.2.)

5B.4. *The root-folding criterion for T_{dP} .* The criterion below is stated in terms of the short and long roots of T_{dP} : Φ^2 denotes the **short roots** (norm -2) and Φ^4 the **long roots** (norm -4 with divisor 2), in the k -root sense of the root systems material.

Lemma 5B.5 (Root folding criterion).

Let $\Phi(T_{\text{dP}})$ be the root system of T_{dP} and $I = -I_{\text{En}}$ the induced involution on T_{dP} whose fixed lattice is T_{En} .

The folded roots $\beta_{[v]} \in \Phi\left(T_{\text{dP}}^{(I)}\right)$ arise in exactly one of the following ways:

1. $v \in \Phi^2(T_{\text{dP}})$ and $\beta_{[v]} \in \Phi(T_{\text{En}})$;
2. $v \in \Phi^4(T_{\text{dP}})$ and $\beta_{[v]} \in \Phi(T_{\text{En}})$; or
3. $v \in \Phi^2(T_{\text{dP}}) \cap I(v)^{\perp T_{\text{dP}}}$, and so $\beta_{[v]} := v + I(v) \in \Phi^4(T_{\text{En}})$ is the sum of orthogonal roots in $\Phi^2(T_{\text{dP}})$.

This criterion governs how roots in the boundary lattices at the 0-cusps of $F_{(2,2,0)}$ descend or combine when passing to the I -invariant sublattices corresponding to $F_{\text{En},2}$.

5B.5. Mirror moves.

Definition 5B.6 (Mirror moves)

A **mirror move** is a lattice-theoretic operation governed by the existence of a primitive isotropic vector $\eta \in T$ of a specified type and splitting:

- **Odd/simple:** $\text{div}_T(\eta) = 1$, splitting as $T \cong U \oplus K$. The new invariants are $(r-2, a, 1)$.
- **Even, ordinary:** $\text{div}_T(\eta) = 2$ and η^* is ordinary, splitting as $T \cong U(2) \oplus K$. The new invariants are $(r-2, a-2, 1)$.
- **Even, characteristic:** $\text{div}_T(\eta) = 2$ and η^* is characteristic, splitting as $T \cong I_{1,1}(2) \oplus K$. The new invariants are $(r-2, a-2, 0)$.

The move replaces T with $\bar{T}_\eta = \eta^{\perp T}/\eta$, navigating Nikulin's pyramid of 2-elementary lattices to compute cusp diagrams when $\Gamma = \text{O}^+(T)$.

5C. Toroidal and semitoroidal compactifications.

Remark 5C.1 (Orientation): This section collects the general definitions of the compactification types used throughout the monograph – toroidal, semitoroidal, and KSBA compactifications, together with generalized Coxeter semifans and recognizable divisors. The material here is background and vocabulary: it fixes the constructions in their natural generality. The Coble-specific application of these constructions – the KSBA stable pair, the restricted ramification semifan, and the comparison between the two compactifications – is developed in the Stable Limits section; see in particular ?? 8C.10 and ?? 8C.13.

5C.1. *Toroidal compactifications.***Definition 5C.2**

A **toroidal compactification** $\overline{F}_\Gamma^{\Sigma^\bullet}$ refines the singular cusps of the Baily–Borel compactification by choosing an admissible rational polyhedral fan Σ_I for each cusp I .

Locally, the boundary is modeled on toric varieties X_{Σ_I} . This resolves the singularities of $\overline{F}_\Gamma^{\text{BB}}$, providing a proper algebraic variety with a divisorial boundary (often snc).

The choice of fans makes the construction non-canonical. However, it often allows the extension of period maps and universal families over boundary strata. There is a proper Γ -equivariant morphism $\overline{F}_\Gamma^{\Sigma^\bullet} \rightarrow \overline{F}_\Gamma^{\text{BB}}$.

5C.2. *Semitoroidal compactifications.***Definition 5C.3**

Introduced by Looijenga [Loo03], a **semitoroidal compactification** $\overline{F}_\Gamma^{\mathcal{F}^\bullet}$ replaces the strict fans of toroidal compactifications with Γ -admissible **semifans** $\mathcal{F}_I$.

A semifan relaxes the conditions of a fan by not requiring local finiteness or full support. This allows “partial” toroidalization at selected cusps while leaving others untouched or less refined.

Semitoroidal compactifications sit in a tower of proper birational morphisms:

$$\overline{F}_\Gamma^{\Sigma^\bullet} \longrightarrow \overline{F}_\Gamma^{\mathcal{F}^\bullet} \longrightarrow \overline{F}_\Gamma^{\text{BB}}.$$

They are critical for modeling KSBA boundaries where fans may be infinitely generated or accumulate.

Theorem 5C.4.

Any normal compactification admitting a tower

$$\overline{F}_\Gamma^{\Sigma^\bullet} \longrightarrow \overline{F}_\Gamma^{\mathcal{F}^\bullet} \longrightarrow \overline{F}_\Gamma^{\text{BB}}$$

as in definition 5C.3 is isomorphic to a semitoroidal compactification [AE23, Thm. 1].

Remark 5C.5 (Polarized Coble trace restriction): The polarized Coble compactification program does not build a new semifan from scratch; it uses a Coble-specific restriction of the Enriques ramification semifan. Concretely, one takes the **trace** of the Enriques ramification semifan on a Coble hyperplane, together with an admissibility condition selecting which restricted walls survive. This is the semitoroidal side of the Coble comparison problem, recorded here as ?? 8C.10; the trace identity is not yet proved, and is kept explicitly conjectural.

5C.3. *Generalized Coxeter semifans.***Definition 5C.6**

The main geometric application of folded Coxeter–Vinberg diagrams is the construction of semitoroidal compactifications of moduli spaces F_Γ via **generalized Coxeter semifans**.

For a 0-cusp with Coxeter diagram $G(\Gamma_\eta)$, partition the simple roots into $\Phi = \Phi^{\text{rel}} \sqcup \Phi^{\text{irr}}$ [AT17]:

- **Irrelevant roots** (Φ^{irr}): correspond to strata that are contracted in the KSBA stable model over η .
- **Relevant roots** (Φ^{rel}): the active walls where combinatorial types of stable models change.

The **generalized Coxeter semifan** $\mathcal{F}_{\text{gen}}$ is obtained by omitting the walls defined by irrelevant roots [AT17, Def. 4.16]. Its maximal cones are unions of Weyl chambers $g(\bigcup_{h \in W^{\text{irr}}} h(\mathfrak{C}))$.

Remark 5C.7 (Toroidal versus strictly semitoroidal): Unlike classical toroidal compactifications, where all walls of the Weyl chamber are preserved, semitoroidal compactifications allow certain irrelevant roots to be removed. The distinction between toroidal and strictly semitoroidal behavior depends on the order of the irrelevant root subgroup W^{irr} :

- **Toroidal:** $|W^{\text{irr}}| < \infty$ (finite irrelevant subgroups).
- **Strictly semitoroidal:** $|W^{\text{irr}}| = \infty$ (infinite irrelevant subgroups, so the semifan is not locally finite).

Remark 5C.8(Polarized Coble trace picture): For the polarized Coble program, the generalized Coxeter semifan of the Enriques cusp is restricted to a Coble hyperplane: one asks for the trace of the Enriques ramification semifan on that hyperplane, together with an admissibility condition determining which restricted walls survive. Under the proposed restriction, a Coble wall is irrelevant precisely when every Enriques wall restricting to it is already irrelevant; see ?? 8C.10.

5C.4. KSBA compactifications.

Definition 5C.9

The **KSBA compactification** generalizes the Deligne–Mumford compactification of curves to higher dimensions. It compactifies moduli of varieties of log general type by considering **stable slc pairs** (X, D) [KS88; Ale96].

A pair $(X, D = \sum b_j D_j)$ is **KSBA-stable** if:

1. X is a projective demi-normal variety;
2. $0 < b_j \leq 1$;
3. the pair (X, D) has **semi log canonical (slc)** singularities;
4. $K_X + D$ is ample and **Q**-Cartier.

For K-trivial varieties (like K3 or Enriques surfaces), one uses pairs $(X, \varepsilon R)$ for $0 < \varepsilon \ll 1$ and a polarizing divisor R . The KSBA moduli space $\overline{F}_\Gamma$ provides a modular, proper, algebraic compactification where boundary divisors correspond to geometric stable degenerations [AET23; Ale+25].

Remark 5C.10(Polarized Coble application): The Coble-specific stable-pair package extracted from this framework – the descended ramification divisor on the stable quotient surface, together with its KSBA obligations (**Q**-Cartierness, ampleness, and slc control) – is developed in the Stable Limits section rather than restated here. The resulting comparison target between the KSBA and semitoroidal compactifications remains an open program rather than a settled theorem; see ?? 8C.13.

5C.5. Recognizable divisors.

Definition 5C.11

A polarizing divisor R on the generic surface in F_S is **recognizable** if, for any quasipolarized Kulikov degeneration $\mathcal{X} \rightarrow \Delta$, the divisor R extends unambiguously to a flat limit $R_0 \subset \mathcal{X}_0$, unique up to $\text{Aut}^0(\mathcal{X}_0)$ in any other smoothing.

Theorem 5C.12.

If R is recognizable, then the normalization of the KSBA compactification $\overline{F}^R$ is isomorphic to a semitoroidal compactification $\overline{F}_S^{\mathcal{F}_R}$, defined by a specific semifan $\mathcal{F}_R$ [AE23].

Example 5C.13: Two basic examples of recognizable divisors:

1. The **rational curve divisor** $R_{\text{rc}} = \sum R_i$ (the sum of all smooth rational curves in $|L|$).
2. The **ramification divisor** R_ι of a nonsymplectic involution.

5D. Scattone’s method for cusp enumeration.

Remark 5D.1(Orientation): The Cusp Correspondence section computes the specific cusp diagrams of the Coble and Enriques moduli spaces F_{Co} and F_{En} . Here we record the *general* arithmetic machinery underlying those computations, following Scattone [Sca87]: the boundary components of a Baily–Borel compactification are enumerated by the orbits of primitive isotropic subspaces of a lattice under an arithmetic group, and those orbits are separated by two invariants — **divisibility** and the image in the **discriminant group** — via Eichler’s criterion and its transvection refinement. The concrete cusp counts of F_{Co} , F_{En} , and the degree-two K3 space F_2 are instances of this method; we cross-reference them below rather than recomputing them.

5D.1. *The cusp–isotropic-subspace correspondence.*

Remark 5D.2 (Boundary components of the Baily–Borel compactification): Let F_{2d} be the moduli space of $2d$ -polarized K3 surfaces, realized as a quotient $D_{L_{2d}}/\Gamma_{2d}$ of a type IV period domain by an arithmetic group Γ_{2d} acting on a lattice L_{2d} of signature $(2, 19)$. Scattone [Sca87] gives a concrete arithmetic approach to enumerating the 0-cusps of the Baily–Borel compactification $\overline{F_{2d}}^{\text{BB}}$.

The boundary of $\overline{F_{2d}}^{\text{BB}}$ is stratified by Γ_{2d} -orbits of primitive isotropic subspaces of L_{2d} :

1. the **0-cusps** (Type III boundary points) correspond bijectively to Γ_{2d} -orbits of primitive isotropic lines $I \subseteq L_{2d}$;
2. the **1-cusps** (Type II boundary curves) correspond bijectively to Γ_{2d} -orbits of primitive isotropic planes (rank-2 isotropic sublattices) $J \subseteq L_{2d}$.

Each 0-cusp records a degeneration of the underlying K3 surfaces, whose limiting Hodge structure is encoded by the lattice $I^\perp/I$; likewise each 1-cusp carries the boundary lattice $J^\perp/J$. Thus the enumeration of boundary components reduces to a purely lattice-theoretic orbit problem for primitive isotropic sublattices of L_{2d} .

Remark 5D.3 (Scattone’s 1987 memoir): Scattone’s 1987 memoir [Sca87] carries out this classification lattice-theoretically, with essentially no Hodge-theoretic machinery. Its three technical ingredients are:

- **Discriminant-form analysis:** separating boundary components by their discriminant quadratic forms;
- **Isotropic-sublattice classification:** the bijective correspondence, above, between boundary components and Γ_{2d} -orbits of primitive isotropic sublattices of the period lattice L_{2d} of signature $(2, 19)$, on which Γ_{2d} acts;
- **Arithmetic group theory:** Eichler’s criterion together with the properties of the group $O^+(L_{2d})$.

For $d = 1$ the classification uses primitive embeddings into the 24 Niemeier lattices to distinguish the four types of rank-17 boundary lattices; see remark 5D.11 and example 5D.12.

5D.2. *Divisibility and the discriminant group.* The two invariants that separate isotropic orbits are the divisibility of a primitive vector and its image in the discriminant group. Recall from the Lattice Theory section that the **divisibility** $\text{div}_L(v)$ of $v \in L$ is the positive generator of the ideal $\beta_L(v, L) \subseteq \mathbf{Z}$, and that $v^* := v/\text{div}_L(v)$.

Proposition 5D.4.

Let L be a nondegenerate lattice and $v \in L$ an arbitrary (not necessarily isotropic) vector. Then $v^* := v/\text{div}_L(v) \in L^\vee$ is primitive in the dual lattice, and its image in the discriminant group $A_L := L^\vee/L$ has order $\text{div}_L(v)$.

In particular $\text{div}_L(v)$ divides $|A_L| = |\text{disc}(L)|$.

Remark 5D.5: Consequently the pair

$$(\text{div}_L(v), [v^*] \in A_L)$$

is a well-defined isometry invariant attached to any primitive v , refining its length v^2 . Eichler’s criterion asserts that, in the presence of two hyperbolic summands, these invariants are *complete*.

5D.3. *Eichler’s criterion.*

Theorem 5D.6 (Eichler’s criterion).

Let $L \xrightarrow{\sim} U^2 \oplus M$ be an even lattice containing two orthogonal copies of the hyperbolic plane U , and let $v, w \in L$ be primitive vectors. Then v and w lie in the same $O^*(L)$ -orbit (where $O^*(L)$ is the stable orthogonal group, acting trivially on A_L) if and only if

$$v^2 = w^2 \quad \text{and} \quad [v^*] = [w^*] \in A_L.$$

Equivalently, the $O^*(L)$ -orbit of a primitive vector is determined by its length together with the pair $(\text{div}_L(v), v/\text{div}_L(v) \bmod L)$ in the discriminant group [Eic74; Sca87].

Theorem 5D.7 (Scattone's condition for Eichler transvection orbits).

Let L be an even lattice admitting a splitting $L \xrightarrow{\sim} U^2 \oplus M$ with M an arbitrary lattice, and fix $k \in \mathbf{Z}$. Write $L[k] := \{x \in L \mid x^2 = k\}$. Then for $v, w \in L[k]$,

$$v \sim_{E(L)} w \iff [v^*] = [w^*] \in A_L,$$

where $E(L)$ is the **Eichler transvection group** and $\sim_{E(L)}$ denotes $E(L)$ -equivalence [Sca87].

Remark 5D.8: Theorem 5D.7 is the engine that separates orbits of isotropic vectors: it reduces the orbit question to computing the image $[v^*]$ in the *finite* discriminant group A_L . Since $E(L) \leq O(L)$ is generated by Eichler transvections, the criterion also refines theorem 5D.6 by pinning the equivalence to the transvection subgroup on each fixed length level $L[k]$.

5D.4. *Orbit classification in $T = U \oplus \overline{T}_\eta$.* Sterk's refinement extends the transvection method to isotropic vectors of arbitrary divisibility, and is the form directly applied to the Enriques and Coble transcendental lattices in the Cusp Correspondence section.

Theorem 5D.9 (Orbit classification (Sterk 1991)).

Fix an even lattice T with a splitting $T = U \oplus \overline{T}_\eta$, where $\overline{T}$ is negative-definite and even. Let $\eta_1, \eta_2 \in T$ be primitive isotropic vectors satisfying

1. $\eta_1^2 = \eta_2^2 = k$,
2. $\text{div}_T(\eta_1) = \text{div}_T(\eta_2) = p > 0$,
3. $\eta_1 \equiv \eta_2 \pmod{pT}$.

Then there exists an isometry $\phi \in O^*(T)$ with $\phi(\eta_1) = \eta_2$ [Ste91].

Remark 5D.10: Theorem 5D.9 generalizes the Eichler transvection method of theorem 5D.7 to classify primitive isotropic vectors by the triple (length, divisibility, residue mod pT). In particular it shows that all divisibility-one isotropic vectors of a given length in T_{En} lie in a single $O^*(T_{\text{En}})$ -orbit; this is the statement used to collapse the divisibility-one cusps in the Cusp Correspondence computations.

5D.5. *Enumeration via Niemeier lattices.*

Remark 5D.11 (Niemeier embeddings for $d = 1$): For the degree-two case $d = 1$, Scattone [Sca87] realizes the boundary data through the 24 **Niemeier lattices** — the even, negative-definite, unimodular lattices of rank 24 [CS10]. The reduction expresses the enumeration of the relevant boundary components as a count of orbits of primitive embeddings into each Niemeier lattice, up to the orthogonal group of that lattice, with the orthogonal complement recording the boundary lattice $\overline{T}_J$.

Scattone realizes the Type II boundary data through the 24 Niemeier lattices as follows: each Type II boundary lattice $\overline{T}_J = J^\perp/J$ is an even negative-definite lattice of rank 17 (for $d = 1$), which by Nikulin's embedding theory [Nik80] admits a primitive embedding into an even unimodular lattice of rank 24; the boundary components are enumerated by classifying these embeddings up to the orthogonal group of the Niemeier lattice. Scattone does *not* embed the hyperbolic plane U into a Niemeier lattice — impossible, since U is indefinite — and no even unimodular negative-definite lattice of rank 22 occurs.

5D.6. *Example: the cusp diagram of F_2 .*

Example 5D.12 (Cusp diagram of F_2 (Scattone)): For $d = 1$, there is exactly one Γ_2 -orbit of primitive isotropic lines in T_2 , yielding a unique 0-cusp (Type III) in $\overline{F}_2^{\text{BB}}$ [Sca87].

There are exactly four Γ_2 -orbits of primitive isotropic planes in T_2 , yielding four Type II boundary curves. The corresponding boundary lattices $\overline{T}_J$ are rank-17 negative-definite lattices, distinguished by their root sublattices:

$$A_1 \oplus E_8^{\oplus 2}, \quad E_7 \oplus D_{10}, \quad A_1 \oplus D_{16}, \quad A_{17}.$$

All four Type II curves meet transversely at the unique Type III point.

Remark 5D.13: Example 5D.12 is the cusp diagram of the *degree-two $K3$* moduli space F_2 , and is distinct from the Coble and Enriques cusp diagrams computed in the Cusp Correspondence section: there the transcendental lattice $T_{\text{Co}} = (11, 11, 1)_1$ produces a single 0-cusp incident to a single 1-cusp via the

mirror-move algorithm, with divisibility data $(2, 2)$. We record the F_2 numbers here only as the illustrative instance of Scattone's general method and do not reuse them for the Coble computation.

5D.7. *Type III degenerations: the Friedman–Scattone analysis.*

Remark 5D.14 (Friedman–Scattone 1986): The geometric counterpart to the arithmetic Type III cusps is the Friedman–Scattone [FS86] analysis of Type III degenerations of K3 surfaces, carried out through **mixed Hodge structure** theory. Its ingredients are:

- **Limiting mixed Hodge structures** in the sense of Steenbrink–Schmid;
- **Monodromy weight filtrations** attached to a nilpotent monodromy operator N with $N^3 = 0$ and $N^2 \neq 0$ (the Type III condition);
- the **Clemens–Schmid exact sequence** and the **weight spectral sequence**, which for K3 surfaces degenerates integrally at E_2 .

The resulting classification identifies Type III degenerations with triangulations of S^2 subject to combinatorial constraints; the dual complex of the degeneration then determines the lattice-theoretic invariants, giving a precise dictionary between the geometric degenerations and their monodromy representations. This Hodge-theoretic picture matches, on the boundary of $\overline{F}_{2d}^{\text{BB}}$, the 0-cusps enumerated arithmetically by the isotropic-line correspondence above.

6. DEGENERATIONS

6A. Kulikov models and limiting mixed Hodge structures.

Remark 6A.1: We collect the local structure theory attached to a one-parameter degeneration of K3 surfaces: the Kulikov models that provide a well-behaved semistable representative of a degeneration, the trichotomy of such models by the nilpotency of the log monodromy operator, and the limiting mixed Hodge structure whose monodromy weight filtration records the same trichotomy on the Baily–Borel boundary.

6A.1. *Kulikov degenerations.*

Definition 6A.2 (Kulikov models)

A *Kulikov model* is a degeneration $\mathcal{X} \rightarrow \Delta$ of K3 surfaces where:

1. The total space $\mathcal{X}$ is regular (smooth as a threefold).
2. The central fiber $\mathcal{X}_0$ is a reduced simple normal crossings (snc) divisor.
3. The relative dualizing sheaf is trivial: $\omega_{\mathcal{X}/\Delta} \cong \mathcal{O}_{\mathcal{X}}$.

Such a model is *semistable* by conditions (1) and (2) and *K-trivial* by condition (3).

Remark 6A.3 (Existence and attribution): The Kulikov model is the semistable, K -trivial representative constructed for degenerations of K3 (and Enriques) surfaces in [Kul77], with the K -trivialization of the semistable reduction supplied for surfaces of trivial canonical bundle by [PP81].

Definition 6A.4 (Type I, II, III Kulikov degenerations)

Let T be the unipotent Picard–Lefschetz monodromy of a Kulikov model and let $N = \log T$ be the associated *log monodromy* operator, a nilpotent endomorphism. Depending on the nilpotency index of N , Kulikov models are classified into three types:

- **Type I** ($N = 0$): $\mathcal{X}_0$ is a smooth K3 surface.
- **Type II** ($N^2 = 0, N \neq 0$): $\mathcal{X}_0$ is a chain of surfaces glued along elliptic curves. The dual complex is an interval $\mathbf{IAD}^1$.
- **Type III** ($N^3 = 0, N^2 \neq 0$): $\mathcal{X}_0$ is a union of rational surfaces glued along rational curves, forming an anticanonical cycle on each component. The dual complex is a sphere $\mathbf{IAS}^2$.

Remark 6A.5 (The Type III dual complex): The identification of the Type III central fiber as a union of rational surfaces whose dual complex triangulates a sphere is the subject of [FS86].

6A.2. *The monodromy weight filtration.*

Remark 6A.6 (The weight filtration by type): The **weight filtration** $W_\bullet$ on $H^2(\mathcal{X}_t; \mathbf{C})$ (weight 2, centered at $n = 2$) reflects the three types, by the nilpotency of N . For Type I ($N = 0$), only Gr_2^W is non-trivial: a pure weight-2 Hodge structure. For Type II ($N \neq 0, N^2 = 0$), the non-trivial graded pieces

are $\mathrm{Gr}_1^W, \mathrm{Gr}_2^W, \mathrm{Gr}_3^W$ (weights 1, 2, 3), with $\mathrm{Gr}_0^W = \mathrm{Gr}_4^W = 0$. For Type III ($N^2 \neq 0, N^3 = 0$), the non-trivial graded pieces are $\mathrm{Gr}_0^W, \mathrm{Gr}_2^W, \mathrm{Gr}_4^W$ (weights 0, 2, 4), of Hodge–Tate type.

Theorem 6A.7 (Limiting mixed Hodge structure and degenerations).

For a one-parameter degeneration of polarized K3 surfaces over a punctured disk, the unipotent monodromy operator $T \in \mathrm{O}(T_{2d})$ and its nilpotent logarithm $N = \log T$ induce a canonical monodromy weight filtration $W^\bullet$.

The boundary components of $\overline{F_{2d}}^{\mathrm{BB}}$ are classified by the nilpotency index of N :

- **Type II degenerations (1-cusps):** $N \neq 0, N^2 = 0$. The weight filtration is 3-step ($W_0 \subset W_1 \subset W_2 \subset W_3$), and $\mathrm{Gr}_2^W = I^\perp/I = \overline{T}_I$ carries a pure polarized Hodge structure of weight 2.
- **Type III degenerations (0-cusps):** $N^2 \neq 0, N^3 = 0$. The weight filtration is of maximal length. These correspond to normal crossing varieties whose dual complex is a triangulation of S^2 .

Remark 6A.8 (Attribution of the weight filtration): The canonical monodromy weight filtration attached to the nilpotent operator $N = \log T$ is that of [Sch73].

Remark 6A.9 (Indexing of the weight filtration): The apparent discrepancy between the two accounts above is not a matter of indexing convention but a distinction between Kulikov types. For Type II the non-trivial graded pieces sit at weights 1, 2, 3, with $\mathrm{Gr}_0^W = 0$, matching the 3-step chain $W_0 \subset W_1 \subset W_2 \subset W_3$ of theorem 6A.7. The even-weight pieces at 0, 2, 4 occur instead for Type III, which is of Hodge–Tate type.

Remark 6A.10 (Source notes carried no citations):

6B. Anticanonical pairs, charge, and MMP singularities.

Remark 6B.1 (Orientation): The components of a Type III degeneration are anticanonical pairs, and the combinatorics of such degenerations is controlled by a single additive invariant, the *charge*. We collect the definitions here, together with the singularity classes of the minimal model program that govern which limits are admissible in KSBA moduli.

6B.1. Anticanonical pairs and their charge.

Definition 6B.2

An **anticanonical pair** (V, D) consists of a smooth projective rational surface V together with a reduced effective snc divisor D such that

$$K_V + D \sim_{\mathbb{Q}} 0.$$

Anticanonical pairs are also known as **log Calabi–Yau surfaces**.

Definition 6B.3

The **charge** of an anticanonical pair (V, D) , with $D = \sum_j D_j$ its decomposition into irreducible components, measures the deviation of (V, D) from being toric. It is defined by

$$Q(V, D) := 12 - \sum_j (D_j^2 + 3).$$

For a toric surface V with $D = \partial V$ its toric boundary, the charge vanishes:

$$Q(V, \partial V) = 0.$$

Proposition 6B.4 (Charge under blowup).

The charge behaves as follows under blowups of an anticanonical pair:

- **Corner blowups**, at nodes of D (points where two components of D meet), preserve the charge.
- **Interior blowups**, at smooth points of D (points lying on a single component of D), increase the charge by 1.

Fig. 4 is migrated a raster talk figure should be redrawn native tikzed diagram and its casing recode with the surrounding notation ($F_{\mathrm{Co},2}$ v figure's $F_{\mathrm{co},2}$). The map $F_{\mathrm{Co},2} \rightarrow F_{\mathrm{C}}$ codes the open qu flagged above.

Theorem 6B.5 (Friedman–Morrison charge theorem).

Let $\mathcal{X} \rightarrow (C, 0)$ be a Type III Kulikov degeneration of K3 surfaces with central fiber

$$\mathcal{X}_0 = \bigcup_{i=1}^n V_i,$$

and for each component set

$$D_i := V_i \cap \overline{(\mathcal{X}_0 \setminus V_i)},$$

so that each (V_i, D_i) is an anticanonical pair. Then the sum of the charges of all components is exactly 24:

$$\sum_{i=1}^n Q(V_i, D_i) = 24.$$

This imposes severe constraints on the possible combinatorial types of degenerations.

6B.2. Singularities in the minimal model program.**Definition 6B.6**

Let (X, D) be a normal pair with $K_X + D$ $\mathbf{Q}$ -Cartier, and let $f: Y \rightarrow X$ be a log resolution. The **discrepancy** $a(E, X, D)$ of a divisor E over X is defined by

$$K_Y + D_Y = f^*(K_X + D) + \sum_E a(E, X, D) E.$$

Definition 6B.7

With discrepancies as in definition 6B.6, the pair (X, D) has the following classes of singularities, according to the values taken by $a(E, X, D)$ over all divisors E over X :

- **Terminal:** $a(E, X, D) > 0$.
- **Canonical:** $a(E, X, D) \geq 0$.
- **Kawamata log terminal (klt):** $a(E, X, D) > -1$.
- **Log canonical (lc):** $a(E, X, D) \geq -1$. This is the maximal class in which the minimal model program works.

The definitions of these singularity classes follow [KM98].

Definition 6B.8

For non-normal varieties one has the following notions.

- A variety X is **demi-normal** if it is reduced, satisfies Serre's condition S_2 , and has at worst normal crossings in codimension 1.
- A pair (X, D) is **semi-log canonical (slc)** if X is demi-normal and the normalization, taken with the conductor divisor, (X^ν, D^ν) is log canonical.

Being slc is the condition required for limits in KSBA moduli spaces.

6C. Divisor models and dlt degenerations.

Remark 6C.1: This section collects three models used to describe degenerations of K3 and Enriques surfaces and their KSBA stable limits: the *dlt* (divisorially log terminal) models of stable involution pairs, the *divisor models* of a degeneration of pairs, and the *half-divisor models* obtained by an involution quotient. The three are related: a half-divisor model is the quotient of a divisor model by an involution, and the dlt models supply the birational models on whose strata the associated combinatorial data live.

6C.1. *dlt models and involution pairs.* Here a *dlt model* refers to a relative divisorially-log-terminal model of the degeneration supplying these boundary strata; the definition below is of the associated *stable involution pair* (X, D, ι) .

Definition 6C.2 (dlt models and involution pairs)

For Enriques surface degenerations we study *stable involution pairs* (X, D, ι) , where X is the K3 cover

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limit, D is the boundary, and ι is the involution. The canonical condition

$$K_X + D \sim 0$$

holds.

The quotient $Y = X/\iota$ has branch divisor B and image boundary $C = \pi(D)$, where $\pi: X \rightarrow Y$ is the quotient map. The pair (Y, C) is a log del Pezzo surface of index at most 2, satisfying

$$K_X + D + \varepsilon R = \pi^* \left(K_Y + C + \frac{1+\varepsilon}{2} B \right),$$

where R is the ramification divisor of π .

Remark 6C.3 (Dual complex topology): The topology of the dual complex of the central fiber Y'_0 distinguishes the degeneration types:

- Generic Type III (e.g. Coxeter type $\tilde{E}_8 + \tilde{E}_8$): the dual complex of Y'_0 is a projective plane $\mathbf{RP}^2$, with the involution acting antipodally on S^2 .
- Other Type III / Type II (e.g. $\tilde{E}_8 + \tilde{D}_{10}$): the dual complex is a 2-disk $\mathbf{D}^2$, with the involution acting as a reflection on S^2 .
- Special Type II cases: the dual complex reduces to a segment.

6C.2. Divisor models.

Definition 6C.4 (Divisor model)

A *divisor model* for a degeneration $\pi: \mathcal{X} \rightarrow C$ of K3 (or Enriques) surfaces is a degeneration of pairs $(\mathcal{X}, \mathcal{R}) \rightarrow C$ where:

1. $\mathcal{R}$ is a Cartier divisor, effective on all fibers.
2. For $t \neq 0$, $\mathcal{R}_t$ is semiample.
3. $\mathcal{R}$ is disjoint from the singular strata of the central fiber $\mathcal{X}_0$.

The geometry is encoded by an *integral-affine divisor* $R_{\text{IA}} \subset \Gamma(\mathcal{X}_0)$, which assigns weights to the double curves reflecting the intersection with $\mathcal{R}_0$, satisfying a balancing condition at every vertex.

6C.3. Half-divisor models.

Definition 6C.5 (Half-divisor model)

A *half-divisor model* is a pair $(\mathcal{Z}, \mathcal{R}_{\mathcal{Z}})$ over a base curve C that arises as the quotient of a divisor model $(\mathcal{X}, \mathcal{R}) \rightarrow C$ by a fixed-point-free involution τ under which $\mathcal{R}$ is τ -invariant, $\tau^*\mathcal{R} = \mathcal{R}$. The class $\mathcal{R}$ descends to $\mathcal{Z}$ only up to the 2-torsion twist $K_{\mathcal{Z}}$ (the two $\mathbf{Z}/2$ -linearizations of the invariant $\mathcal{R}$ differ by the sign character of τ); this is why $\mathcal{R}_{\mathcal{Z}}$ is Weil but not Cartier, while $2\mathcal{R}_{\mathcal{Z}}$ is Cartier.

This naturally models limits of polarized Enriques surfaces, where the polarization does not descend to a Cartier divisor on the Enriques quotient.

Boundary strata for Enriques stable limits are modeled by pairs $(\mathcal{Z}_0, \frac{1}{2}\mathcal{R}_{\mathcal{Z}_0})$ with $K_{\mathcal{Z}_0} + \frac{1}{2}\mathcal{R}_{\mathcal{Z}_0}$ ample and slc singularities.

Remark 6C.6 (Half-divisor models from the Enriques involution): Equivalently, a half-divisor model arises when a divisor model $(\mathcal{X}, \mathcal{R})$ admits an Enriques involution ι_{En} preserving $\mathcal{R}$; the quotient is

$$(\mathcal{Z}, \mathcal{R}_{\mathcal{Z}}) = (\mathcal{X}, \mathcal{R})/\iota_{\text{En}}.$$

Its KSBA stable limit can be computed directly from the relative Proj of its section ring.

These two subsections are probably too much for the introduction, and may not end up being relevant anyway?

6D. Halphen Surfaces.

Remark 6D.1: It is well-known (e.g. as in [DZ99] and [CDL25, Ch.5 §6]) that a Halphen surface of index n is a relative minimal elliptic surface S with F_1 the divisor class of a fibre satisfying $F_1 = -nK_S$. By [DZ99, §2.4], an index 1 Halphen surface admits a section and has no multiple fibres, and an index 2 Halphen surface has a unique double fibre $2F_1$. By [DK25, Prop. 9.1.4], any terminal Coble surface of K3 type is isomorphic to one of the following:

- A blowup of a Halphen surface of index 1 at two singular points of two simple reduced fibres of types $\tilde{A}_k, \tilde{A}_m$ where $k + m \leq 8$, or
- A blowup of a Halphen surface of index 2 at one singular point of a single simple reduced fibre of type $\tilde{A}_n$ where $n \geq 8$

This is used in [DK25, Cor. 9.1.5] to show that $K_S^2 \geq -10$, and since $K_S^2 = -n$, this shows that such Coble surfaces have at most ten boundary components.

Of most immediate relevance to us is the case considered in [Cob19] where $n = 1$. Following [DK25, Ex. 9.1.7], the image $W := V(f_6) := \eta(C)$ is an irreducible plane sextic with 10 A_1 singularities⁸. One can find a cubic $V(f_3)$ passing through nine of them and form a pencil $V(\lambda f_6 + \mu f_3^2)$ such that the minimal resolution of its basepoints is a Halphen surface of index 2. The proof of [DK25, Prop. 9.1.8] indicates that the moduli space of general Halphen surfaces of index 2 has dimension 9.

The following is [CD12, Prop. 3.1]:

Lemma 6D.2 (Cobles to Halphens).

Let X be a Coble surface and $\pi_E : X \rightarrow Y$ be the blowing down of a (-1) -curve E . Then

- Y is a Halphen surface of index 2;
- C is the proper transform of the fiber F containing $y_0 = \pi_E(E)$;
- the fiber F is irreducible, and y_0 is its unique singular point.

Conversely, the blow-up of a singular point of an irreducible non-multiple fiber of a Halphen surface of index 2 is a Coble surface.

Remark 6D.3: Why introduce Halphen surfaces? I conjecture that the geometric construction(s) above yield some kind of correspondence between moduli spaces of Coble surfaces with n boundary components and the various moduli spaces of index 2 Halphen pencils on the $g = 0$ line of Nikulin's triangle diagram of 2-elementary lattices. Regarding the coarse spaces as period domains attached to lattices, the lattices match up precisely; see table 2. Applications of mirror moves indicate that the cusp diagrams would correspondingly coincide as well.

6E. Morrison's degenerations.

Remark 6E.1 (On Morrison's degenerations): As explained in [Mor81] and the appendix of [Sha81], a **flower** is a pair $(\mathbf{P}^2, C)$ where $C^2 = 4$ is a conic, while a **pot** is a pair (V, D) where V is a rational surface and $D = \sum C_i$ is a sum of rational curves satisfying $C_i \cdot C_j = -4\delta_{ij}$. A **stalk** is a pair $(\Sigma_4, s_0 + s_\infty)$ where Σ_4 is a rational ruled surface with sections s_0, s_∞ satisfying $s_0^2 = 4$ and $s_\infty^2 = -4$. A **stalk assembly** is a flower glued to some number of stalks along boundary double curves, and a **flower pot** is a pot glued to stalk assemblies along double curves.

By [Mor81, Lem. 7.1, 7.2], some degenerations of Coble surfaces correspond to flower pot degenerations $\mathcal{X}$, since the period of central fiber $\mathcal{X}_0$ of such a degeneration is contained in the Coble divisor $\mathcal{H}_{-2} \subseteq D_{\text{En}}/\Gamma_{\text{En}}$ and $2K_{\mathcal{X}} \neq 0$. By [Ale+25], flowers are the semistable resolutions of the S_2 -quotients of the ADE singularities in the total space of the smoothing. The pots are components of the dlt models of [Ale+25] along which flowers and stalks are attached. Flower pots correspond to type (i.b) in the classification [Mor81, Cor. 6.2], which by [Ale+25, Rmk. 4.12] are Type I semistable degenerations of Enriques surfaces with a klt singularity.

7. CUSP CORRESPONDENCE

7A. Cusp Diagrams.

Remark 7A.1: We recall the mirror move algorithm from [AE22]. We have Nikulin's 2-elementary diagram:

⁸ A_1 or A_2 singularities.

egrated note lists among its aliases body gives no def- of dlt (divisorial minimal); definition here rather than l.

source notes de- he relationship models to the program (below); not discuss the ship to Kulikov explicitly. If a comparison is l here, supply it.

Note that v_0 corresponds to a Type III boundary, while C_0 corresponds to a type II boundary. It is easily verified that the Coxeter diagram $G_{(9,9,1)_1}$ at v_0 has precisely one maximal parabolic subdiagram, corresponding to a finite-index root lattice of type B_7 . We note that by [AE22, §5], such isotropic vectors are unique up to $O(T_{\text{Co}})$, and so we can choose representatives:

- $v_0 = e'$,
- $v_1 = 2h + \alpha_1 + \alpha_2$.

Calculations verify that $v_0^2 = v_1^2 = 0$, that $v_1 \in v_0^\perp T_{\text{Co}} / \langle v_0 \rangle$, and that $v_0 v_1 = 0$, and thus $J := \langle v_0, v_1 \rangle$ is an admissible choice of an isotropic plane. We further note that $\text{div}_{T_{\text{Co}}}(v_0) = \text{div}_{T_{\text{Co}}}(v_1) = 2$, which will be an important invariant for establishing a correspondence to cusps of other moduli spaces. For an isotropic plane J , we denote the divisibilities of the constituent generating vectors as a tuple (d_1, d_2) , and in this convention we have $\text{div}_{T_{\text{Co}}}(v_0, v_1) = (2, 2)$.

By [CDL25, Prop. 5.4.6], there is an open embedding $F_{(11,11,1)} \hookrightarrow F_{(10,10,0)}$, i.e. $F_{\text{Co}} \hookrightarrow F_{\text{En}}$, realizing F_{Co} as the coarse space of marked Coble surfaces with $n = 1$, where n is the number of boundary components in $C = C_1 + \dots + C_n$. The image is an open subset of a closed irreducible subset of $\mathcal{H}_{-2}/\Gamma_{\text{En}}$. By [CDL25, Thm. 5.8.2], the coarse space of F_{Co} is a rational variety, and since F_{En} is quasiprojective, so too is F_{Co} . Moreover, $\partial \overline{F}_{\text{En}}^{\text{BB}}$ consists of F_{Co} and two modular curves X and $X_0(2)$ by [CDL25, Thm. 5.9.8], and the closure of $\mathcal{H}_{-2}$ contains the modular curve X .

Remark 7A.2: We note the divisibilities of the v_i under various lattice embeddings:

TABLE 3. Divisibilities of the isotropic vectors v_0, v_1 under the embeddings of T_{Co} .

Coble Vector	Representative	$\text{div}_{T_{\text{Co}}}$	$\text{div}_{T_{\text{En}}}$	$\text{div}_{T_{\text{dP}}}$
v_0	e'	2	2	2
v_1	$2h + \alpha_1 + \alpha_2$	2	2	1

More concisely:

TABLE 4. Images of the isotropic vectors of T_{Co} and their divisibilities.

Lattice	Image of v_0	Image of v_1	Divisibility
T_{Co}	e'	$2h + \alpha_1 + \alpha_2$	$(2, 2)$
T_{En}	e'	$2e + 2f + \alpha_1 + \alpha_2$	$(2, 2)$
T_{dP}	e'	$2e + 2f + \alpha_1 + \tilde{\alpha}_1 + \alpha_2 + \tilde{\alpha}_2$	$(2, 1)$

The divisibilities in T_{dP} can be seen as follows: the image of v_0 in T_{dP} is $e' \in U(2)$ and $e' f' = 2$, while $\text{div}_{T_{\text{dP}}}(v_1) = 1$ follows from the fact that $v_1 \alpha_3 = 1$.

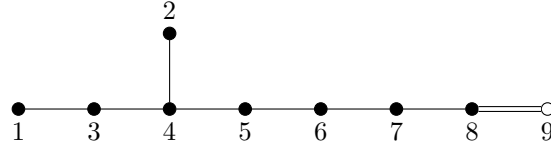
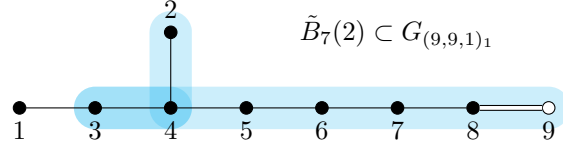
Remark 7A.3: As further proof that the cusp diagram of F_{Co} is correct, we can use the theory of Coxeter diagrams. Given an isotropic vector $e \in L$ a lattice of signature $(2, n)$, the lattice $e^\perp / \langle e \rangle$ is a hyperbolic lattice equipped with a root system R_e with a Coxeter diagram G_e . Generally, when e corresponds to a 0-cusp in a Baily-Borel compactification, the adjacent 1-cusps correspond precisely to maximal parabolic subdiagrams of G_e . The cusp diagram of F_{Co} suggests that the 0-cusp v_0 should have a Coxeter diagram G_{v_0} with precisely one maximal parabolic subdiagram. One can run Vinberg's algorithm to determine the Coxeter diagram for v_0 , and it is a straightforward check to determine that there is indeed a unique maximal parabolic subdiagram of the form $\tilde{B}_7(2)$; see fig. 11.

7B. Coble cusps. Applying the mirror moves from section 7A, we obtain the cusp diagram for F_{Co} , the moduli space of unpolarized Coble surfaces, shown in fig. 9.

$$\begin{array}{ccc}
\mathbb{D}^2, \text{div} = 2, v_1 = e' & \text{div} = (2, 2), v_2 = 2h + \alpha_1 + \alpha_2 & \\
\langle 2 \rangle \oplus E_8(2) & \longrightarrow & A_1^{\oplus 7} \\
(9, 9, 1)_1 & & (7, 7, 1)_0
\end{array}$$

FIGURE 9. Cusp diagram for $F_{\text{Co}} = F_{(11,11,1)}$ where $T_{\text{Co}} = \langle 2 \rangle \oplus E_{10}(2)$.

The corresponding Coxeter diagrams are computed in [AN06] and [Ale+25], and shown in fig. 10. Only the maximal parabolic subdiagrams of $(9, 9, 1)$ are relevant when determining 1-cusps, and these are shown in fig. 11.

FIGURE 10. The Coxeter diagram $G_{(9,9,1)_1} = G_{\langle 2 \rangle \oplus E_8(2)}$.FIGURE 11. The unique maximal parabolic subdiagram $\tilde{B}_7(2)$ of $(9, 9, 1)_1$, corresponding to single one-cusp $(7, 7, 1)_0$ in F_{Co} .

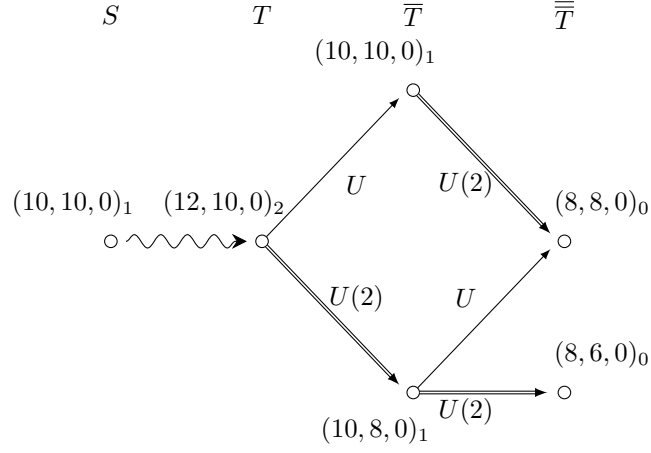
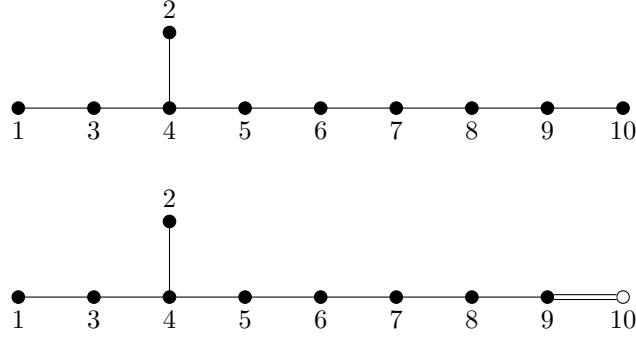
We note that $G_{(9,9,1)_1}$ has precisely one maximal parabolic subdiagram of the form $\tilde{B}_7(2)$, verifying that there is only one adjacent 1-cusp to the unique 0-cusp in F_{Co} . We also note the correspondence between the maximal parabolic $\tilde{B}_8(2)$ in F_{En} , and $\tilde{B}_7(2)$ in F_{Co} , and furthermore that $G_{(10,8,0)_1}$ can be obtained from $G_{(9,9,1)_1}$ by inserting one new node. This correspondingly transforms the $\tilde{B}_7(2)$ subdiagram of $G_{(9,9,1)_1}$ into the $\tilde{B}_8(2)$ subdiagram of $G_{(10,8,0)_1}$.

7C. Enriques cusps. We recall from [Ale+25] the cusp diagram for F_{En} , the moduli space of unpolarized Enriques surfaces, in fig. 12 and the corresponding Coxeter diagrams in fig. 14.

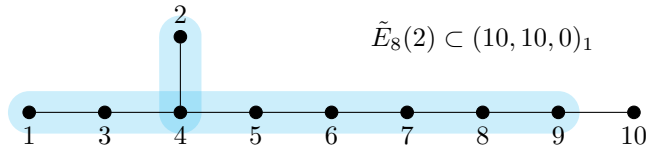
$$\begin{array}{ccccccc}
\mathbf{RP}^2, \text{div} = 1, e & \text{div} = (1, 2), e, e' & \mathbb{D}^2, \text{div} = 2 & \text{div} = (2, 2), e', 2e + 2f + 2\bar{\alpha}_1 & & & \\
E_{10}(2) & \longrightarrow & E_8(2) & \longleftarrow & U \oplus E_8(2) & \longrightarrow & A_1^{\oplus 8*} \\
(10, 10, 0)_1 & & (8, 8, 0)_0 & & (10, 8, 0)_1 & & (8, 6, 0)_0
\end{array}$$

FIGURE 12. Cusp diagram for $F_{\text{En}} = F_{(10,10,0)}$ corresponding to $T_{\text{En}} = U \oplus E_{10}(2)$.

This can be recovered using the mirror move algorithm:

FIGURE 13. The mirror moves for $S_{\text{En}} = (10, 10, 0)_1$.FIGURE 14. Coxeter diagrams $G_{(10,10,0)_1} = G_{E_{10}(2)}$ and $G_{(10,8,0)_1} = G_{U \oplus E_8(2)}$.

The maximal parabolic subdiagrams for $(10, 10, 0)$ and $(10, 8, 0)$ are shown in fig. 15 and fig. 16 respectively.

FIGURE 15. The unique maximal parabolic $\tilde{E}_8(2)$ in $(10, 10, 0)_1$, corresponding to the one-cusp $(8, 8, 0)_0$ in F_{En} .

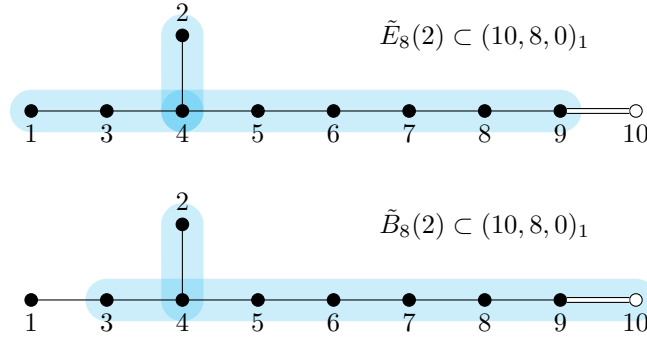


FIGURE 16. The two maximal parabolics $\tilde{E}_8, \tilde{B}_8$ in $(10, 8, 0)_1$ corresponding to the 1-cusps $(8, 8, 0)_0$ and $(8, 6, 0)_0$ respectively.

7D. Sterk and K3 cusps.

Remark 7D.1: We recall Sterk's cusp diagram for $F_{\text{En},2}$:

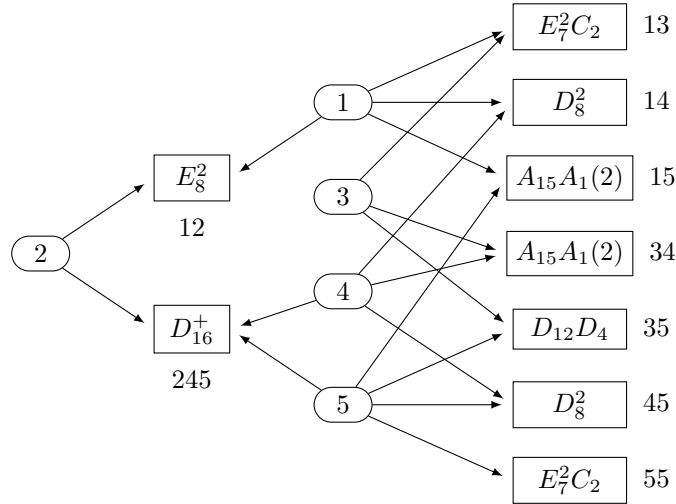


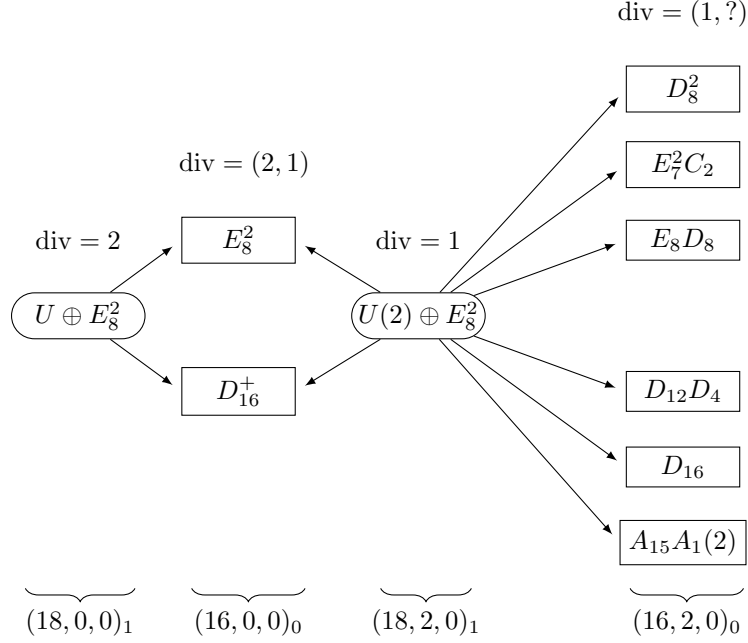
FIGURE 17. Sterk's cusp diagram for $F_{\text{En},2}$.

We have the following divisibilities in various lattices:

TABLE 5. Isotropic vectors in $F_{\text{En},2}$ and their divisibilities.

Sterk Cusp	Vector	$\text{div}_{T_{\text{En}}}$	$\text{div}_{T_{\text{K3}}}$
1	e	1	1
2	e'	2	2
3	$e' + f' + \bar{\alpha}_8$	2	1
4	$2e' + f' + \bar{\alpha}_1$	2	1
5	$2e + 2f + \bar{\alpha}_1$	2	1

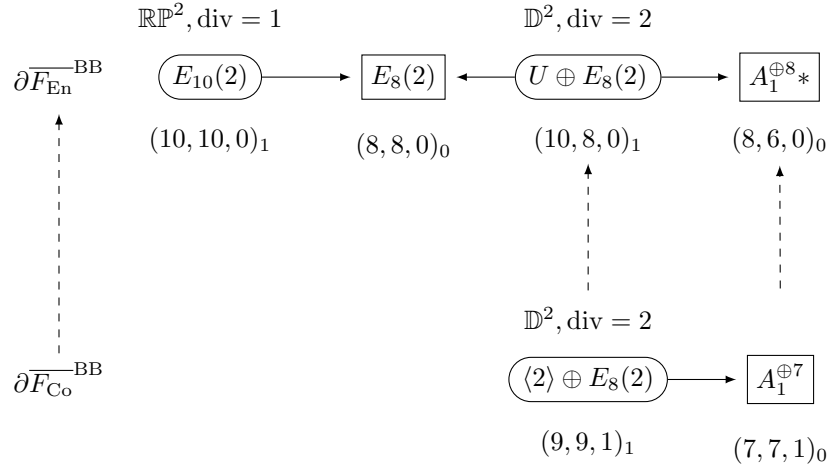
Remark 7D.2: We recall the cusp diagram for $F_{(2,2,0)}$:

FIGURE 18. Cusp diagram for $F_{(2,2,0)}$.

7E. Enriques to Coble correspondence.

Theorem 7E.1.

The embedding $\eta : F_{\text{Co}} \rightarrow F_{\text{En}}$ induces the correspondence on boundary cusps of the Baily-Borel compactifications shown in fig. 19.

FIGURE 19. Cusp correspondence $F_{\text{Co}} \rightarrow F_{\text{En}}$.

We prove the cusp correspondence by comparing divisibilities of isotropic vectors at the corresponding 0-cusps and 1-cusps in both moduli spaces.

Lemma 7E.2.

Any $v \in T_{\text{Co}}$ satisfies $\text{div}_{T_{\text{Co}}}(v) = 2$ if there exists any $v' \in T_{\text{Co}}$ for which $v \cdot v' \neq 0$.

Proof.

Let $M_{T_{\text{Co}}}$ be the Gram matrix for T_{Co} , and note that since

$$T_{\text{Co}} = \langle 2 \rangle \oplus E_{10}(2) = (\langle 1 \rangle \oplus E_{10})(2),$$

every entry in $M_{T_{\text{Co}}}$ is 2-divisible. Thus one can write $M_{T_{\text{Co}}} = 2A$ for some integral matrix A . Letting $v' \in T_{\text{Co}}$ be any other vector not orthogonal to v , we have

$$v \cdot v' = \langle v, M_{T_{\text{Co}}} v' \rangle = \langle v, 2Av' \rangle = 2\langle v, Av' \rangle \neq 0,$$

where $\langle \cdot, \cdot \rangle$ is the standard Euclidean inner product. Thus 2 divides $v \cdot v'$, and since $\text{div}_{T_{\text{Co}}}(v) \in \{1, 2\}$ for any $v \in T_{\text{Co}}$, the result follows. $\blacksquare$

Lemma 7E.3.

Fixing notation,

$$\begin{aligned} v_1 &:= e' & w_1 &:= \eta(v_1) = \tilde{e}' \\ v_2 &:= 2h + \alpha_1 + \alpha_2 & w_2 &:= \eta(v_2) = 2\tilde{e} + 2\tilde{f} + \tilde{\alpha}_1 + \tilde{\alpha}_2 \\ J &:= \langle v_1, v_2 \rangle & \tilde{J} &:= \langle w_1, w_2 \rangle \end{aligned}$$

We then have $\text{div}_{T_{\text{En}}}(w_1) = 2$.

Proof.

By lemma 7E.2, we have in particular that v_1 has divisibility 2 in T_{Co} ; moreover it is isotropic. Since there is a unique Γ_{Co} -orbit of isotropic vectors in T_{Co} , one can associate v_1 to the unique 0-cusp of F_{Co} , so that

$$(v_1)^{\perp T_{\text{Co}}} / v_1 \cong (9, 9, 1) \cong \langle 2 \rangle \oplus E_8(2).$$

Since $\tilde{e}' \tilde{f}' = 2$, and $\tilde{e}'$ is orthogonal to the remaining generators of T_{En} , we have

$$\text{div}_{T_{\text{En}}}(w_1) := \text{div}_{T_{\text{En}}}(\eta(e')) = \text{div}_{T_{\text{En}}}(\tilde{e}') = 2.$$

Explicitly, writing $T_{\text{En}} = U \oplus U(2) \oplus E_8(2)$, one can check the pairing of $\tilde{e}'$ against an arbitrary vector. Write $x + y + z \in U \oplus U(2) \oplus E_8(2)$; then

$$\tilde{e}' \cdot (x + y + z) = \tilde{e}' \cdot y \in \{0, 2\}$$

since $y \in U(2)$ and $\tilde{e}' \in (U \oplus E_8(2))^{\perp T_{\text{En}}}$. Thus $\beta_{T_{\text{En}}}(\tilde{e}', T_{\text{En}}) = \beta_{T_{\text{En}}}(\tilde{e}', U(2)) = 2\mathbb{Z}$ since $\tilde{e}' \cdot \tilde{f}' = 2$. $\blacksquare$

Lemma 7E.4.

The 0-cusp $(9, 9, 1)_1$ in F_{Co} maps to the 0-cusp $(10, 8, 0)_1$ in F_{En} .

Proof.

The cusp correspondence follows from computing the lattice $w_1^{\perp T_{\text{En}}} / w_1$ under the primitive embedding η , since $\eta(v_1) = w_1$ and v_1 is the isotropic vector corresponding to $(9, 9, 1)_1$ in F_{Co} :

$$\frac{(\tilde{e}')^{\perp T_{\text{En}}}}{\langle \tilde{e}' \rangle} = \frac{\langle \tilde{e}, \tilde{f} \rangle \oplus \langle \tilde{e}' \rangle \oplus \langle \tilde{\alpha}_1, \dots, \tilde{\alpha}_8 \rangle}{\langle \tilde{e}' \rangle} \cong \langle \tilde{e}, \tilde{f} \rangle \oplus \langle \tilde{\alpha}_1, \dots, \tilde{\alpha}_8 \rangle \cong U \oplus E_8(2) \cong (10, 8, 0)_1.$$

Alternatively, by [AE22, Prop. 5.5], the isomorphism type of $w_1^{\perp T_{\text{En}}} / w_1$ is determined by $\text{div}_{T_{\text{En}}}(w_1)$; by lemma 7E.3 $\text{div}_{T_{\text{En}}}(w_1) = 2$. Since the divisibility of the isotropic vector at the Enriques 0-cusp $(10, 8, 0)_1$ is also 2 and the two Enriques 0-cusps are distinguished by divisibility, the correspondence follows. $\blacksquare$

Lemma 7E.5.

The 1-cusp $(7, 7, 1)_0$ in F_{C_0} maps to the 1-cusp $(8, 6, 0)_0$ in F_{E_n} .

Proof.

The results follows from verifying that the unique orbit J of a primitive isotropic plane in T_{C_0} satisfies $J^\perp_{T_{C_0}}/J \cong (7, 7, 1)$, and identifying the isomorphism type of its image $\tilde{J}^\perp_{T_{E_n}}/\tilde{J}$ in T_{E_n} . One checks that both v_2 and w_2 are isotropic, and $v_2 \in v_1^\perp_{T_{C_0}}/v_1$, and so J and $\tilde{J}$ define isotropic planes in T_{C_0} and T_{E_n} respectively. By [AE22, Prop. 5.5, Lem. 5.9], it suffices to show

$$\operatorname{div}_{w_1^\perp_{T_{E_n}}/w_1}(w_2) := \operatorname{div}_{U \oplus E_8(2)}(2\tilde{e} + 2\tilde{f} + \tilde{\alpha}_1 + \tilde{\alpha}_2) = 2,$$

since the isomorphism type of $\tilde{J}^\perp/\tilde{J}$ is uniquely determined by the isomorphism type of $w_2^\perp_{T_{E_n}}/w_2$ in $w_1^\perp_{T_{E_n}}/w_1$, which is in turn uniquely determined by the characterization of w_2 as odd, even ordinary, or even characteristic in $w_1^\perp_{T_{E_n}}/w_1$, which by lemma 7E.4 is isomorphic to $U \oplus E_8(2)$. One checks directly in coordinates: let $x + y \in U \oplus E_8(2)$ and consider its pairing with w_2 :

$$(x + y) \cdot w_2 = (x + y) \cdot (2\tilde{e} + 2\tilde{f} + \tilde{\alpha}_1 + \tilde{\alpha}_2) = 2x \cdot (\tilde{e} + \tilde{f}) + y \cdot (\tilde{\alpha}_1 + \tilde{\alpha}_2),$$

where we've used orthogonality relations. We note that the first term is evidently even, while the second term is even because all pairings are either zero or divisible by two in the $E_8(2)$ summand of $U \oplus E_8(2)$. ■

Remark 7E.6: This can also be proved by considering the divisorial presentation of F_{C_0} as $\mathcal{H}_{-2}/O^+(T_{E_n})^*$. If J' is an isotropic plane in T_{E_n} generated by x_1, x_2 where both x_i are orthogonal to a (-2) -vector in T_{E_n} , then the 1-cusp corresponding to J' will be contained in the closure of $\mathcal{H}_{-2}$. One checks that cusp $(8, 6, 0)_0$ is associated to the isotropic plane

$$J' = \langle e', 2e + 2f + 2\tilde{\alpha}_1 \rangle \subset T_{E_n}$$

where both e' and $2e + 2f + 2\tilde{\alpha}_1$ are orthogonal to the (-2) -vector $e - f \in T_{E_n}$. On the other hand, cusp $(8, 8, 0)$ does not satisfy this property – the vector $e \in T_{E_n}$ is not orthogonal to any (-2) -vectors in T_{E_n} , and the isotropic plane at this cusp must include e as a generator.

Remark 7E.7: The two 1-cusps $(8, 8, 0)$ and $(8, 6, 0)$ in F_{E_n} are isomorphic to the modular curves $X_0(2)$ and $X := \mathbb{H}/\operatorname{SL}_2(\mathbb{Z})$ respectively, and by [CDL25, Cor. 5.9.10] the 1-cusp $(7, 7, 1)$ in F_{C_0} is isomorphic to X . This can additionally be verified by [AE22, Prop. 5.13]: the 1-cusp $(7, 7, 1)$ in T_{C_0} is incident to exactly one 0-cusp, as is the 1-cusp $(8, 6, 0)$ in T_{E_n} , and thus the corresponding modular curves are both isomorphic to X . We conjecture that general correspondences on 1-cusps must preserve the isomorphism types of the corresponding modular curves, yielding an alternative proof of lemma 7E.5.

Lemma 7E.8.

Let $\tilde{w}_i$ be the images of v_i in T_{dP} under the embedding described in lemma 4G.2. Then

$$\tilde{w}_1^\perp_{T_{dP}}/\tilde{w}_1 \cong (18, 0, 0)_1 \cong U \oplus E_8^2$$

$$\tilde{w}_2^\perp_{T_{dP}}/\tilde{w}_2 \cong (16, 0, 0)_0$$

The first identification is forced: $(18, 0, 0)_1$ is the unique even unimodular lattice of its (indefinite) signature. The second lattice is even, unimodular, and negative-definite of rank 16, and is thus determined only up to the two such isometry classes.

Thus the embedding $F_{C_0} \hookrightarrow F_{(2,2,0)}$ induces the following maps on cusps:

$$(9, 9, 1)_1 \mapsto (18, 0, 0)_1$$

$$(7, 7, 1)_0 \mapsto (16, 0, 0)_0$$

Proof.

The map on 0-cusps follows from identifying $\tilde{w}_1 = e' \in T_{\text{dP}} = \langle e, f \rangle \oplus \langle e', f' \rangle \oplus \langle \alpha_1, \dots, \alpha_8 \rangle \oplus \langle \tilde{\alpha}_1, \dots, \tilde{\alpha}_8 \rangle$ and noting

$$\begin{aligned} \frac{(e')^{\perp T_{\text{dP}}}}{e'} &\cong \frac{\langle e, f \rangle \oplus \langle e' \rangle \oplus \langle \alpha_1, \dots, \alpha_8 \rangle \oplus \langle \tilde{\alpha}_1, \dots, \tilde{\alpha}_8 \rangle}{e'} \\ &\cong \langle e, f \rangle \oplus \langle \alpha_1, \dots, \alpha_8 \rangle \oplus \langle \tilde{\alpha}_1, \dots, \tilde{\alpha}_8 \rangle \\ &\cong U \oplus E_8^2. \end{aligned}$$

To determine the map on 1-cusps, it suffices to determine

$$\text{div}_{U \oplus E_8^2}(\tilde{w}_2), \quad \tilde{w}_2 = 2e + 2f + \alpha_1 + \alpha_2 + \tilde{\alpha}_1 + \tilde{\alpha}_2.$$

However, observing that $\tilde{w}_2 \cdot \alpha_3 = 1$, we immediately obtain that the divisibility is one. Thus $\tilde{w}_2$ is odd, and we apply case (a) of [AE22, Thm. 5.10]. ■

Conjecture 7E.9.

The negative-definite lattice $\tilde{w}_2^{\perp T_{\text{dP}}} / \tilde{w}_2 \cong (16, 0, 0)_0$ of lemma 7E.8 is isometric to $E_8^{\oplus 2}$, and not to the other even negative-definite unimodular lattice of rank 16, namely D_{16}^+ .

Remark 7E.10: The invariants $(r, a, \delta) = (16, 0, 0)$ do not by themselves determine the isometry class, since the even negative-definite unimodular lattices of rank 16 are exactly $E_8^{\oplus 2}$ and D_{16}^+ . A proof of ?? 7E.9 would follow from showing that $\tilde{w}_2^{\perp T_{\text{dP}}} / \tilde{w}_2$ inherits the $E_8 \oplus E_8$ splitting of $T_{\text{dP}} = U \oplus U(2) \oplus E_8^2$.

Remark 7E.11: By the cusp correspondence in [Ale+25], this matches cusp $(9, 9, 1)_1$ in T_{Co} with Sterk cusp 2 as the folding of $(18, 0, 0)_1$ by the horizontal symmetry of its Coxeter diagram. This means that the corresponding IAS and Kulikov models will be disc type, and correspond to the folding involution of Sterk 2.

8. STABLE LIMITS

8A. Geometric preliminaries.

Remark 8A.1: Following [Dol17, §5.1], Coble surfaces can be obtained by degenerating (X, τ) a K3 with a fixed-point-free involution to a K3 surface (X_0, τ_0) where the fixed locus of τ_0 is a smooth rational curve. The resulting quotient X_0 / τ_0 is a Coble surface.

A certain linear system $|2f_i + 2f_j|$ defines a degree 2 map onto a quartic del Pezzo surface $\phi_{ij} : S \rightarrow D$ with 4 A_1 singularities, as in the case of Enriques surfaces, but this map is never finite in the Coble case. The deck transformation of ϕ_{ij} is a biregular automorphism of S . The f_i come from an *isotropic sequence* and are obtained by modifying the basis $e_0, \dots, e_{10}$ of $K_S^{\perp \text{Num}(S)} \cong E_{10}$ where e_0 is the preimage of a hyperplane class and $e_1, \dots, e_{10}$ are the classes of exceptional curves in the blowup of a plane sextic at 10 A_1 singularities.

Remark 8A.2: We recall the objects whose stable limits are taken. A Coble surface is a smooth rational projective surface S with $|-K_S| = \emptyset$ but $|-2K_S| \neq \emptyset$ [DM20; DK25]. It is *terminal of K3 type* when $|-2K_S|$ contains a reduced divisor $C = C_1 + \dots + C_n$ of disjoint smooth rational curves with $C_i^2 = -4$; the C_i are the *boundary components*, and one has

$$n = -K_S^2, \quad n \leq 10$$

[DM20; DK25]. In the case $n = 1$ studied here, the anti-bicanonical divisor is a single smooth rational curve C with $C^2 = -4$; this is the curve that reappears as the contracted curve on the stable model.

Remark 8A.3: Such a terminal Coble surface is *basic rational*: it admits a birational morphism to $\mathbf{P}^2$ obtained by blowing up $N = 9 + n$ points [DK25]. For $n = 1$ one recovers the classical Coble surface by blowing up the ten A_1 -singularities of an irreducible rational plane sextic, whose proper transform then lies in $|-2K_S|$ [Cob19; Cob29; CDL25]. Coble surfaces are the anti-bicanonical analogue, among Cremona-special point configurations, of the unnodal Halphen surfaces [CD12], and every terminal Coble surface of K3 type is tied to both a K3 double cover and an index-2 Halphen surface [DK25; CD12].

Remark 8A.4: The polarization enters through the degree-2 numerically polarized Enriques picture into which the Coble locus embeds. A degree-2 numerically polarized Enriques surface is a pair $(Z, [\mathcal{L}_Z])$ with

$[\mathcal{L}_Z] \in \text{Num}(Z)$ an ample class of degree 2; the system $|\mathcal{L}_Z^{\otimes 2}|$ is basepoint-free and realizes Z as a double cover $\rho: Z \rightarrow W$ of a quartic del Pezzo surface W with singularities of type $4A_1$ or $A_3 + 2A_1$, branched along a divisor $B \subset W$ [CDL25]. This is the finite analogue of the map ϕ_{ij} above, which in the Coble case fails to be finite. The ramification divisor $R_Z = \rho^{-1}(B)$ is ample, $\mathbf{Q}$ -Cartier, and lies in $|\mathcal{L}_Z^{\otimes 2}|$, so $(Z, \varepsilon R_Z)$ is log canonical for small $\varepsilon > 0$ [CDL25]. It is the Coble descent of this ramification divisor that supplies the stable-pair boundary of the KSBA stable limits.

8B. Integral affine structures.

Remark 8B.1: Following [Ale+25], a **Kulikov model** is a K -trivial semistable model $\mathcal{X} \rightarrow (C, 0)$ of a degeneration of K3 surfaces over a pointed curve C . For each such degeneration, one can define the dual complex of the central fiber $\Gamma(\mathcal{X}_0)$. For Type II degenerations of K3 surfaces the dual complex is an interval $\mathbb{D}^1$, and for Type III it is an integral affine S^2 with singularities of total charge 24. The additional data of an integral affine polarization $R_{\text{IA}} \subset \Gamma(\mathcal{X}_0)$ describes the KSBA stable limit of a degeneration $(\mathcal{X}^*, \varepsilon \mathcal{R}^*)$. The geometry of such a degeneration is depicted in fig. 20: the family $\mathcal{X}$ is fibred over a curve C in the moduli space $\mathcal{M}$, and the stable limit is the fiber over the point where C meets the boundary. For Enriques (and hence Coble) surfaces, we take the corresponding dlt models $\mathcal{Z} := \mathcal{X}/\iota_{\text{En}}$ and half-divisor models $(\mathcal{Z}, \mathcal{R}_{\mathcal{Z}}) := (\mathcal{X}, \mathcal{R})/\iota_{\text{En}}$ where $\mathcal{X} \rightarrow (C, 0)$ and $(\mathcal{X}, \mathcal{R})$ are Kulikov and divisor models of their K3 covers.

Definition 8B.2

The dual complex $\Gamma(\mathcal{X}_0)$ of a Type III Kulikov model carries a canonical *singular integral affine structure*: away from a finite singular set its charts map to $\mathbf{R}^2$ with transition functions in $\text{GL}_2(\mathbb{Z}) \ltimes \mathbf{R}^2$, and its singularities correspond to the components of positive charge, the non-toric anticanonical pairs. The total charge is 24, which constrains the number and type of singularities; for instance one may have 24 singularities of type I_1 .

Definition 8B.3

Given a monodromy invariant λ with barycentric coordinates $\ell_i = \lambda \cdot \alpha_i$, the *Symington polytope* $P(\lambda)$ is the integral affine polygon determined by these coordinates [Sym02]. Gluing two copies along their boundary,

$$B(\lambda) = P(\lambda) \cup P(\lambda)^{\text{op}},$$

produces the integral affine sphere realizing the dual complex $\Gamma(\mathcal{X}_0)$; the equator along which the two copies are glued supports the integral affine polarization R_{IA} [AE23].

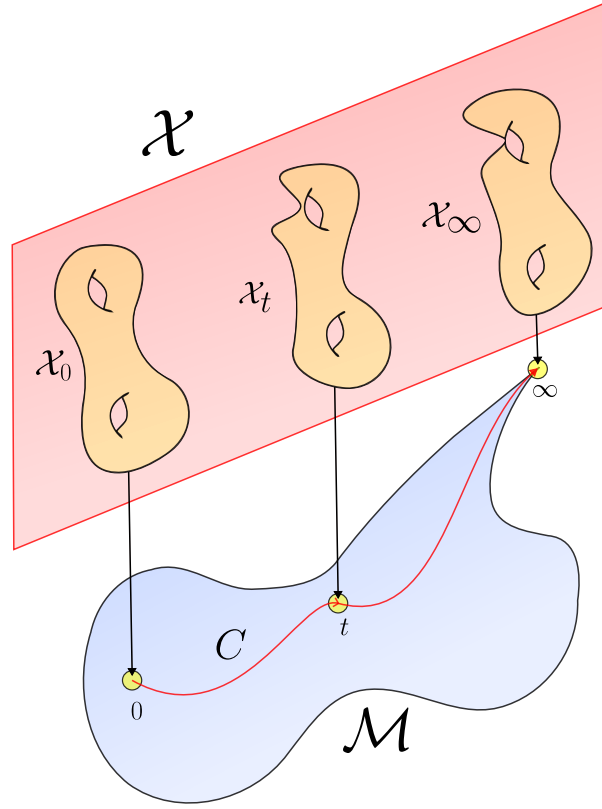


FIGURE 20. A one-parameter family $\mathcal{X} \rightarrow C$ of surfaces over a curve $C \subseteq \mathcal{M}$ in the moduli space, with fibers $\mathcal{X}_0$ and $\mathcal{X}_t$ over interior points and the limit $\mathcal{X}_\infty$ over the boundary point ∞ .

Remark 8B.4: The following is a representation of a Type II degeneration – it is a chain of surfaces whose dual complex is an interval $\mathbb{D}^1$, where the ends V_1 and V_n are rational and the remaining V_i are isomorphic to $E \times \mathbf{P}^1$ for a fixed elliptic curve E . The intersections $V_i \cap V_{i+1}$ are double curves isomorphic to E .

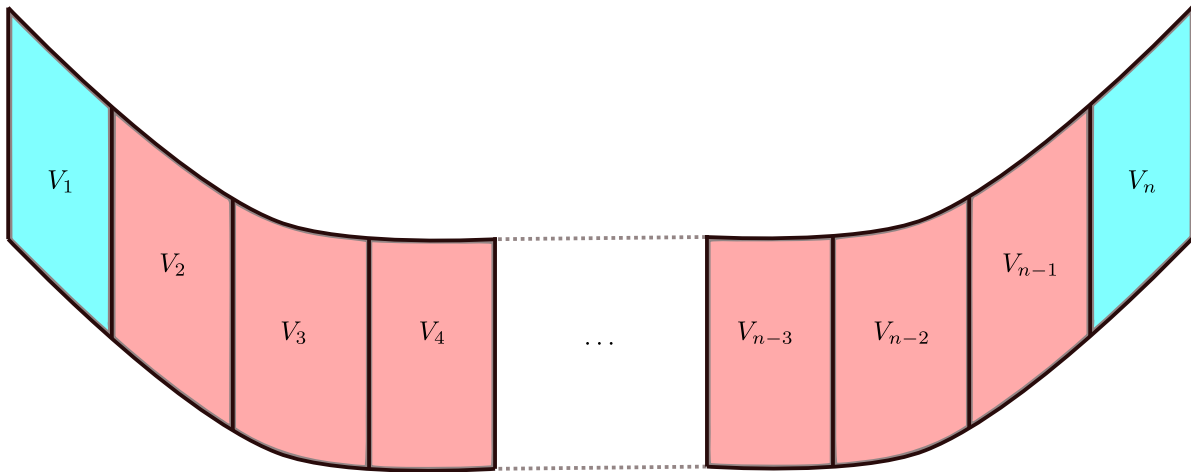


FIGURE 21. A Type II Kulikov degeneration.

A Type III degeneration can be represented by a triangulation of S^2 with singularities, depicted as follows:

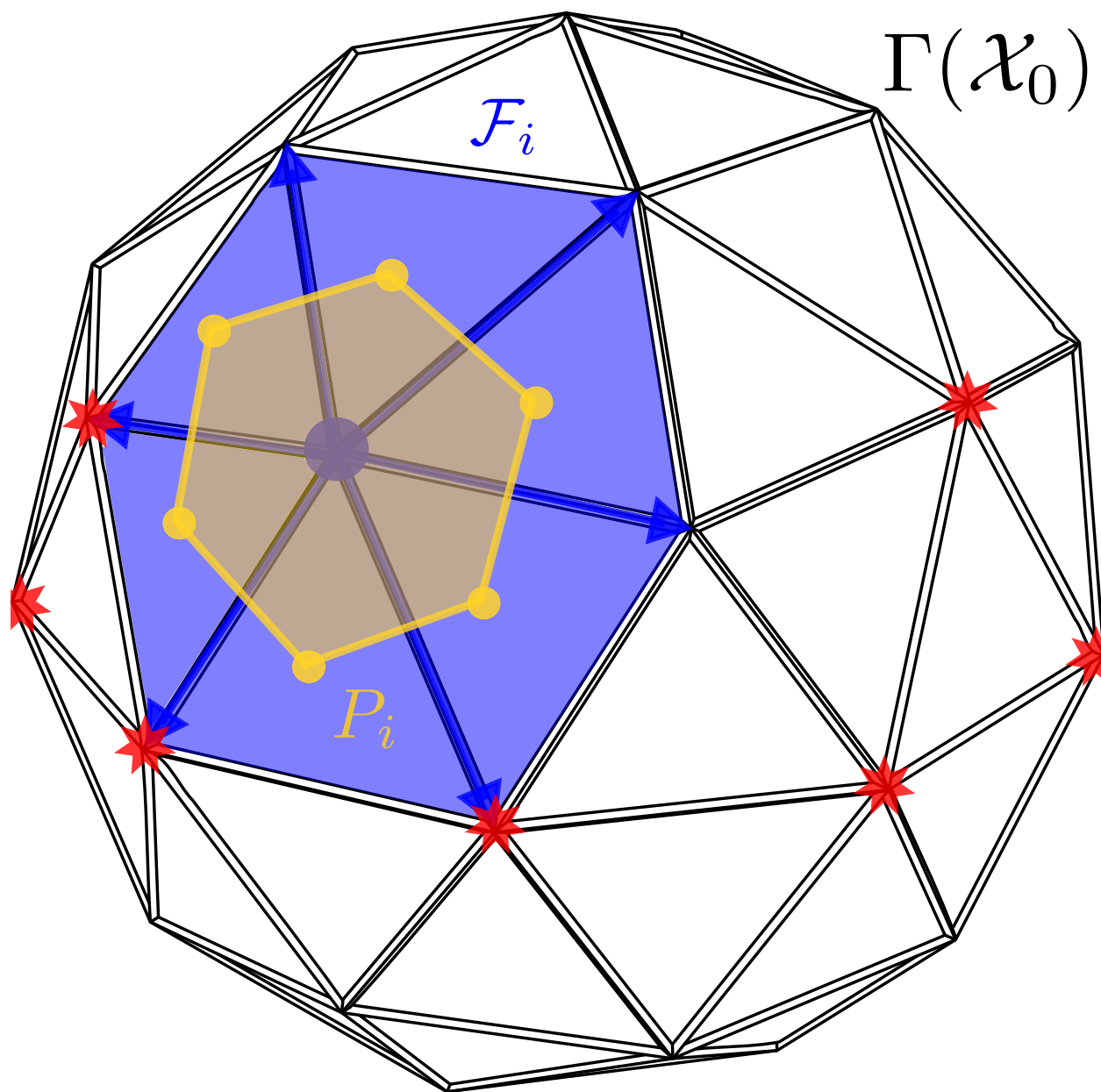


FIGURE 22. A triangulated integral affine sphere.

Remark 8B.5: The following is a combinatorial representation of a Kulikov model for Sterk 2.

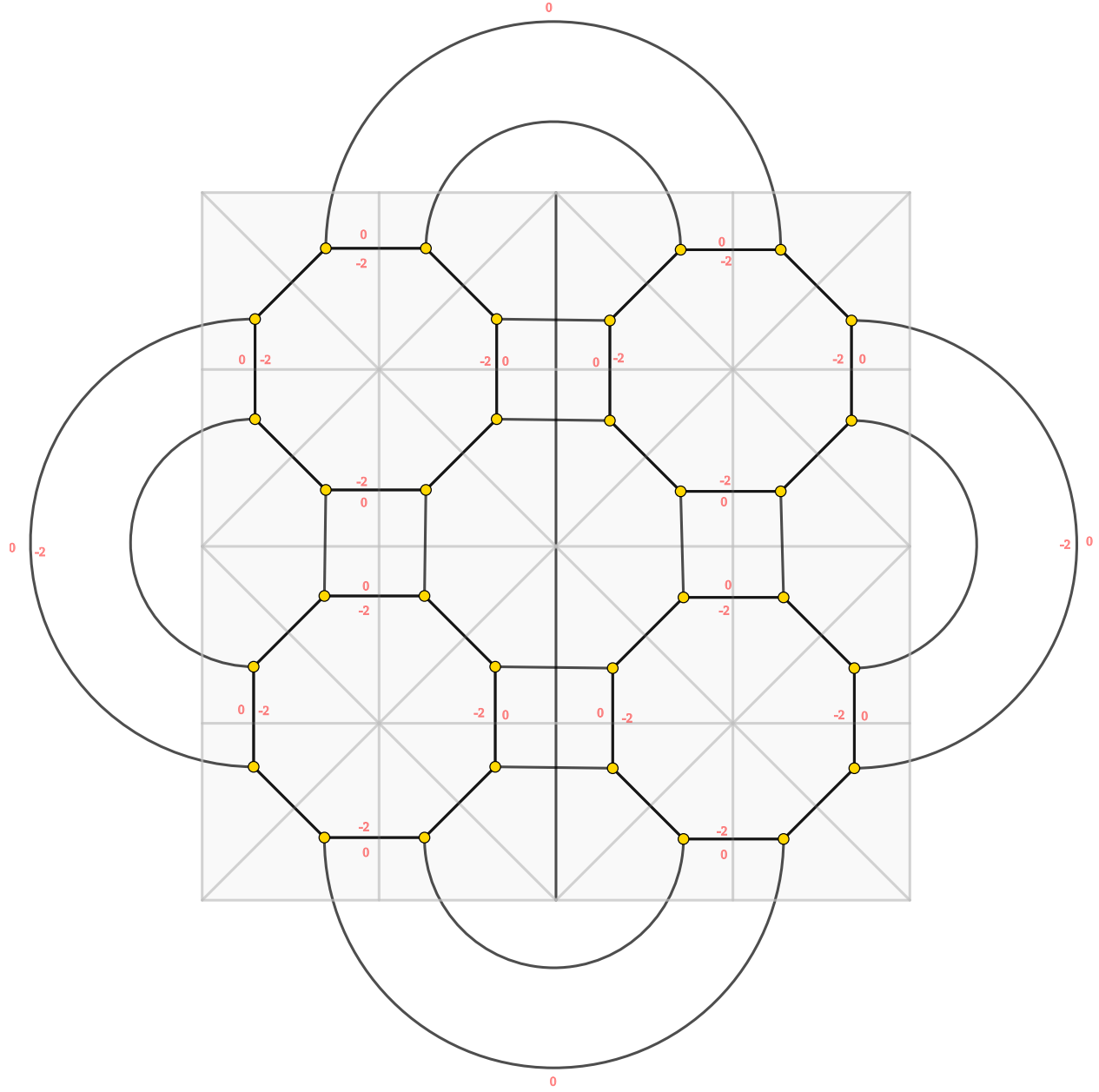


FIGURE 23. A combinatorial Kulikov model for Sterk 2.

8B.1. *The Sterk 2 integral affine structure.*

The figures below were drawn for this construction but their accompanying text has not been written; they are collected here so that the artwork is not orphaned. Each caption states only what the picture shows.

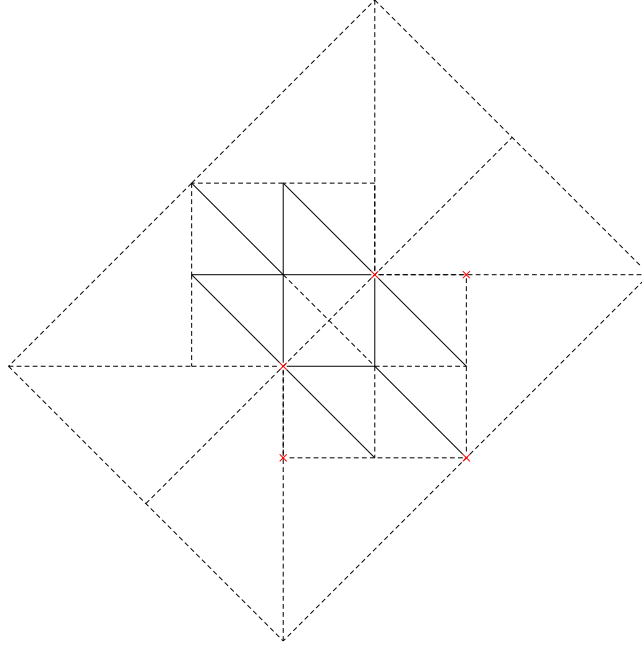


FIGURE 24. An integral affine structure for Sterk 2 drawn in the plane: five integral affine singularities are marked $\times$, solid segments carry the triangulation and dashed segments the boundary of the region.

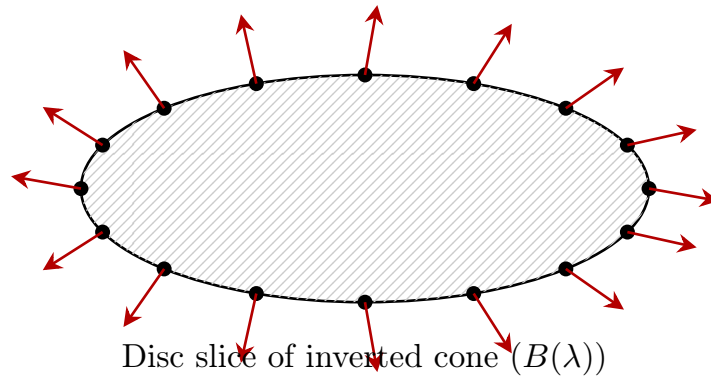


FIGURE 25. The disc slice $B(\lambda)$ of the inverted cone, with the outward rays at its boundary points.

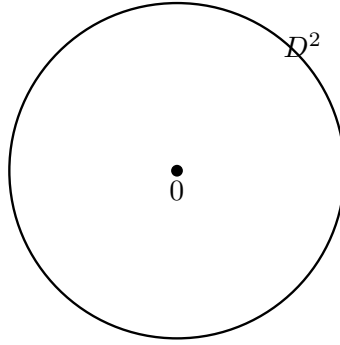
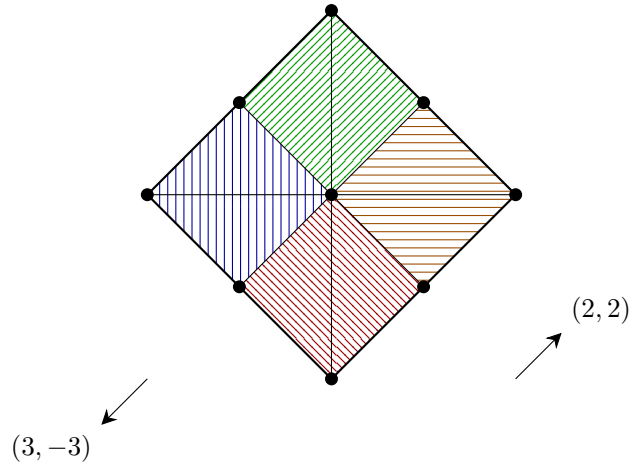
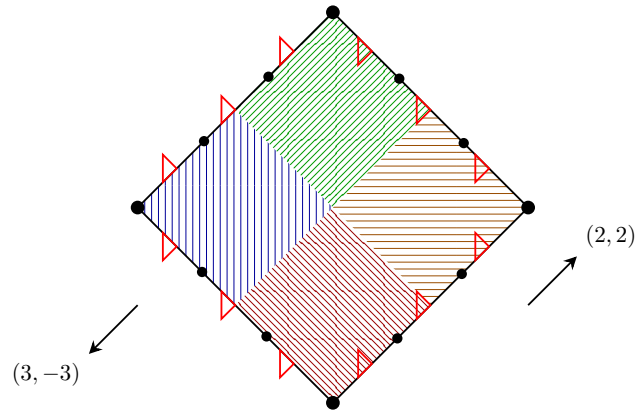
FIGURE 26. The integral affine disc $\mathbb{D}^2$.FIGURE 27. A triangulated integral affine polytope with its charge distribution, in the directions $(2, 2)$ and $(3, -3)$.

FIGURE 28. The same polytope after Symington surgeries, marked in red along the boundary.

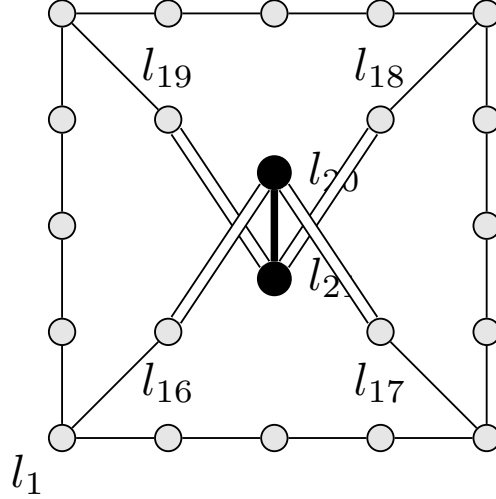


FIGURE 29. The 16-gon with its boundary lines $\ell_1, \ell_{16}, \dots, \ell_{21}$ labelled.

Remark 8B.6: We leverage the theory of [Ale+25; AE22; AE23; AET23; ABE22].

8B.2. *Marked-root structures for the Coble locus.*

Remark 8B.7: For Enriques and Coble surfaces the integral affine data is built on the K3 cover first and only then folded downstairs. One starts from the K3 monodromy (Coxeter) data behind the integral affine sphere, imposes the Coble condition by marking a root r of zero length,

$$\lambda \cdot r = 0,$$

and only afterwards passes to the folded Enriques data and the Coble hyperplane. The marked root records the vanishing cycle producing the node on the K3 cover; it is what distinguishes the Coble integral affine structure from an ordinary Enriques boundary structure.

Question 8B.8

Does there exist an equivariant triangulation of the integral affine sphere compatible with the marked root r ? Absent such a triangulation, the marked-root prescription remains a construction principle rather than a finished combinatorial model.

Remark 8B.9: Under the cusp correspondence, the Coble 0-cusp corresponds to Sterk cusp 2, realized as the folding of the cusp $(18, 0, 0)_1$ by the horizontal symmetry of its Coxeter diagram [Ale+25]. The associated integral affine and Kulikov models are therefore of *disc* type, matching the folding involution of Sterk 2; the Coble boundary is expected to produce disc-type integral affine structures rather than sphere- or $\mathbf{RP}^2$ -type limits, in agreement with the flowerpot degenerations of Morrison [Mor81; Ale+25]. The combinatorial disc-type model of fig. 23 and the integral affine structure for Sterk 2 of fig. 24 illustrate this case.

8C. KSBA stable limits.

Remark 8C.1: We describe the KSBA stable limits of Coble surfaces.

Remark 8C.2: We give an example of an integral affine structure for a degeneration of Coble surfaces.

Both sections above are outlines: the KSBA stable limits and the worked integral affine example remain to be written.

8C.1. *The polarized Coble locus and its branches.*

Remark 8C.3: The stable limits below compactify the polarized Coble locus, which we realize inside the moduli space $F_{\text{En},2}$ of degree-2 numerically polarized Enriques surfaces: one takes the Coble Heegner divisor cut out inside $F_{\text{En},2}$ and normalizes it. Because the polarized arithmetic group $\Gamma_{\text{En},2}$ has finite index in the full Enriques group Γ_{En} , a single unpolarized root orbit can split into several polarized orbits. Until root-orbit uniqueness is established for $\Gamma_{\text{En},2}$, the polarized Coble locus is therefore described as a union of *branchwise* quotients, one attached to each orbit of admissible Coble roots, rather than as a single global quotient.

Remark 8C.4: Each branch carries its own arithmetic group. Fixing an admissible Coble root δ – a primitive (-2) -vector in the polarized Enriques period lattice – the group governing the corresponding branch is the image in $O(\delta^\perp)$ of the stabilizer of $\mathbb{Z}\delta$ inside $\Gamma_{\text{En},2}$; this is the minimal group making the period-domain inclusion

$$\mathbb{D}(\delta^\perp) \hookrightarrow \mathbb{D}(T_{\text{En}})$$

equivariant. Since $T_{\text{Co}} \cong \delta^\perp T_{\text{En}}$ for a (-2) -vector δ [DK13], this refines the arithmetic group $\Gamma_{\text{Co},2}$ of the moduli summary, and the two descriptions agree exactly when the polarized root orbit is unique.

Question 8C.5

Is the orbit of admissible Coble roots under $\Gamma_{\text{En},2}$ unique, so that the branchwise polarized Coble locus collapses to a single normalized divisor? Two routes to an affirmative answer are available: an arithmetic double-coset computation for the polarized subgroup $\Gamma_{\text{En},2}$, or a geometric argument that the D_4 -symmetry of the Horikawa model acts transitively on the torus-fixed-point branches.

8C.2. The KSBA stable pair.

Remark 8C.6: The ambient degree-2 Enriques picture is settled. For a degree-2 numerically polarized Enriques surface $(Z, [\mathcal{L}_Z])$, the ramification divisor R_Z of the associated double cover of a quartic del Pezzo surface is ample, $\mathbf{Q}$ -Cartier, and lies in the polarizing system, so the pair $(Z, \varepsilon R_Z)$ is log canonical for $0 < \varepsilon \ll 1$ and $F_{\text{En},2}$ admits a KSBA compactification $\overline{F}_{\text{En},2}$ [CDL25; Ale+25]. The Coble stable pair is the descent of this picture along the quotient by the Enriques involution ι_{En} , which for Coble surfaces is *not* fixed-point free.

Remark 8C.7: The intended KSBA boundary object is a pair

$$(\overline{S}, \varepsilon R_{\overline{S}}), \quad 0 < \varepsilon \ll 1,$$

where $\overline{S}$ is the stable quotient surface and $R_{\overline{S}}$ is the descended ramification divisor. The divisor is identified; establishing that this pair is KSBA stable remains a research program with the following open obligations:

- that $R_{\overline{S}}$ is $\mathbf{Q}$ -Cartier;
- that $R_{\overline{S}}$ is ample – expected to follow by descending the ample ramification divisor from the K3 cover, that is, by pulling back $K_{\overline{S}} + \varepsilon R_{\overline{S}}$ to εR_X with R_X coming from the $(2,2)$ -divisor on $Y = \mathbf{P}^1 \times \mathbf{P}^1$, rather than by asserting stability of the quotient directly;
- that $(\overline{S}, \varepsilon R_{\overline{S}})$ is slc, for which one proposed route runs through the du Val singularities on the K3 cover and a finite quasi-étale quotient in codimension one;
- and a controlled account of how the anti-bicanonical (-4) -curve is seen on the smooth resolution versus on the stable model.

Conjecture 8C.8.

On the stable quotient $\overline{S}$, an A_1 -node of the K3 cover fixed by ι_{En} descends to a cyclic quotient singularity of type $\frac{1}{4}(1,1)$, and the anti-bicanonical (-4) -curve on the smooth Coble resolution is the curve contracted to this point. In the Horikawa model on $Y = \mathbf{P}^1 \times \mathbf{P}^1$ with $\tau(x,y) = (-x,-y)$ [Hor77], the local input producing the A_1 -node on the double cover is a τ -invariant $(4,4)$ -curve passing through a τ -fixed point with nondegenerate quadratic term.

Remark 8C.9: The local singularity package of ?? 8C.8 is central to the program, but it is currently a migrated research claim rather than a proven statement; it is precisely the input awaited by the slc and ampleness verifications above.

8C.3. The restricted ramification semifan.

Conjecture 8C.10.

The semitoroidal model of the polarized Coble locus is obtained by restricting the Enriques ramification semifan of the degree-2 compactification problem to the hyperplane cut out by the Coble root, and keeping exactly those walls whose relative interiors meet the Coble positive cone. Under this restriction, a Coble wall is irrelevant precisely when every Enriques wall restricting to it is already irrelevant.

The migrated note carry no citations dlt/divisor/half-d apparatus tracks Alexeev–Engel–G Schaffler degree-2 Enriques program (c use of the dlt mod of [AEGS25] in Morrison degener section); attach th tended references confirmed rather asserting them he

Attribution unver fied: Prop. 9.1.8 a published comput $h^1(\Theta_{V/k}) = 10$ neither it nor its mentions the dim of Halphen modul cate the intended

Remark 8C.11: Proving that this restriction defines the semitoroidal fan requires showing that no extra roots appear after restriction, that no essential Enriques wall collapses or restricts trivially, and that running Vinberg’s algorithm on the restricted lattice is not conflated with a proof of the fan itself.

8C.4. *Comparison with the KSBA compactification.*

Remark 8C.12: The KSBA stable limits sit inside the K3 stable-pair family of the degree- $(2, 2, 0)$ problem via the embeddings of lemma 4G.3. The proposed comparison proceeds by restricting the universal K3 stable-pair family over $F_{(2,2,0)}$ to the Coble Noether-Lefschetz locus $\mathbb{D}(r^\perp)$, extending the Enriques involution over the stable limits by uniqueness of KSBA limits, descending the ramification divisor, and matching the induced boundary stratification against the restricted ramification semifan of ?? 8C.10.

Conjecture 8C.13.

After normalization, the KSBA compactification of the polarized Coble locus agrees with the semitoroidal compactification induced by the restricted ramification semifan.

Conjecture 8C.14.

The stable quotient remembers the marked Coble root. Geometrically, this memory is carried by the $\frac{1}{4}(1, 1)$ singularity of ?? 8C.8 – equivalently, by the contracted anti-bicanonical (-4) -curve on the resolution – so that degenerations differing only by their marked root are not identified. Without this memory the restricted semifan would be too fine for the actual KSBA boundary, and the comparison of ?? 8C.13 would fail.

Remark 8C.15: ?? 8C.13 remains open on four counts: root-orbit uniqueness (?? 8C.5), the ramification-semifan restriction identity (?? 8C.10), the no-moduli-loss statement (?? 8C.14), and the exact cusp enumeration. The boundary dictionaries and cusp tables appearing in preliminary work remain unverified pending the restriction theorem and an explicit cusp computation.

8C.5. *Boundary cusp data.*

Remark 8C.16: The boundary is organized by cusp pairs marked with a Coble root. A 0-cusp is modeled by an orbit of a pair (I, r) consisting of an isotropic line I and a compatible Coble root r , and a 1-cusp by an orbit of a pair (J, r) consisting of an isotropic plane J and the same root, with incidence recorded by the containment $I \subset J$ preserving r . A root r is *admissible* at a cusp when it lifts to a primitive (-2) -root of the ambient Enriques lattice lying in the designated Coble orbit; this is the datum that promotes plain Enriques cusp data to polarized Coble cusp data, and it must be preserved along the incidence $I \subset J$.

Question 8C.17

What is the precise admissibility test for Coble roots at a cusp, formulated against the folded K3-to-Enriques Coxeter data? Sterk cusps 3 and 5 are the delicate cases where additional reflection data may intervene.

Remark 8C.18: Any actual cusp count must reduce to explicit lattice-orbit work – through Sterk’s representatives (five 0-cusps and nine 1-cusps for the Enriques space [Ste91]) together with their stabilizers, or direct period-domain enumeration – and discriminant-form shortcuts suggest candidates but do not by themselves prove the cusp diagram. One durable exclusion is nonetheless available: since primitive isotropic vectors of T_{C_0} pair evenly in the ambient Enriques lattice, they have divisibility 2 (lemma 7E.2), so the divisibility-one Sterk cusp 1 does not occur on the polarized Coble boundary and only the divisibility-two Sterk cusps 2–5 are in play. This is consistent with the cusp correspondence of theorem 7E.1, under which the unique Coble 0-cusp corresponds to a divisibility-two Enriques cusp.

9. THE MODULI SPACE OF DEGREE-2 ENRIQUES SURFACES

9A. The moduli space $F_{\text{En},2}$: main results. Here we collect the principal structural results for the KSBA compactification $\overline{F}_{\text{En},2}$ of the moduli space $F_{\text{En},2}$ of degree-2 numerically polarized Enriques surfaces: the identification of the normalized compactification with a semitoroidal model, the realization of $F_{\text{En},2}$ as a normalization inside the degree- $(2, 2, 0)$ K3 moduli space, and the enumeration of the five 0-cusps together with their folded Coxeter data. These describe the *ambient* degree-2 Enriques picture, into which the polarized Coble locus is later cut by an admissible root (cf. ?? 8C.10).

9A.1. *The main theorem.***Theorem 9A.1 (Compactification of $F_{\text{En},2}$).**

Let $\overline{F_{\text{En},2}}$ denote the KSBA compactification of the moduli space $F_{\text{En},2}$ of numerically polarized Enriques surfaces of degree 2. Let $\mathcal{F}_\bullet = \{\mathcal{F}_k\}_{k=1}^5$ be the collection of folded semifans for the 0-cusps.

1. The normalization $\overline{F_{\text{En},2}}^\nu$ is isomorphic to the semitoroidal compactification $\overline{F_{\text{En},2}}^{\mathcal{F}_\bullet}$ [Ale+25, Thm. 1].
2. This compactification is toroidal over cusps 2, 4, their adjacent 1-cusps, and cusp 35, and strictly semitoroidal over all other cusps.
3. The isomorphism is established via an intermediate normalization B^ν of the Zariski closure of the Noether–Lefschetz locus inside the K3 compactification $\overline{F_{(2,2,0)}}$ [Ale+25, Sec. 6].

Remark 9A.2: This theorem records the settled ambient degree-2 Enriques picture: the normalized KSBA compactification of $F_{\text{En},2}$ coincides with an explicit semitoroidal model built from five folded semifans, one per 0-cusp. The proof runs through the K3 moduli space $F_{(2,2,0)}$ of the degree- $(2,2,0)$ problem, identifying $\overline{F_{\text{En},2}}^\nu$ with a normalization of the closure of the relevant Noether–Lefschetz locus; see lemma 9A.3 for the corresponding period-domain statement.

Check that this is Hirzebruch surface with only part of boundary.

9A.2. *Normalization inside $F_{(2,2,0)}$.***Lemma 9A.3 (Normalization of $F_{\text{En},2}$).**

There exists a closed subscheme $X \subset F_{(2,2,0)}$ such that $F_{\text{En},2}$ is canonically isomorphic to the normalization of X .

Proof.

This is established via an algebraic morphism of period domains

$$\Psi : F_{\text{En},2} \rightarrow F_{(2,2,0)}$$

induced by the unique lattice embedding $\tilde{\Psi} : T_{\text{En}} \hookrightarrow T_{\text{dP}}$ defined by

$$\tilde{\Psi}(u_1, u_2, v) := (u_1, u_2, v, v).$$

Restricting Ψ to its scheme-theoretic image X yields a finite, birational map from the normal variety $F_{\text{En},2}$ to X , which by Zariski’s Main Theorem exhibits $F_{\text{En},2}$ as the normalization of X . ■

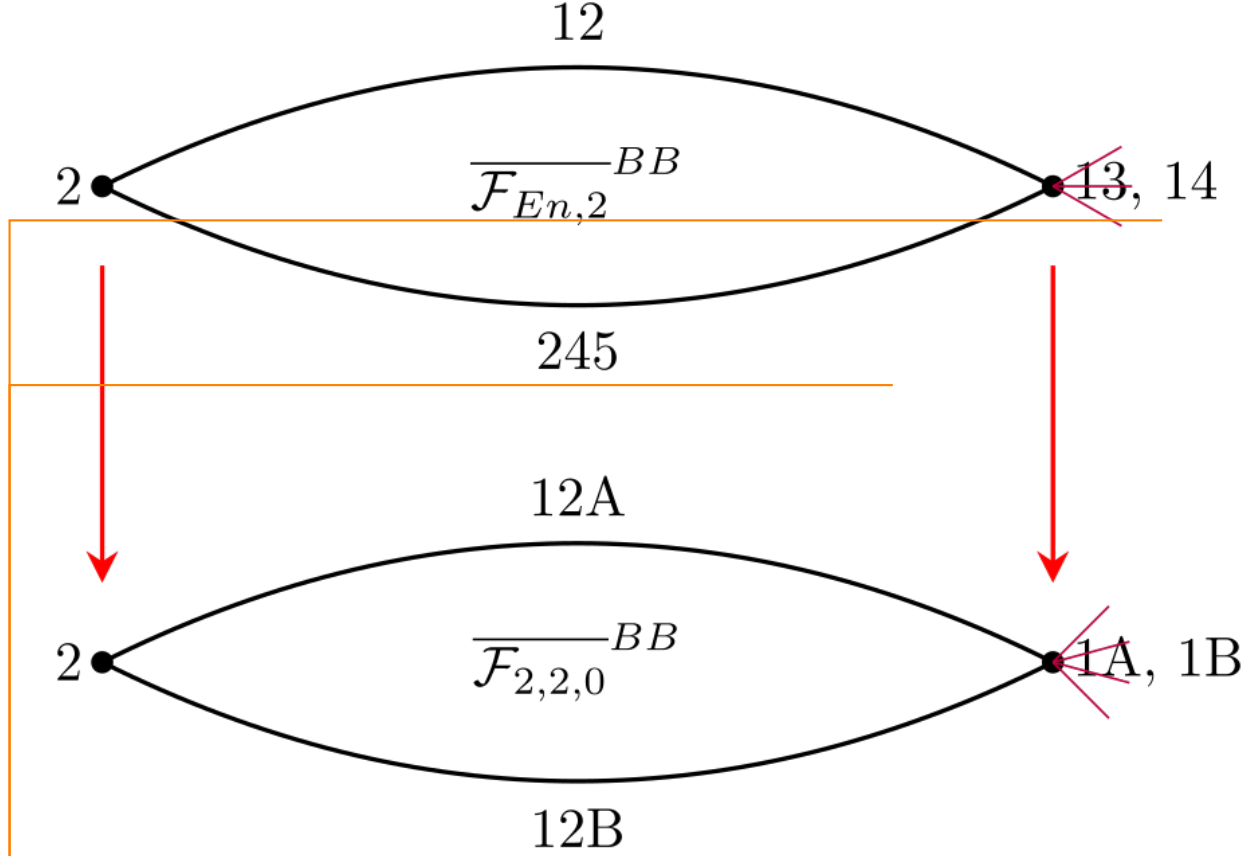


FIGURE 30. The induced correspondence of Bailly–Borel boundaries under $\Psi: F_{\text{En},2} \rightarrow F_{(2,2,0)}$: the cusps of $\overline{F_{\text{En},2}}^{\text{BB}}$ (top) map to the cusps of $\overline{F_{(2,2,0)}}^{\text{BB}}$ (bottom).

9A.3. The five 0-cusps.

Example 9A.4 (The five 0-cusps of $\overline{F_{\text{En},2}}$): The boundary of the KSBA compactification $\overline{F_{\text{En},2}}$ is stratified by 27 rays across five 0-cusps. For each 0-cusp we record the topological type of the reduced dual complex $\Gamma(Z_0)$, the number of Type II and Type III rays, and the integral-affine-structure (IAS) involution.

1. **Cusp 1:** Semitoroidal. $\Gamma(Z_0) = \mathbf{RP}^2$. 2 Type II rays, 0 Type III. IAS involution: 180° rotation + flip hemispheres.
2. **Cusp 2:** Toroidal. $\Gamma(Z_0) = \mathbf{D}^2$. 2 Type II rays, 7 Type III. IAS involution: vertical flip of both hemispheres.
3. **Cusp 3:** Semitoroidal. $\Gamma(Z_0) = \mathbf{D}^2$. 2 Type II rays, 7 Type III. IAS involution: diagonal flip.
4. **Cusp 4:** Toroidal. $\Gamma(Z_0) = \mathbf{D}^2$. 4 Type II rays, 7 Type III. IAS involution: horizontal flip.
5. **Cusp 5:** Semitoroidal. $\Gamma(Z_0) = \mathbf{D}^2$. 3 Type II rays, 0 Type III. IAS involution: flip hemispheres.

Remark 9A.5: This enumeration is the boundary data underlying the folded semifans $\mathcal{F}_\bullet$ of theorem 9A.1: each 0-cusp carries a reduced dual complex, a partition of its rays into the Type II (adjacent 1-cusp) and Type III (deeper) strata, and the involution of its integral affine structure that folds the covering K3 data onto the Enriques data. The precise per-cusp ray counts and IAS involutions are migrated from the working notes and, as with the cusp tables discussed in ?? 8C.13, should be regarded as provisional pending an independent cusp computation.

9A.4. Folded Coxeter diagrams of the five cusps.

Remark 9A.6 (Folded Coxeter diagrams of $F_{\text{En},2}$): The five 0-cusps of $F_{\text{En},2}$ are expected to correspond to five distinct orbits of primitive isotropic vectors in T_{En} , each realized as a folded image of a Coxeter diagram for $F_{(2,2,0)}$ under the involution $I = -I_{\text{En}}$ (the root-folding criterion of lemma 5B.5, in the sense of definition 5B.3):

1. η_1 : Divisibility 1, derived from $\tilde{\eta}_1$ via 180° rotation. (Boundary lattice: $U(2) \oplus E_8(2)$.)
2. η_2 : Divisibility 2, derived from $\tilde{\eta}_2$ via vertical reflection. (Boundary lattice: $U \oplus E_8(2)$.)
3. η_3 : Divisibility 2, derived from $\tilde{\eta}_1$ via diagonal reflection + root swap. (Boundary lattice: $U \oplus E_8(2)$.)
4. η_4 : Divisibility 2, derived from $\tilde{\eta}_1$ via horizontal reflection. (Boundary lattice: $U \oplus E_8(2)$.)
5. η_5 : Divisibility 2, derived from $\tilde{\eta}_1$ via 8 commuting reflections. (Boundary lattice: $U \oplus E_8(2)$.)

Remark 9A.7: The folded chamber $\mathfrak{C}^I = \mathfrak{C} \cap \overline{T}_{\eta, \mathbf{R}}^{I=1}$ has walls defined by the roots descending from the covering domain (cf. the classical foldings of example 5B.4).

The polarized Coble boundary problem uses this folded Enriques data only as ambient input. The extra marked-root refinement – the admissibility test for Coble roots at a cusp and the restriction of the ramification semifan to the polarized Coble locus – is recorded separately in ?? 8C.17 and ?? 8C.10.

9B. KSBA and semitoroidal compactifications of $F_{\text{En},2}$.

Remark 9B.1 (Orientation): This section records the degree-2 Enriques side of the KSBA–semitoroidal comparison: the compactification of $F_{\text{En},2}$, the moduli space of degree-2 numerically polarized Enriques surfaces. Here the comparison is a theorem, obtained by reducing to the recognizable-divisor machinery of Alexeev–Engel (theorem 5C.12, theorem 5C.4); the ambient degree-2 Enriques KSBA compactification is itself settled [Ale+25; CDL25]. This is the established counterpart of the polarized *Coble* comparison, which remains an open program: see ?? 8C.13 for the Coble isomorphism target and ?? 8C.10 for the Coble semifan-restriction analogue of the descent below.

9B.1. The isomorphism theorem.

Theorem 9B.2 (Isomorphism between KSBA and semitoroidal compactifications of $F_{\text{En},2}$).

Let $F_{\text{En},2}$ be the moduli space of degree-2 numerically polarized Enriques surfaces, and let $\overline{F}_{\text{En},2}$ be its KSBA compactification. There is an isomorphism

$$\left(\overline{F}_{\text{En},2}^{\mathcal{F}_\bullet} \right)^\nu \xrightarrow{\sim} \left(\overline{F}_{\text{En},2} \right)^\nu$$

where $(-)^^\nu$ denotes normalization. The left-hand side is the semitoroidal compactification corresponding to an explicit collection

$$\mathcal{F}_\bullet = \{ \mathcal{F}_1, \Sigma_2, \mathcal{F}_3, \Sigma_4, \mathcal{F}_5 \}$$

of semifans, one for each 0-cusp of the Baily–Borel compactification $\overline{F}_{\text{En},2}^{\text{BB}}$, and the right-hand side is the KSBA compactification.

Remark 9B.3 (Fan versus strict-semifan bookkeeping): The five cusps of $\overline{F}_{\text{En},2}^{\text{BB}}$ match Sterk’s five 0-cusps of the Enriques period space [Ste91], one semifan per cusp. Among the entries of $\mathcal{F}_\bullet$, the even-indexed entries Σ_2, Σ_4 are honest fans, while the odd-indexed entries $\mathcal{F}_1, \mathcal{F}_3, \mathcal{F}_5$ are strict semifans (in the sense of definition 5C.6, i.e. with infinite irrelevant subgroup, so not locally finite).

Proof.

The proof reduces to the recognizable-divisor theorem of Alexeev–Engel [AE23]: for a recognizable divisor R , the normalization of the KSBA compactification of stable K3 pairs $(X, \varepsilon R)$ is isomorphic to a semitoroidal compactification (theorem 5C.12, and the more general tower criterion of theorem 5C.4). The relevant polarizing divisor here is the ramification divisor R_ℓ of the nonsymplectic Enriques involution, which is recognizable (example 5C.13). Applying the theorem produces a semitoroidal compactification on the normalization of $\overline{F}_{\text{En},2}$, and identifying the resulting semifan cusp-by-cusp gives the explicit collection $\mathcal{F}_\bullet$, one semifan per 0-cusp. The passage from the ambient K3 picture to the Enriques space is the descent of ?? 9B.5, which supplies the folded semifans $\mathcal{F}_k$ and their boundary stratification. ■

Remark 9B.4 (Role of the normalization): The normalization $(-)^{\nu}$ is a technical condition standard in KSBA compactifications: taking a Zariski closure can introduce non-normal points where distinct degenerations are identified, producing a non-separated stack. Since the normalization morphism is finite, birational, and relatively smooth in codimension one, it confines the worst singularities to high-codimension sub-loci, which is what makes the isomorphism above an isomorphism of normal varieties.

9B.2. Descent of semitoroidal data.

Conjecture 9B.5 (Descent of semifans to B^{ν}).

The normalization $B^{\nu} \rightarrow B$ of the Noether–Lefschetz closure B yields a normal projective variety. The semitoroidal structure on $\overline{F}_{(2,2,0)}$ defined by the ramification semifan $\mathcal{F}_{\text{ram}}$ restricts to B^{ν} . Imposing the involution constraints on this restricted structure produces a collection of *folded semifans* $\mathcal{F}_k$.

Remark 9B.6 (Folded semifans and the boundary stratification): The folded semifans $\mathcal{F}_k$ determine the semitoroidal compactification $\overline{F}_{\text{En},2}^{\mathcal{F}\bullet}$ appearing in theorem 9B.2, and they define the combinatorial stratification of its boundary, which maps directly onto the KSBA strata. The ambient degree- $(2,2,0)$ K3 picture and the Noether–Lefschetz locus enter through the stable-pair family and the locally closed embeddings of lemma 4G.3; the descent here is the Enriques (involution-quotient) analogue of the Coble semifan restriction of ?? 8C.10.

9B.3. The polarized Coble analogue.

Remark 9B.7: The polarized Coble compactification records the same comparison as an open program rather than a theorem: after normalization the KSBA compactification of the polarized Coble locus is conjectured to agree with the semitoroidal compactification induced by the *restricted* ramification semifan (?? 8C.13, ?? 8C.10). Its extra difficulties are exactly the ones absent from the Enriques theorem above: branchwise root data, the ramification-semifan restriction identity, and the no-moduli-loss problem (?? 8C.14).

9C. The classifying map and the Enriques involution.

Remark 9C.1 (Orientation): The results collected here are the intended culmination of the compactification program for $F_{\text{En},2}$: the classifying morphism ϕ from the normalized Noether–Lefschetz closure B^{ν} to the KSBA compactification of the polarized moduli space should be shown to be an isomorphism, so that the geometric KSBA limits are identified with the explicitly constructible semitoroidal boundary. The three ingredients are a *global extension of the Enriques involution* over B , a *finiteness statement* for ϕ , and the *application of Zariski’s Main Theorem* that combines them.

These are the program’s own in-progress steps rather than settled theorems. They are stated conjecturally throughout, and they are exactly parallel to the open comparison of ?? 8C.13 and to the incomplete normalization argument of theorem 4G.4, whose migrated proof sketch records the author’s remark that finiteness is *still unclear*. Zariski’s Main Theorem itself is a classical, proven theorem; only its application to ϕ is conjectural here.

9C.1. Global extension of the Enriques involution.

Conjecture 9C.2 (Global extension of the Enriques involution).

Over the Noether–Lefschetz closure $B \subseteq \overline{F}_{(2,2,0)}$ of definition 2C.5, there is a universal KSBA family $(\mathcal{X}_B, \varepsilon\mathcal{R}_B)$. The fixed-point-free Enriques involution ι_{En} on the smooth fibers is expected to extend uniquely to a global involution on the entire family $\mathcal{X}_B$.

This global involution should preserve the slc pair structure, preserve the ramification divisor $\mathcal{R}_B$, and commute with the global del Pezzo involution ι_{dP} . The quotient $(\mathcal{X}_B, \varepsilon\mathcal{R}_B)/\iota_{\text{En}}$ would then form a flat family of stable Enriques pairs over B .

Remark 9C.3 (Intended mechanism): The uniqueness of the extension is the expected consequence of the separatedness of KSBA stable limits: on the smooth locus the involution is the fixed-point-free deck transformation ι_{En} of the canonical cover, and a fiberwise automorphism of a family of stable pairs extends across the boundary by uniqueness of the limit [Ale+25; Kol23]. This is the same extension-by-uniqueness step invoked in the proposed KSBA-to-semitoroidal comparison (?? 8C.13), where the Enriques involution is extended over the stable limits and the ramification divisor descended. The source note asserted the extension,

is migrated says "5", read alongside and 4. This is tent with the five-umeration of ex-A.4, where cusps are recorded ly semitoroidal sps 2 and 4 are). "cusp 35" is ced verbatim e source note; whether it is a "cusps 3, 5", ct 1-cusp label, s-transcription, oncle with the /semitoroidal the cusp enumera-

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the preservation of the slc structure and of $\mathcal{R}_B$, the commutation with ι_{dP} , and the flatness of the quotient family, without proof; each remains an obligation of the program.

9C.2. Finiteness of the classifying map.

Conjecture 9C.4 (*Finiteness of the classifying map*).

The classifying morphism

$$\phi: B^\nu \rightarrow \overline{F_{\text{En},2}}$$

from the normalization B^ν of the Noether–Lefschetz closure is finite.

Remark 9C.5 (*Proposed combinatorial argument*): The source note proposes to establish finiteness by a combinatorial comparison of semifans. The normal KSBA compactification induces its own semifans $\mathcal{G}_k$, which are coarsenings of the folded Coxeter semifans $\mathcal{F}_k$. The map is finite if and only if no coarsening occurs, i.e. $\mathcal{G}_k = \mathcal{F}_k$ for all k .

If a strict coarsening occurred, then distinct maximal degenerations – differing by a double curve – would be identified in the target. The proposed input is the KSBA principle that the dual complex uniquely identifies maximally degenerate limits; granting this, no such identification can occur, and hence no coarsening occurs.

This argument is presented as the program’s intended route, not as a completed proof.

9C.3. Zariski’s Main Theorem and the classifying isomorphism.

Theorem 9C.6 (*Zariski’s Main Theorem (classical)*).

Let $f: X \rightarrow Y$ be a birational, finite morphism between normal varieties, with Y proper. Then f is an isomorphism.

Remark 9C.7: theorem 9C.6 is a classical theorem [Har10]; a finite birational morphism onto a normal variety is an isomorphism. It is invoked here only as an external tool; the content below – that the classifying map ϕ satisfies its hypotheses – is the program’s own claim.

Conjecture 9C.8 (*The classifying map is an isomorphism*).

For the Enriques compactification, the classifying map

$$\phi: B^\nu \rightarrow \overline{F_{\text{En},2}}^\nu$$

is expected to satisfy the four hypotheses of theorem 9C.6:

1. **Birational:** ϕ is an isomorphism on the interior $F_{\text{En},2}$.
2. **Finite:** by the proposed “no coarsening” semifan argument of ?? 9C.4.
3. **Normal:** both spaces are normal by construction (B^ν as a normalization, $\overline{F_{\text{En},2}}^\nu$ likewise).
4. **Proper:** the KSBA moduli stacks are proper [Ale+25; Kol23].

Granting these, Zariski’s Main Theorem would give that ϕ is an isomorphism, identifying the geometric KSBA limits with the explicitly constructible semitoroidal boundary.

Remark 9C.9 (*Status of the hypotheses*): Of the four inputs to ?? 9C.8, properness and normality are the ambient structural facts of the KSBA and Baily–Borel constructions, while birationality on the interior and finiteness are the load-bearing steps: finiteness is the still-open ?? 9C.4, and the extension of the Enriques involution (?? 9C.2) is what makes the universal quotient family – and hence ϕ – available over the boundary in the first place. The isomorphism statement therefore inherits the open status of ?? 9C.2?? 9C.4, and is the compactification-side counterpart of the semitoroidal comparison ?? 8C.13. This conclusion also refines the incomplete normalization statement theorem 4G.4, whose migrated proof leaves both the stabilizer identity and finiteness unestablished.

10. OPEN PROBLEMS

10A. Open problems and technical gaps.

Remark 10A.1 (*Orientation*): This section records the known open problems and computational obstacles of the Coble moduli program. Each entry is a genuine gap between the results established elsewhere in this document and the fully rigorous, coordinate-explicit statements one would want for parity with the

The header records “rays” in total, but per-cusp counts sum to 2+9+9+11+3 = 32. II+Type III rays. discrepancy is explained by Type I rays being shared between adjacent 0-cusps (each incident to 3 0-cusps), so that total of 27 counts distinguishes Type II rays; confirmed intended bookkeeping.

Enriques degree-2 program of [Ale+25]. The problems are grouped by theme: lattice-theoretic verifications, the enumeration of isotropic orbits and cusps together with the reflection-group combinatorics, the *dlt* and KSBA stable models, and the foundational equations and computational checks. None of the statements below is claimed to be settled; several cross-reference partial results already in the text, and in each case the *residue* that remains open is stated explicitly.

10A.1. Lattice-theoretic gaps.

Remark 10A.2 (Open problem: isometry class and genus of the Coble lattices): The invariants (r, a, δ) of the geometric lattice $S_{\text{Co}} \cong (11, 11, 1)_1$ (signature $(1, 10)$, Gram matrix $\text{diag}(2, -2, \dots, -2)$, equivalently $\langle -2 \rangle \oplus E_{10}(2)$) and of the transcendental lattice $T_{\text{Co}} = S_{\text{Co}}^{\perp L_{\text{K3}}} \cong (11, 11, 1)_2$ (signature $(2, 9)$) are recorded in the Coble lattice table (table 2), and both satisfy $q_{S_{\text{Co}}} \cong q_{T_{\text{Co}}} \cong (\mathbf{Z}/2\mathbf{Z})^{11}$ with $q_{S_{\text{Co}}} = -q_{T_{\text{Co}}} \bmod 2\mathbf{Z}$. What is not yet formally established is the *isometry-class verification* and the *genus decomposition*: whether the genus of T_{Co} contains a unique isometry class. Concretely:

- compute the Gram matrices of S_{Co} and T_{Co} and verify their (r, a, δ) invariants and genus cardinality via Nikulin’s classification (the $r > a$ check for a 2-elementary lattice) [Nik80];
- confirm that $r > a$ forces a unique class in the genus, so that the reduction of isotropic-orbit questions to the discriminant form $q_{T_{\text{Co}}}$ (theorem 5D.9) is justified.

Remark 10A.3 (Open problem: explicit primitive embedding matrices): The chain of primitive embeddings

$$T_{\text{Co}} \hookrightarrow T_{\text{En}} \hookrightarrow T_{\text{dP}} \hookrightarrow L_{\text{K3}}$$

is used throughout (lemma 4G.2, lemma 4G.1), and Nikulin’s apparatus [Nik80, Prop. 1.14.4, 1.15.2] is the intended tool for the uniqueness of the primitive embedding $T_{\text{Co}} \hookrightarrow T_{\text{En}}$ and for the surjectivity of $O(L) \rightarrow O(T_{\text{Co}})$. What remains is to *exhibit the embedding matrices in coordinate bases*: derive the explicit primitive-embedding matrices realizing the chain above, and verify their primitivity directly rather than by invocation. The same coordinate deficit affects the derivation of the explicit equations for C and X (below): the invariants are listed, but the primitivity of the lattice embeddings lacks a rigorous derivation in terms of coordinate bases.

Conjecture 10A.4 (Open problem: the horizontal folding involution and its eigenspaces).

There is an orthogonal involution θ on the K3 lattice $L_{\text{K3}} \cong U^3 \oplus E_8^2$ whose invariant and coinvariant sublattices are

$$L_{\text{K3}}^{\theta} \cong T_{\text{Co}}, \quad L_{\text{K3}}^{-\theta} \cong S_{\text{Co}},$$

and whose action on roots induces the *horizontal folding* of the $(18, 0, 0)_1$ Coxeter diagram, as a functorial consequence of the lattice isometry (cf. definition 5B.3, lemma 5B.5, and the invariant/coinvariant construction of definition 4F.2). The involution should swap the polarization generators between sectors, $h_{\text{En}} \leftrightarrow h_{\text{Co}}$. The open technical content is to *write down the 22×22 matrix of θ on the standard basis of $U^3 \oplus E_8^2$* and to verify the isometry classes of its ± 1 eigenspaces by the (r, a, δ) comparison of proposition 4F.3, confirming $L_{\text{K3}}^{+} \cong T_{\text{Co}}$, $L_{\text{K3}}^{-} \cong S_{\text{Co}}$ (2-elementary check), and that the induced embedding $T_{\text{Co}} \hookrightarrow T_{\text{En}}$ is primitive.

10A.2. Isotropic orbits, cusps, and reflection groups.

Question 10A.5 (Open problem: enumeration of isotropic orbits by Sterk’s technique)

How many orbits of primitive isotropic vectors does T_{Co} have under $O(T_{\text{Co}})$, $O^*(T_{\text{Co}})$, and the arithmetic group Γ_{Co} ? Sterk’s technique [Ste91] determines these by analyzing the orbits of the images (lifts) in the discriminant group $A_{T_{\text{Co}}} \cong (\mathbf{Z}/2\mathbf{Z})^{11}$ under $O(q_{T_{\text{Co}}})$: for a 2-elementary lattice with $r > a$ the genus contains a unique class and $O(T) \rightarrow O(q_T)$ is surjective [Nik80], so a primitive isotropic vector v with $\text{div}(v) = d$ is determined up to $O(T)$ by the tuple $(\text{div}(v), \bar{v} \in A_T, v^2 = 0)$. The number of such orbits in $A_{T_{\text{Co}}}$ is not yet computed. The open work is to:

- enumerate the isotropic vectors of $A_{T_{\text{Co}}}$ and compute their orbits under $O(q_{T_{\text{Co}}})$;
- lift these orbits to T_{Co} using Sterk’s lifting theorems and verify that exactly one $O^*(T)$ -orbit exists in divisibility 2;
- verify that the $O(T)$ -orbits coincide with the Γ_{Co} -orbits, so that the Baily–Borel 0-cusp is unique.

That every primitive isotropic $v \in T_{\text{Co}}$ has $\text{div}_{T_{\text{Co}}}(v) = 2$ is already known (lemma 7E.2), which fixes the divisibility datum but not the orbit count.

Remark 10A.6 (*Open problem: explicit generators of Γ_{Co} and uniqueness of the 1-cusp*): The arithmetic group governing the Coble locus is the stabilizer of the polarization h_{Co} inside $\text{O}(T_{\text{En}})$, cut further by the horizontal folding involution θ ,

$$\Gamma_{\text{Co}} = \text{Stab}_{\text{O}(T_{\text{En}})}(h_{\text{Co}}) \cap Z_{\text{O}(T_{\text{En}})}(\theta),$$

where in the Enriques sector $h_{\text{En}} = e + f$ is the degree-2 polarization ($h^2 = 2$) and h_{Co} is induced by the hyperplane class E_0 . An explicit representation of Γ_{Co} by *matrix generators* is presently a stub. The open work is to:

- compute the stabilizer/centralizer intersection in the Enriques (equivalently L_{K3}) lattice to produce a minimal set of matrix generators for Γ_{Co} , as the intersection of the reflection group $W(T)$ with the centralizer $Z(\theta)$ and the stabilizer of the primitive vector h_{Co} ($h_{\text{Co}}^2 = 2$);
- verify the *uniqueness of the 1-cusp* by enumerating all $\text{O}(T_{\text{Co}})$ -orbits of primitive isotropic planes $J \subset T_{\text{Co}}$, computing the negative-definite quotient $J^\perp/J$ for each, and confirming isometry with $A_1^{\oplus 7}$.

This refines, at the level of orbits and generators, the 1-cusp correspondence $(7, 7, 1)_0 \mapsto (8, 6, 0)_0$ established in lemma 7E.5.

Conjecture 10A.7 (*Open problem: uniqueness of the maximal parabolic subdiagram*).

The reflection group $W(S_{\text{Co}})$ acts on the period domain, and the 0-cusp $(9, 9, 1)_1$ is described by a maximal parabolic subdiagram of the Coxeter diagram $G_{S_{\text{Co}}}$. The claim underlying “one 0-cusp” is that $\tilde{B}_7(2)$ is the *only* maximal parabolic subdiagram of the highly connected 10-node diagram $G_{S_{\text{Co}}}$. Establishing this requires a subdiagram search on the 10×10 Gram matrix of roots to identify all maximal parabolic configurations and confirm that $\tilde{B}_7(2)$ is unique. (The underlying finite root system is of type B_7 , matching the double-edge B/C -type diagram at the 0-cusp rather than the simply-laced A_7 .)

Remark 10A.8 (*Open problem: hyperbolic-quotient derivation of the cusp correspondence*): The cusp correspondence between the Coble cusps $(9, 9, 1)$, $(7, 7, 1)$ and their Enriques predecessors is established in theorem 7E.1 (via the divisibility computations of lemma 7E.4 and lemma 7E.5). The migrated notes propose an independent *hyperbolic-quotient* derivation that would place the correspondence on a self-contained lattice footing rather than resting on the alignment of invariants:

- compute the isometry type of the hyperbolic quotient $e^\perp/e$ directly for a primitive isotropic $e \in T_{\text{Co}}$;
- use $\text{div}_{T_{\text{Co}}}(e)$ to pin down the unique $\text{O}(T_{\text{Co}})$ -orbit and verify that the quotient matches the Enriques signatures $(10, 8, 0)_1$ for 0-cusps and $(8, 6, 0)_0$ for 1-cusps.

The residue that is genuinely open is the *orbit-uniqueness input*: the proof of lemma 7E.3 presently *assumes* a unique Γ_{Co} -orbit of isotropic vectors in T_{Co} , which is exactly the enumeration left open above.

10A.3. *dlt and KSBA stable models.*

Remark 10A.9 (*Open problem: Coble-specific dlt models and pot geometry*): The degree-2 Enriques program modernizes Morrison’s flowerpots [Mor81] into *dlt* stable pairs, but the *Coble-specific dlt models* are not yet detailed. The open geometric content is:

- **Coble pot geometry.** Define the explicit stable pair $(\mathcal{V}, \mathcal{D})$ whose “Pot” component is a rational Coble surface, carrying a $\frac{1}{4}(1, 1)$ cyclic quotient singularity corresponding to the nodes of the rational sextic C . This is the local singularity package already conjectured for the stable quotient in ?? 8C.8, seen here on the *dlt* model (cf. definition 6C.2, definition 6C.4).
- **Stalk assembly.** Describe the transition of the stalk assembly and the integral-affine configuration as the Enriques surface log-collapses onto the discriminant divisor Δ , where the K3 cover becomes nodal.

Remark 10A.10 (*Open problem: monodromy invariants and stable models $B(\lambda)$*): Stable limits of Coble surfaces arise as S_2 -quotients of nodal K3 surfaces, parameterized by a monodromy invariant $\ell \in \mathcal{H}$ (surgery sizes) through the construction $B(\lambda)$ of [Ale+25]. The open work is to:

- determine the map from the transcendental polarization h_{Co} to the discretization ℓ on the dual complex;
- verify the slc stability of the resulting stable pair $(Z, \varepsilon C)$ for specific surgery vectors ℓ .

This is the discrete-datum side of the KSBA program whose stability obligations are recorded in ?? 8C.8 and the surrounding KSBA discussion.

10A.4. *Foundational equations and computational verification.*

Remark 10A.11 (*Open problem: explicit equations for the Coble curve and its K3 cover*): A Coble surface S is the blowup of $\mathbf{P}^2$ at the ten A_1 nodes of an irreducible rational plane sextic

$$C = \{F(x, y, z) = 0\}, \quad F(x, y, z) = \sum_{i+j+k=6} a_{ijk} x^i y^j z^k,$$

subject to the nodal conditions $F(p_m) = \partial_x F(p_m) = \partial_y F(p_m) = \partial_z F(p_m) = 0$ at the ten special positions $p_1, \dots, p_{10} \in \mathbf{P}^2$; the moduli space of such sextics is 9-dimensional, and explicit models may be sought from the Steiner sextic or from index-2 Halphen pencils (cf. lemma 6D.2, lemma 3C.1). The K3 cover is the double cover $X \xrightarrow{2:1} S$ of $\mathbf{P}^2$ branched along C , with equation $w^2 = F(x, y, z)$ in $\mathbf{P}(1, 1, 1, 3)$, whose ten A_1 nodes lie above the positions p_m (at $w = 0$, $[x : y : z] = p_m$). What is missing is a *worked instance*: derive an explicit equation $F(x, y, z) = 0$ for a rational sextic with ten nodes together with the corresponding cover $w^2 = F$, realizing C as the image of $\mathbf{P}^1$ under a degree-6 map $(s : t) \mapsto [f_0 : f_1 : f_2]$. Such an instance would anchor the coordinate derivations demanded by the lattice-embedding and involution problems above.

Remark 10A.12 (*Lattice and orbit anchors for the computations*): The problems above share a small set of concrete starting data, recorded here for reference:

- **Isotropic vectors.** In S_{Co} , primitive isotropic lines are represented by $v = e_0 \pm e_i$ (with $e_0^2 = 2$, $e_i^2 = -2$).
- **Polarization basis.** The degree-2 polarization $h \in T_{\text{En}}$ is $h = e + f$ in a standard U -basis, which must be identified in a basis compatible with θ .
- **Discriminant forms.** $q_{S_{\text{Co}}} : (\mathbb{F}_2)^{11} \rightarrow \mathbf{Q}/2\mathbf{Z}$ and $q_{T_{\text{Co}}} : (\mathbb{F}_2)^{11} \rightarrow \mathbf{Q}/2\mathbf{Z}$, with isometry of complements forcing $q_{S_{\text{Co}}} = -q_{T_{\text{Co}}} \bmod 2\mathbf{Z}$.
- **Sterk orbit lift.** For the 2-elementary T_{Co} , an orbit is determined by the tuple $(\text{div}(v), \bar{v} \in A_{T_{\text{Co}}}, v^2 = 0)$, reducing the search to $\mathbb{F}_2$ -vector-space orbits under $O(q_{T_{\text{Co}}})$.
- **Γ_{Co} generators.** Computed as the intersection of the reflection group $W(T)$ with the centralizer $Z(\theta)$ and the stabilizer of the primitive vector h_{Co} .

11. APPENDIX

11A. Formulae.

Remark 11A.1: Let $f : Y = \text{Bl}_V X \dashrightarrow X$ be the blowup of X along a subvariety V . Then one has the adjunction formula

$$K_Y = f^* K_X + (\text{codim}_V(X) - 1)E.$$

Note that if X is a surface and V is a point, this recovers

$$K_Y = f^* K_X + E.$$

Remark 11A.2: Let $f : Y \rightarrow X$ be a branched cover with branch locus B and ramification locus R . Then

$$K_Y = f^* K_X + R, \quad R := \sum_{p \in X} \text{len}(\Omega_{X/Y, p})[p].$$

11B. Diagrams. Conventions:

1. White vertices: $v^2 = -2$.
2. Black vertices: $v^2 = -4$.

Classical and affine Dynkin diagrams. Table 6 is a table of (labeled) classical and affine Dynkin diagrams.

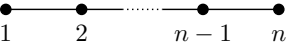
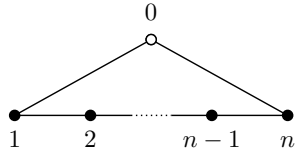
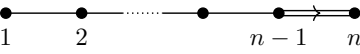
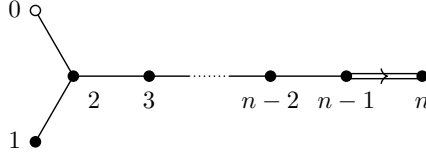
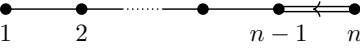
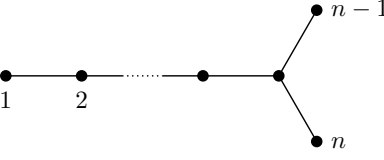
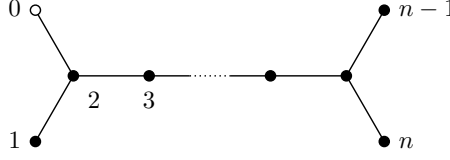
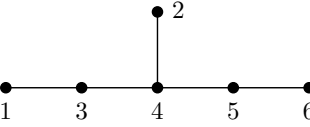
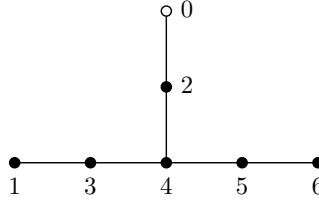
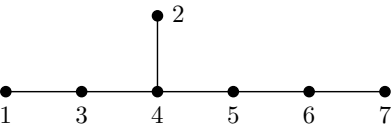
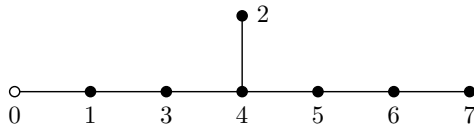
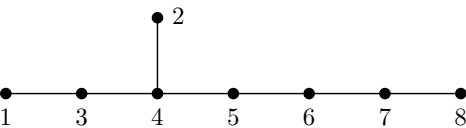
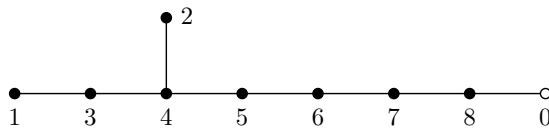
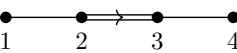
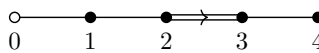
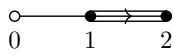
Classical Type	Affine Type
A_n 	$\tilde{A}_n$ 
B_n 	$\tilde{B}_n$ 
C_n 	$\tilde{C}_n$ 
D_n 	$\tilde{D}_n$ 
E_6 	$\tilde{E}_6$ 
E_7 	$\tilde{E}_7$ 
E_8 	$\tilde{E}_8$ 
F_4 	$\tilde{F}_4$ 
G_2 	$\tilde{G}_2$ 

TABLE 6. Classical and Affine Dynkin Diagrams

Mirror move algorithm.

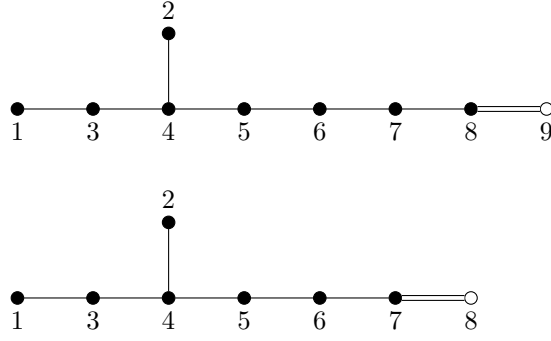
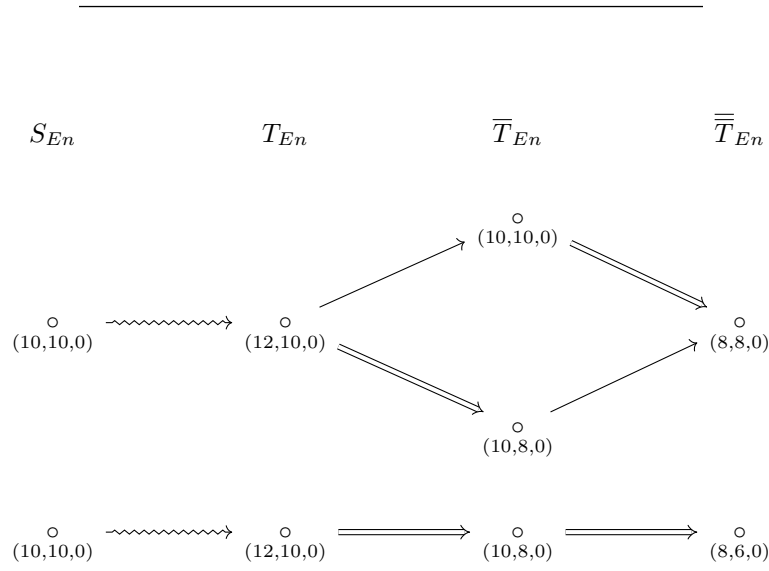
FIGURE 31. Coxeter diagrams for T_{Co} and T_{En} parabolics

FIGURE 32. Mirror moves for the Enriques lattices.

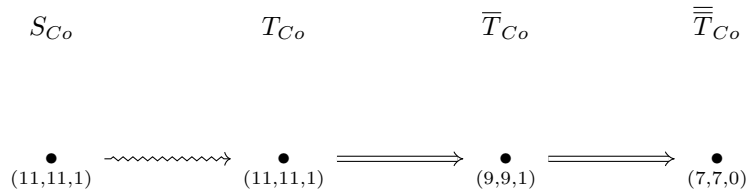
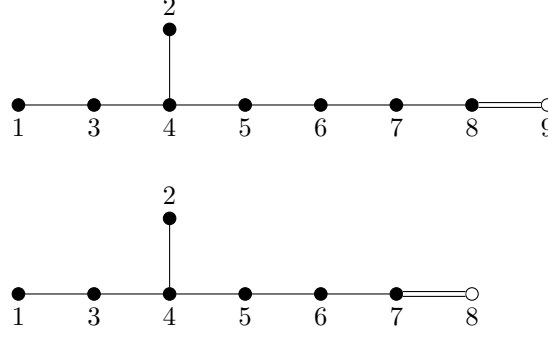


FIGURE 33. Geometric cusp correspondence (Coble)

FIGURE 34. The Coxeter diagram of $L_{Co,1} := \langle 2 \rangle \oplus E_8(2)$.

12. EXTRACTED REFERENCES

12A. **Extracted references.** This appendix imports the per-paper literature-summary notes verbatim from the project knowledge base. Each entry preserves the note’s original “Used here for” and “Project relevance” lines and its abstract as written; only the citation key has been remapped to its canonical build key where the note’s filename key was stale (the remap is stated in each entry). The **Papers Index** navigation note is not imported.

12A.1. *ABE22* — Valery Alexeev, Adrian Brunsate, Philip Engel (2022): *Compactifications of moduli of elliptic K3 surfaces*. Bibkey: ABE22 (imported from note AB21; remapped AB21 $\rightarrow$ ABE22). Cited as [ABE22].

Abstract. KSBA and toroidal compactifications for elliptic K3 surfaces, used in the paper as background for stable-pair compactification techniques.

- **Used here for:** background on stable-pair compactifications of K3 moduli, especially in the IAS / stable-limits context.
- **Project relevance:** supplies KSBA techniques and comparison principles for compactifications of K3 moduli spaces.

12A.2. *AECS25* — Valery Alexeev, Philip Engel, D. Zack Garza, Luca Schaffler (2023): *Compact Moduli of Enriques Surfaces with a Numerical Polarization of Degree 2*. Bibkey: AECS25 (imported from note AECS23; remapped AECS23 $\rightarrow$ AECS25). Cited as [Ale+25].

Abstract. We describe a geometric, stable pair compactification of the moduli space of Enriques surfaces with a numerical polarization of degree 2, and identify it with a semitoroidal compactification of the period space.

Contents of the note (the paper’s table of contents):

- 1. Introduction — 1
- 2. General setup — 3
- 3. Cusps and Coxeter diagrams — 8
- 4. Dlt models via integral-affine structures on the disk and $\mathbb{RP}^2$ — 16
- 5. Toroidal, semitoroidal, and KSBA compactifications — 28
- 6. ABCDE surfaces — 31
- 7. KSBA stable degenerations of Enriques surfaces — 38
- References — 41

12A.3. *AET23* — Valery Alexeev, Philip Engel, Alan Thompson (2023): *Stable pair compactification of moduli of K3 surfaces of degree 2*. Bibkey: AET23 (unchanged). Cited as [AET23].

Abstract. Stable-pair compactification for degree-2 K3 surfaces, providing a direct KSBA model for a key K3 moduli problem adjacent to the Coble and Enriques constructions.

- **Used here for:** the stable-pair side of degree-2 K3 compactifications in the stable-limits discussion.
- **Project relevance:** forms part of the KSBA background for comparing period-domain compactifications with geometric compactifications.

12A.4. *AN06* — Valery Alexeev, Viacheslav V. Nikulin (2006): *Del Pezzo and K3 surfaces*. Bibkey: AN06 (unchanged; not listed in the citation-key audit — used as-is). Cited as [AN06].

Abstract. Reference on del Pezzo and K3 surfaces used for Coxeter diagrams, reflective lattices, and parabolic subdiagram calculations.

- **Used here for:** the Coble Coxeter diagrams and the identification of maximal parabolic subdiagrams.
- **Project relevance:** supports the cusp-diagram computations around $(9, 9, 1)_1$ and the $\tilde{B}_7(2)$ parabolic.

12A.5. *AS15* — Michela Artebani, Alessandra Sarti (2015): *Symmetries of order four on K3 surfaces*. Bibkey: AS15 (unchanged; not remapped in the audit — used as-is. Note: the audit records that at one *citation site* — the Rational Sextic Hurwitz/Lemma 2.2 attribution — the key AS15 was wrong and was corrected to AD18; that correction concerns the site, not this Artebani–Sarti note, whose key is retained). Cited as [AS15].

Abstract. Study of order-4 nonsymplectic symmetries on K3 surfaces, used here for the branched-cover and Hurwitz-type calculations behind the classical Coble sextic construction.

- **Used here for:** the Hurwitz-type calculation that identifies the proper transform of the rational sextic with the anti-bicanonical divisor in the classical Coble construction.
- **Project relevance:** supplies one of the cited inputs behind the sextic-to-Coble transition.

12A.6. *BHO+12* — Blekherman, Hauenstein, Ottem, Ranestad, Sturmfels (2012): *Algebraic boundaries of Hilbert’s SOS cones*. Bibkey: BHO+12 (imported from note BHO+11; remapped BHO+11 $\rightarrow$ BHO+12 per the audit’s year-drift resolution, arXiv 2011 $\rightarrow$ Compositio Math. 148 (2012), Zotero GJZ44LJI. Note: the original note frontmatter carried only the placeholder title “BHO+11”; the author/title/year shown here are supplied from the audit’s resolution of the work. The audit is internally inconsistent about this work — the “RESOLVED” block pins it to BHO+12, while a later verification line states it is “not in Zotero, still unresolved”; the resolved key is used here). Cited as [Ble+12].

Abstract. Appendix reference for the Severi variety $V_{6,10}$ of rational plane sextics with ten nodes.

- **Used here for:** the statement that the nodal rational sextics relevant to the classical Coble construction form a Severi variety $V_{6,10}$ whose generic point has ten A_1 -singularities.
- **Project relevance:** supports the appendix note separating the generic ten-node Coble sextic from more degenerate rational sextics.

12A.7. *CD12* — Serge Cantat, Igor Dolgachev (2012): *Rational surfaces with a large group of automorphisms*. Bibkey: CD12 (unchanged; audit confirms CD12 = Cantat–Dolgachev, live Zotero PMC33V8M). Cited as [CD12].

Abstract. Reference on rational surfaces with large automorphism groups, including Cremona-special point sets, Coble and Halphen surfaces, and moduli consequences.

- **Used here for:** Cremona-special and unnodal Coble surfaces, the Halphen blowdown lemma, and the modular-curve description of Enriques and Coble cusps.
- **Project relevance:** provides geometric structure theorems for rational Coble surfaces and their automorphism groups.

12A.8. *CDL25* — Francois Cossec, Igor Dolgachev, Christian Liedtke (2024): *Enriques surfaces I*. Bibkey: CDL25 (imported from note CDL24; remapped CDL24 $\rightarrow$ CDL25). Cited as [CDL25].

Abstract. Draft reference on Enriques and Coble surfaces used throughout the paper for 2-elementary lattice tables, fixed loci, and Coble/Halphen geometry.

- **Used here for:** the lattice table for Coble surfaces, fixed-locus formulas for the K3 cover involution, boundary-component geometry, and Halphen/Coble results.
- **Project relevance:** this is the main modern reference behind many Coble facts imported into the paper’s introduction and appendix.

12A.9. *Cob19* — Arthur B. Coble (1919): *The ten nodes of the rational sextic and of the Cayley symmetroid*. Bibkey: Cob19 (unchanged). Cited as [Cob19].

Abstract. Classical source for the rational sextic with ten nodes and the associated Coble surface construction.

- **Used here for:** the original plane-sextic construction of the classical Coble surface.
- **Project relevance:** foundational citation for the blowup of the ten nodes of a rational sextic.

12A.10. *Cob29* — Arthur B. Coble (1929): *Algebraic geometry and theta functions*. Bibkey: Cob29 (unchanged). Cited as [Cob29].

Abstract. Classical monograph on point configurations, Cremona geometry, and the background of Coble surfaces.

- **Used here for:** classical background on point configurations and the historical geometry surrounding Coble surfaces.
- **Project relevance:** complements the original sextic-node construction with the broader classical theory.

12A.11. *DK25* — Igor Dolgachev, Shigeyuki Kondo (2024): *Enriques surfaces II*. Bibkey: DK25 (imported from note DK24; remapped DK24 $\rightarrow$ DK25). Cited as [DK25].

Abstract. Draft reference used for terminal Coble surfaces of K3 type, K3 double covers, basic rational surface structure, and Halphen constructions.

- **Used here for:** the double-cover construction $X \rightarrow S$, the structure of terminal Coble surfaces of K3 type, and the Halphen-surface description.
- **Project relevance:** one of the main sources for the paper’s introductory and geometric facts about Coble surfaces.

12A.12. *DM20* — Igor Dolgachev, Dimitri Markushevich (2020): *Lagrangian tens of planes, Enriques surfaces and holomorphic symplectic fourfolds*. Bibkey: DM20 (imported from note DM19; remapped DM19 $\rightarrow$ DM20). Cited as [DM20].

Abstract. Source used for the definition of Coble surfaces and related Enriques-surface geometry.

- **Used here for:** an explicit modern definition of Coble surfaces.
- **Project relevance:** supplies a concise reference point for the paper’s opening definition of the Coble locus.

12A.13. *Dol12* — Igor V. Dolgachev (2012): *Classical algebraic geometry*. Bibkey: Dol12 (unchanged; the audit remaps a specific *site* from Dol12 to Dol17 but does not retire the Dol12 key, which is this note’s Classical Algebraic Geometry reference). Cited as [Dol12].

Abstract. Modern reference on classical algebraic geometry, used here for the geometric degeneration picture of Coble surfaces and the quartic del Pezzo model coming from isotropic sequences.

- **Used here for:** the geometric description of Coble surfaces as degenerations of Enriques/K3 quotients and the linear systems $|2f_i + 2f_j|$ mapping to quartic del Pezzo surfaces.
- **Project relevance:** anchors the source citations for the geometric Coble remarks migrated into the vault.

12A.14. *Dol17* — Igor Dolgachev (2017): *Salem numbers and Enriques surfaces*. Bibkey: Dol17 (unchanged). Cited as [Dol17].

Abstract. Background source on Enriques-surface automorphisms and lattice-theoretic constructions.

- **Used here for:** the geometric remark on degenerating an Enriques K3 cover to the Coble case.
- **Project relevance:** provides background on Enriques involutions and automorphisms that informs the geometric discussion.

12A.15. *DZ99* — Igor Dolgachev, D. Zhang (1999): *Coble rational surfaces*. Bibkey: DZ99 (unchanged). Cited as [DZ99].

Abstract. Foundational paper on Coble rational surfaces, including their relationship to Halphen surfaces and degenerations of Enriques surfaces.

- **Used here for:** the relation between Coble surfaces and Halphen surfaces, and for the broader geometry of Coble rational surfaces.
- **Project relevance:** one of the main classical-modern references for the geometry behind the paper’s moduli problem.

12A.16. *GK20 — Rational plane sextics with worse singularities (appendix reference)*. Bibkey: **GK20** (unchanged; not listed in the citation-key audit — used as-is. Note: the original note frontmatter carried only the placeholder title “GK20” with no author or year, so no fuller title is supplied here). Cited as [\[GK20\]](#).

Abstract. Appendix reference for rational plane sextics with worse singularities, used to note that rational sextics need not lie in the ten-node Severi stratum.

- **Used here for:** the example that rational sextics can have fewer than ten A_1 -points when more degenerate singularities such as A_2 occur.
- **Project relevance:** supports the caveat that the classical Coble construction uses a special generic nodal stratum rather than all rational sextics.

12A.17. *Hor77 — Eiji Horikawa (1978): On the periods of Enriques surfaces. II*. Bibkey: **Hor77** (imported from note **Hor78**; remapped **Hor78** $\rightarrow$ **Hor77**, the global key for “Periods II”). Cited as [\[Hor77\]](#).

Abstract. Classical period-domain source behind the Horikawa model for Enriques and Coble surfaces.

- **Used here for:** the $\mathbf{P}^1 \times \mathbf{P}^1$ model with a τ -invariant anti-bicanonical branch curve.
- **Project relevance:** underlies the Horikawa construction that realizes the Coble locus as a boundary of Enriques moduli.

12A.18. *KK72 — Peter Kiernan, Shoshichi Kobayashi (1972): Satake compactification and extension of holomorphic mappings*. Bibkey: **KK72** (unchanged). Cited as [\[KK72\]](#).

Abstract. Technical compactification result used to extend locally closed period-domain embeddings to Baily-Borel compactifications.

- **Used here for:** extending the lattice-induced embeddings on period spaces to morphisms of Baily-Borel compactifications.
- **Project relevance:** supplies the compactification-theoretic justification for the maps between F_{Co} , F_{En} , and $F_{(2,2,0)}$.

12A.19. *Mor81 — David R. Morrison (1981): Semistable degenerations of Enriques’ and hyperelliptic surfaces*. Bibkey: **Mor81** (unchanged). Cited as [\[Mor81\]](#).

Abstract. Original source for flower, stalk, pot, and flowerpot degenerations of Enriques-type surfaces.

- **Used here for:** the flowerpot description of degenerations expected to model parts of the Coble boundary.
- **Project relevance:** historical source behind the degeneration picture later reformulated in dlt / KSBA language.

12A.20. *Nik80 — V. V. Nikulin (1979): Integer symmetric bilinear forms and some of their geometric applications*. Bibkey: **Nik80** (imported from note **Nik79**; remapped **Nik79** $\rightarrow$ **Nik80**, the global key for the Integer Symmetric Bilinear Forms paper). Cited as [\[Nik80\]](#).

Abstract. Fundamental reference for 2-elementary lattices, discriminant forms, primitive embeddings, and orthogonal-group surjectivity.

- **Used here for:** classification of 2-elementary lattices, uniqueness of primitive embeddings, and surjectivity statements for orthogonal groups.
- **Project relevance:** this is the lattice-theoretic backbone of the Coble, Enriques, and degree-2 K3 embedding arguments.

12A.21. *Nue15 — Howard Nuer (2016): Unirationality of moduli spaces of special cubic fourfolds and K3 surfaces*. Bibkey: **Nue15** (unchanged; the audit maps **Nue16** $\rightarrow$ **Nue15** as the single global entry). Cited as [\[Nue15\]](#).

Abstract. Reference used for the statement that the nodal quotient appearing on the Enriques boundary is a Coble surface.

- **Used here for:** the identification of a quotient with a Coble surface in the Enriques-boundary construction.
- **Project relevance:** provides a supporting citation for the geometric transition from nodal K3 covers to Coble surfaces.

12A.22. *Ols04* — *Martin C. Olsson (2004): Semistable degenerations and period spaces for polarized K3 surfaces.* Bibkey: **Ols04** (unchanged). Cited as [\[Ols04\]](#).

Abstract. Foundational treatment of semistable degenerations and period spaces for polarized K3 surfaces, used here for flowerpot degenerations and their relation to Coble boundary behavior.

- **Used here for:** the flower, stalk, and pot terminology and the Type I / flowerpot degeneration picture applied to Coble surfaces.
- **Project relevance:** provides the older degeneration language that the vault uses to relate the Coble boundary to modern KSBA and IAS viewpoints.

12A.23. *PS71* — *I. I. Pjateckii-Shapiro, I. R. Shafarevich (1971): A Torelli theorem for algebraic surfaces of type K3.* Bibkey: **PS71** (unchanged). Cited as [\[PS71\]](#).

Abstract. Classical Torelli theorem for algebraic K3 surfaces, used to pass from lattice data to period-domain moduli.

- **Used here for:** the Torelli input behind the direct period-domain construction of F_{Co} .
- **Project relevance:** justifies reconstructing the Coble K3 cover from Hodge-theoretic lattice data.

12A.24. *Sha81a* — *Jayant Shah (1981): Projective degenerations of Enriques' surfaces.* Bibkey: **Sha81a** (imported from note **Sha81**; remapped **Sha81** $\rightarrow$ **Sha81a**, distinguishing Shah's Enriques paper from the global **Sha81**). Cited as [\[Sha81\]](#).

Abstract. Projective-degeneration source for Enriques surfaces, cited alongside Morrison in the discussion of flowerpot degenerations.

- **Used here for:** projective degenerations related to the flowerpot picture.
- **Project relevance:** forms part of the classical degeneration background that the paper interprets on the Coble boundary.

12A.25. *Ste91* — *Hans Sterk (1991): Sterk 1991 note.* Bibkey: **Ste91** (imported from note **Ste91-note**; remapped the note's **Ste91-note** filename key $\rightarrow$ the work's citation key **Ste91**, which the audit confirms exists in the global bib). Cited as [\[Ste91\]](#).

Abstract. Supplemental note linking the existing Ste91 extraction to its role in the Coble paper.

- **Used here for:** divisibilities of isotropic vectors, cusp labels, and the identification of Sterk cusp 2.
- **Project relevance:** the Coble cusp map to $(18, 0, 0)_1$ is interpreted through Sterk's cusp classification.

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